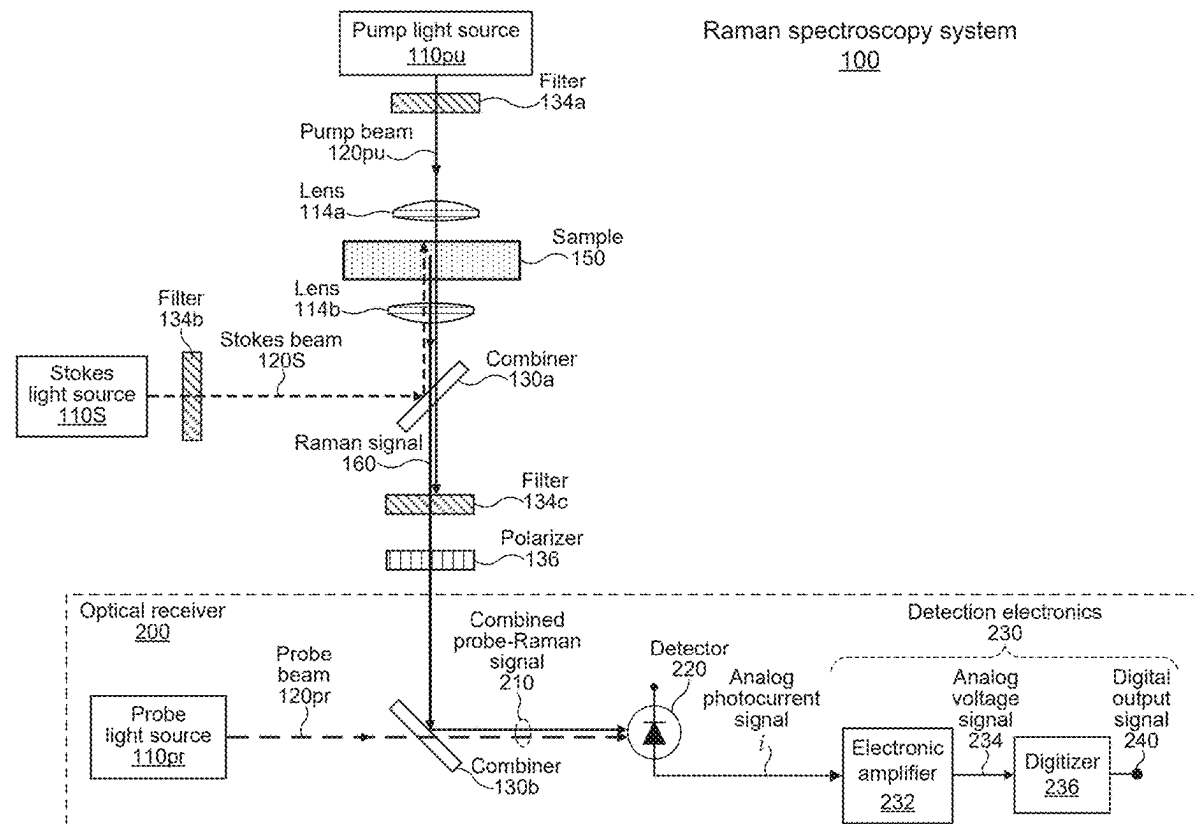




US 20250277745A1

(19) **United States**(12) **Patent Application Publication**
LaChapelle et al.(10) **Pub. No.: US 2025/0277745 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **RAMAN SPECTROSCOPY SYSTEM WITH
COUNTERPROPAGATING PUMP AND
STOKES BEAMS**(52) **U.S. Cl.**
CPC ... **G01N 21/65** (2013.01); **G01N 2201/06113**
(2013.01)(71) Applicant: **Haemanthus, Inc.**, Austin, TX (US)(72) Inventors: **Joseph G. LaChapelle**, Philomath, OR
(US); **Roger S. Cannon**, Oviedo, FL
(US)(21) Appl. No.: **19/020,938**(22) Filed: **Jan. 14, 2025****Related U.S. Application Data**(63) Continuation-in-part of application No. 18/762,412,
filed on Jul. 2, 2024, now Pat. No. 12,203,862.(60) Provisional application No. 63/560,521, filed on Mar.
1, 2024.**Publication Classification**(51) **Int. Cl.**
G01N 21/65 (2006.01)(57) **ABSTRACT**

In one embodiment, a system includes (i) a pump light source configured to produce a pump beam of light at a pump frequency and (ii) a Stokes light source configured to produce a Stokes beam of light at a Stokes frequency, where the pump and Stokes frequencies are offset by a frequency offset Ω . The system also includes one or more optical elements configured to: (i) direct the pump and Stokes beams of light to a sample along counterpropagating beam paths and (ii) collect a Raman signal produced by coherent Raman scattering of the pump and Stokes beams of light at the sample. The system further includes an optical receiver configured to detect the Raman signal. The optical receiver includes a probe light source configured to produce a probe beam of light at a probe frequency, an optical detector, and an electronic circuit.



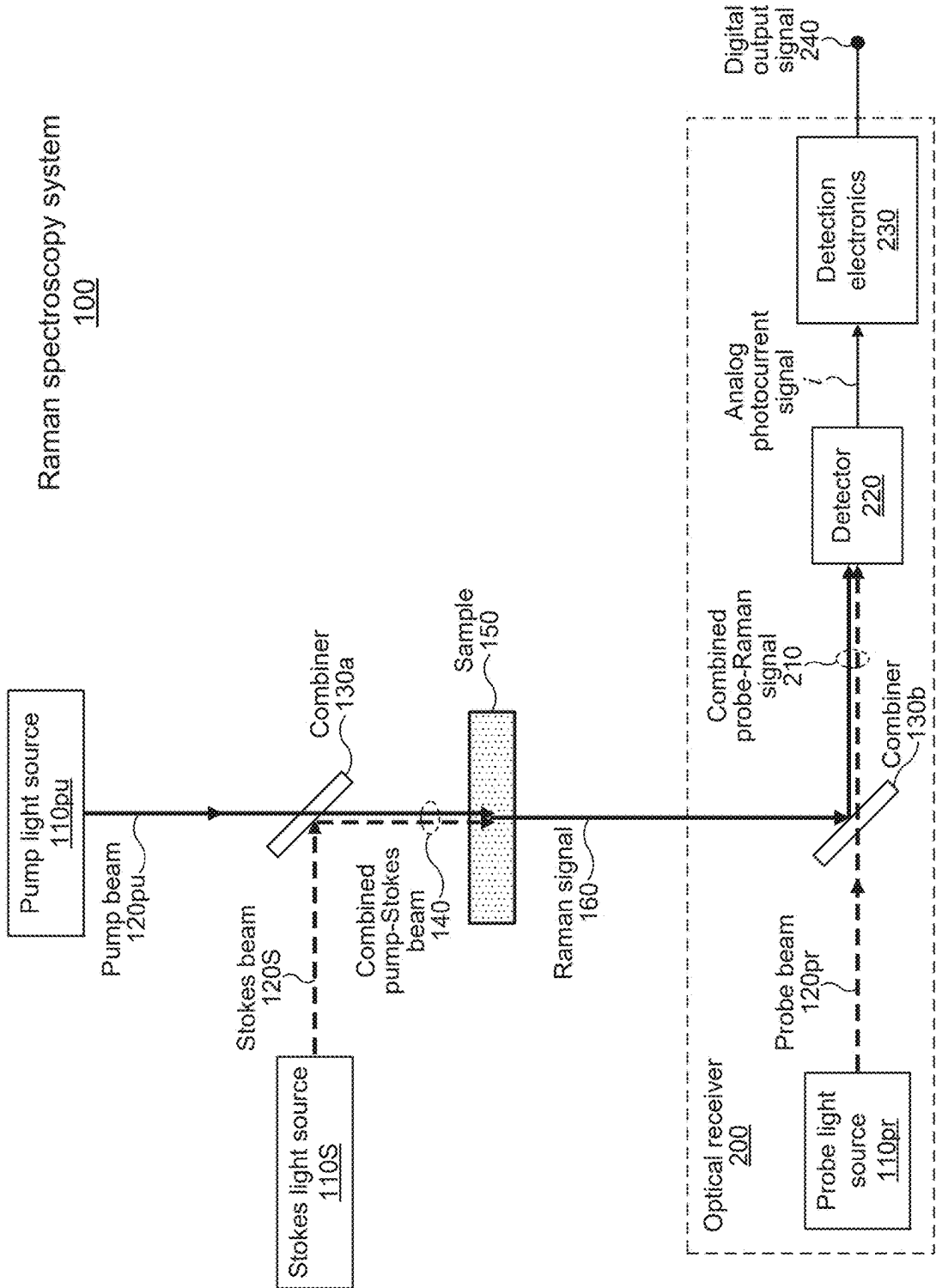


Figure 1

Raman spectroscopy system 100

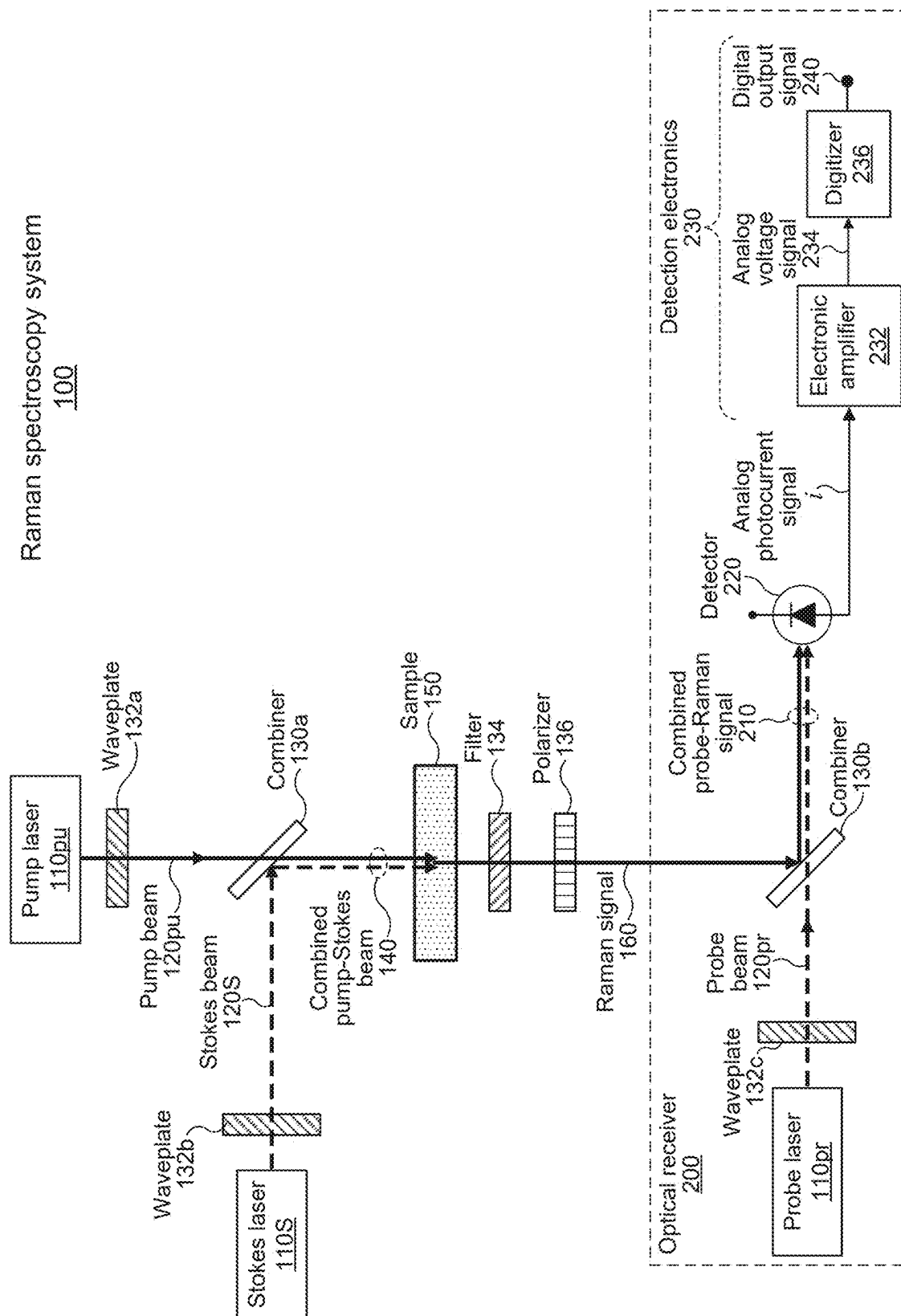


Figure 2

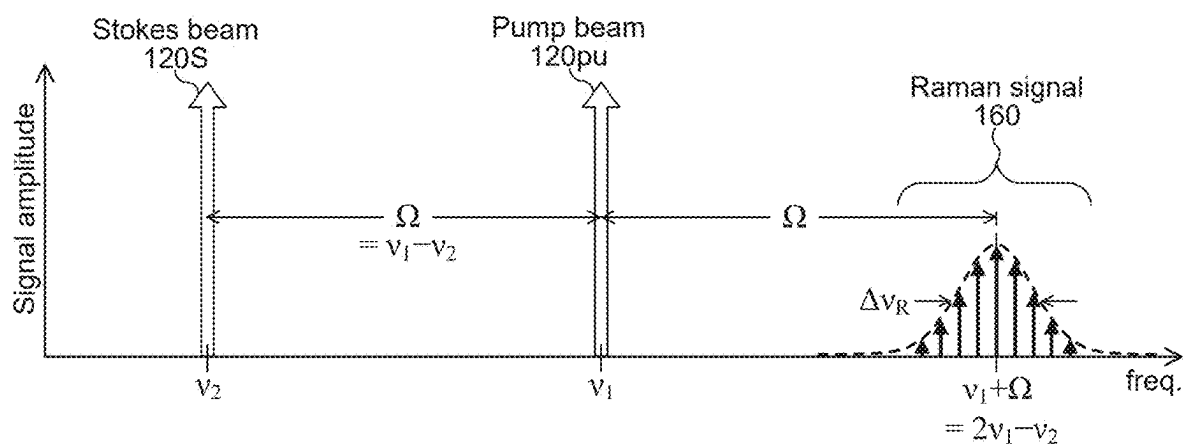


Figure 3

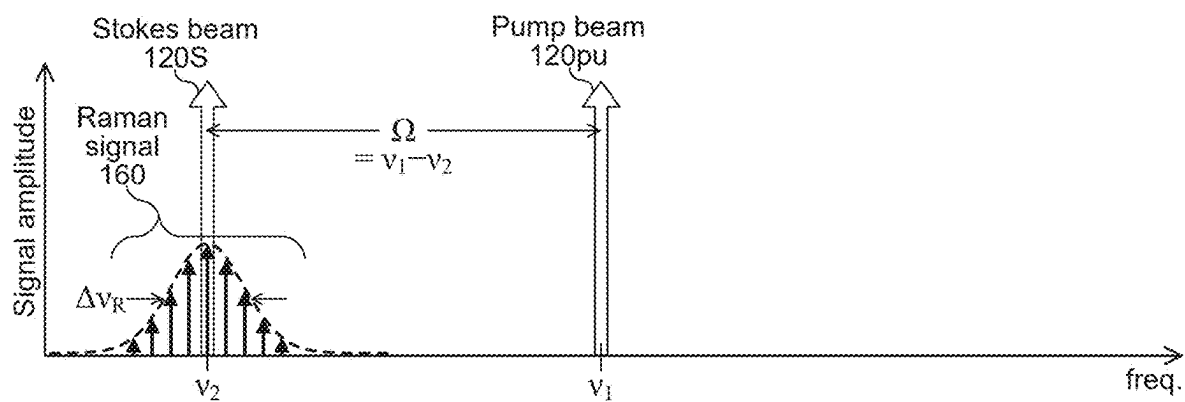


Figure 4

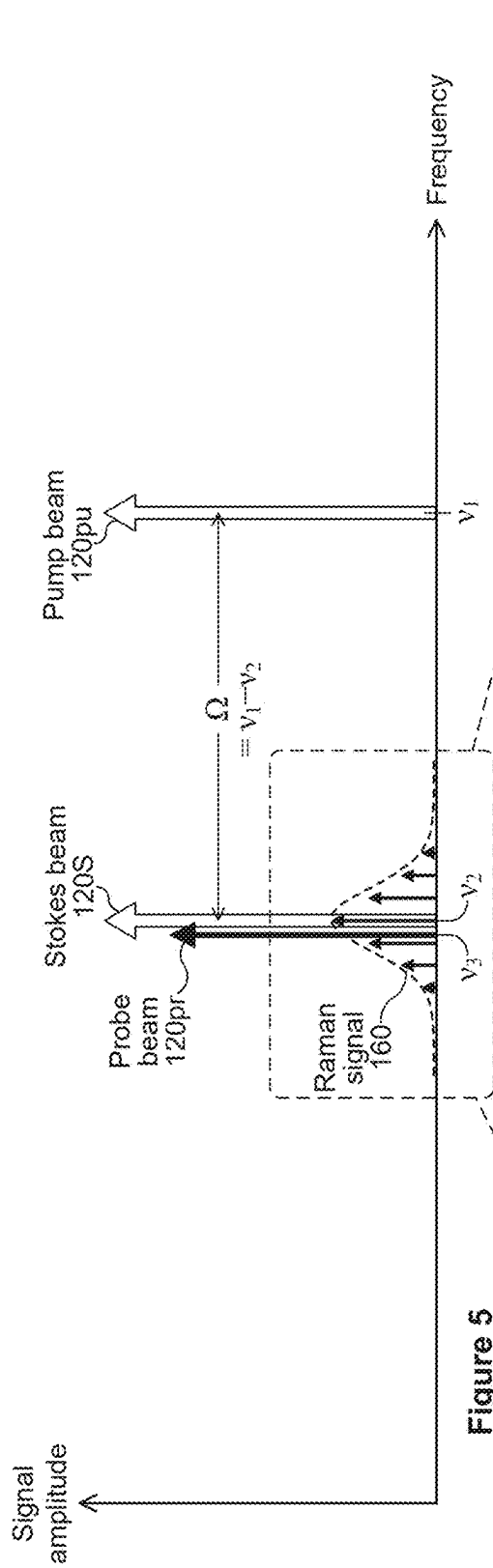


Figure 5

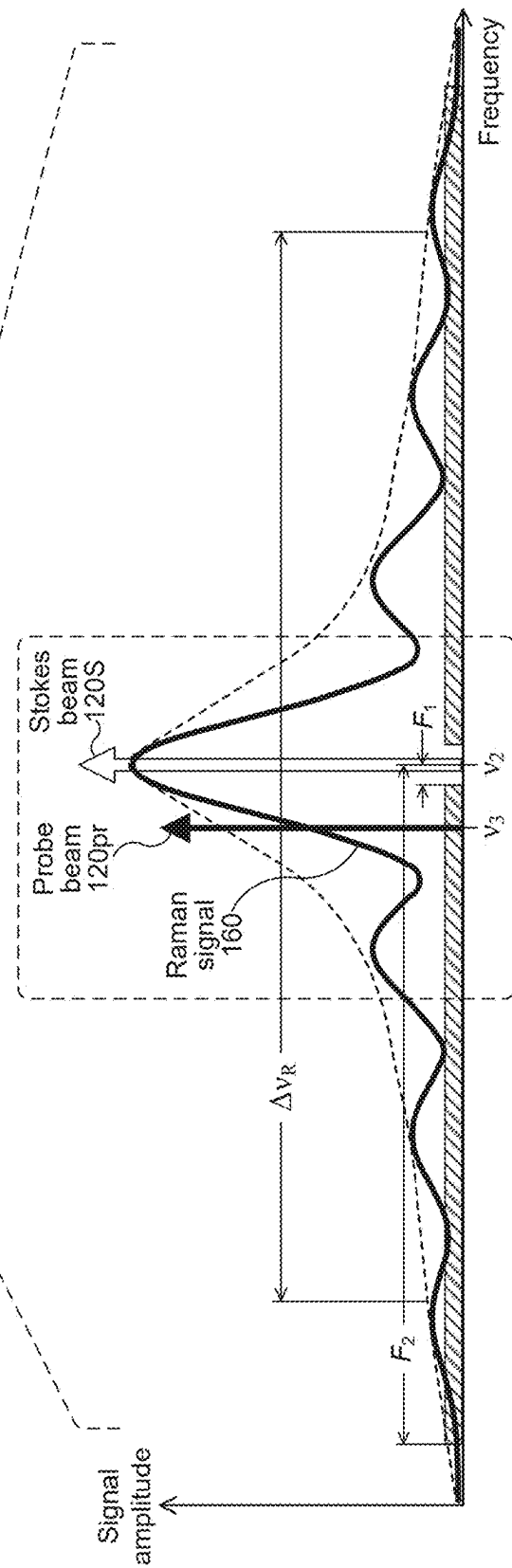


Figure 6

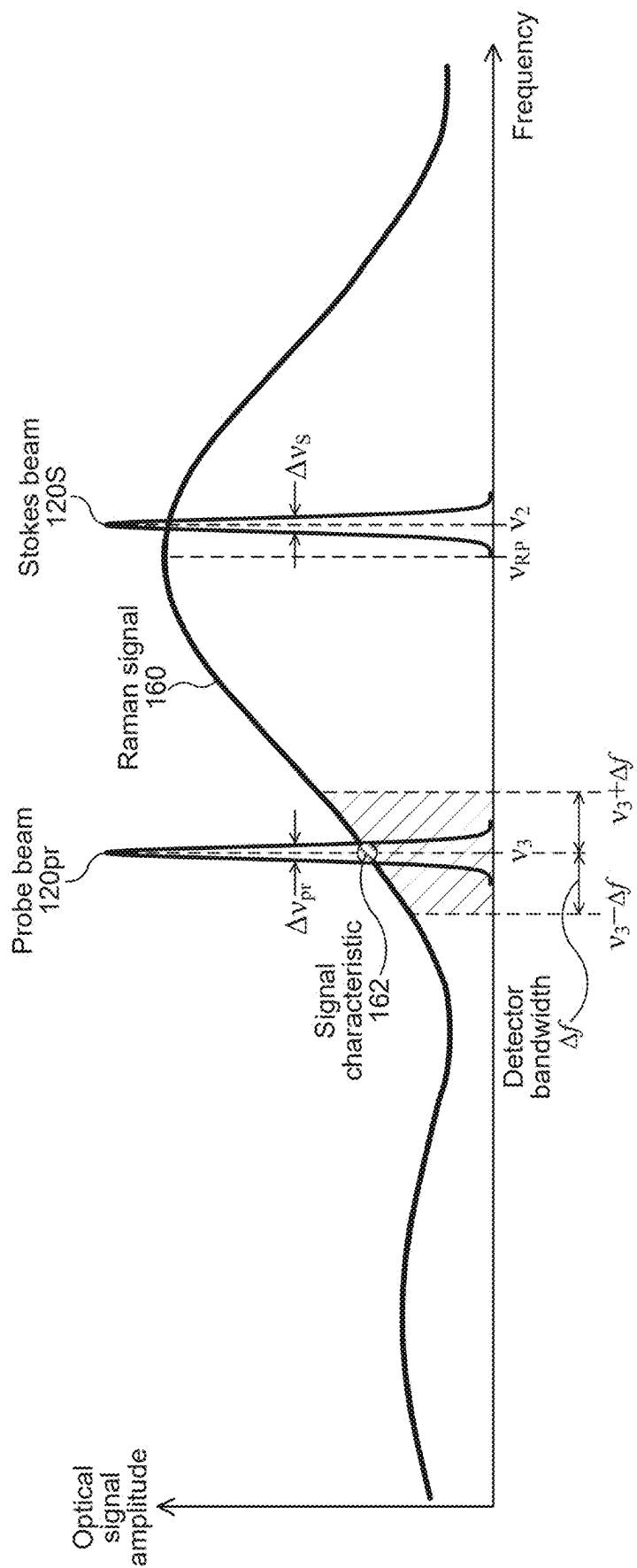


Figure 7

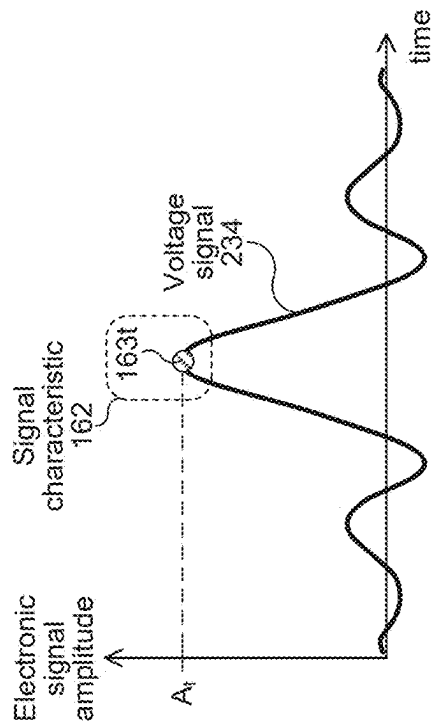


Figure 8

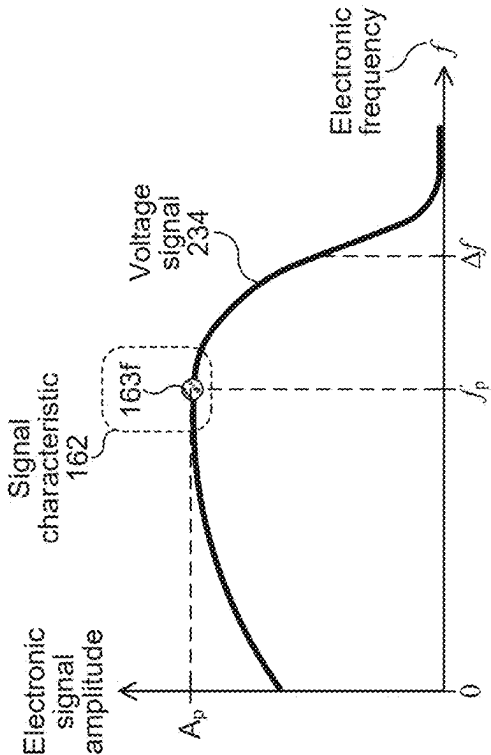


Figure 9

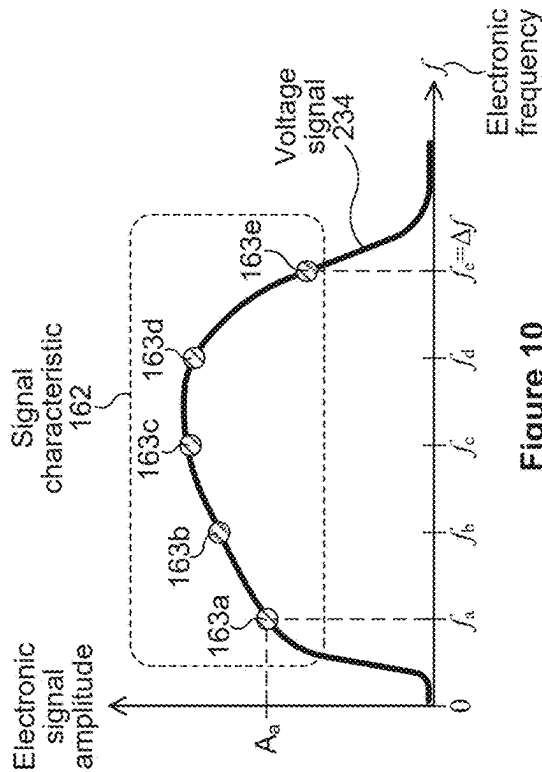


Figure 10

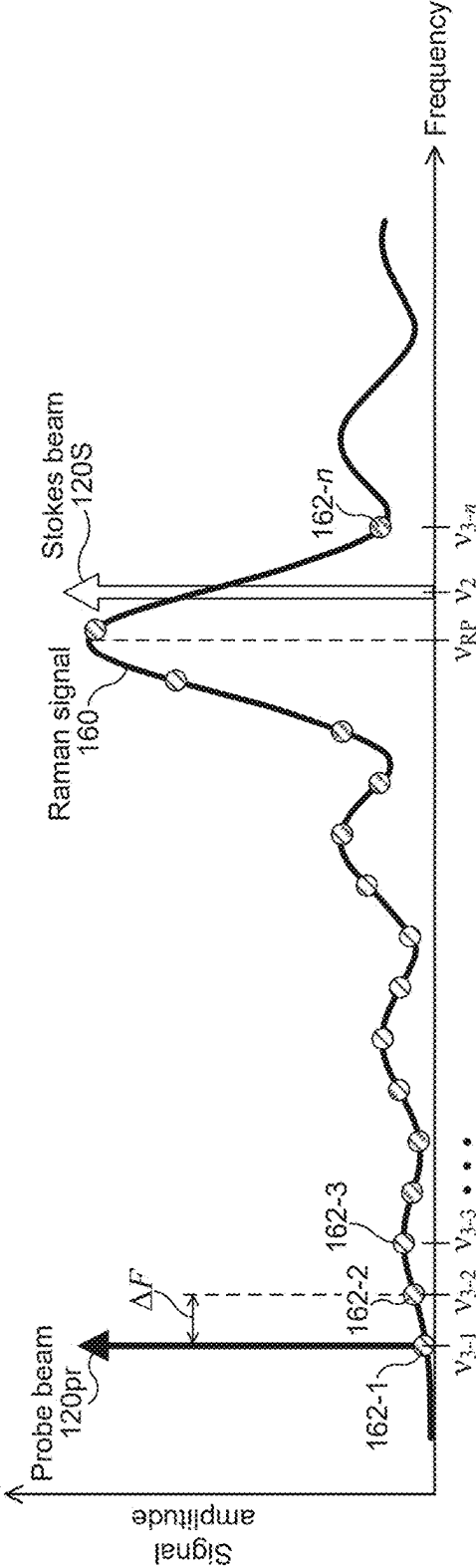


Figure 11

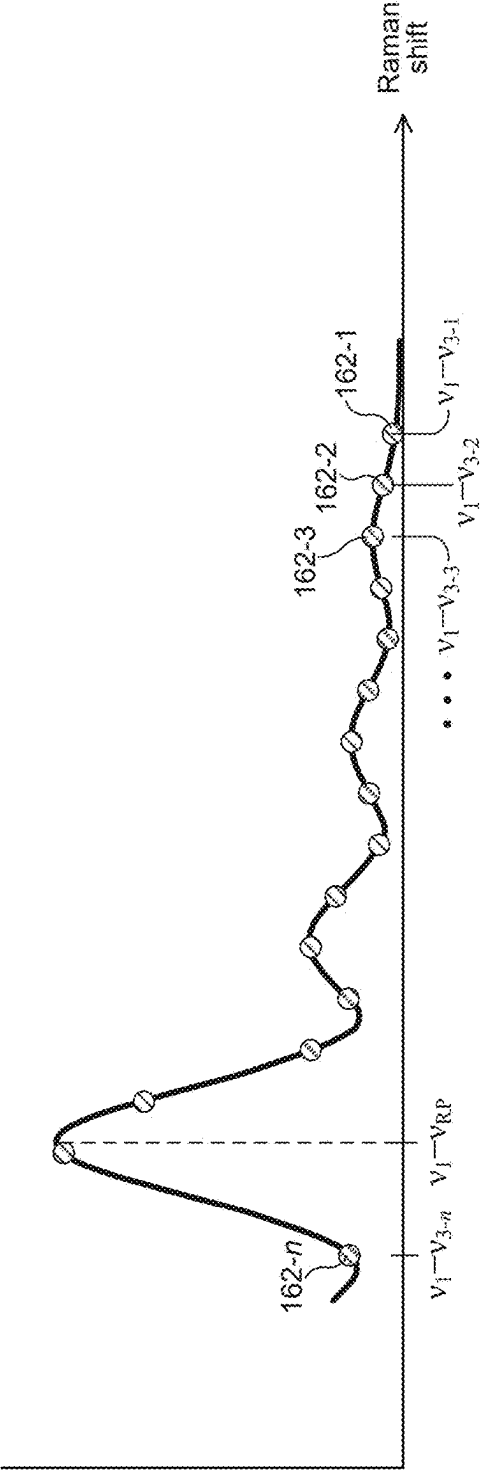


Figure 12

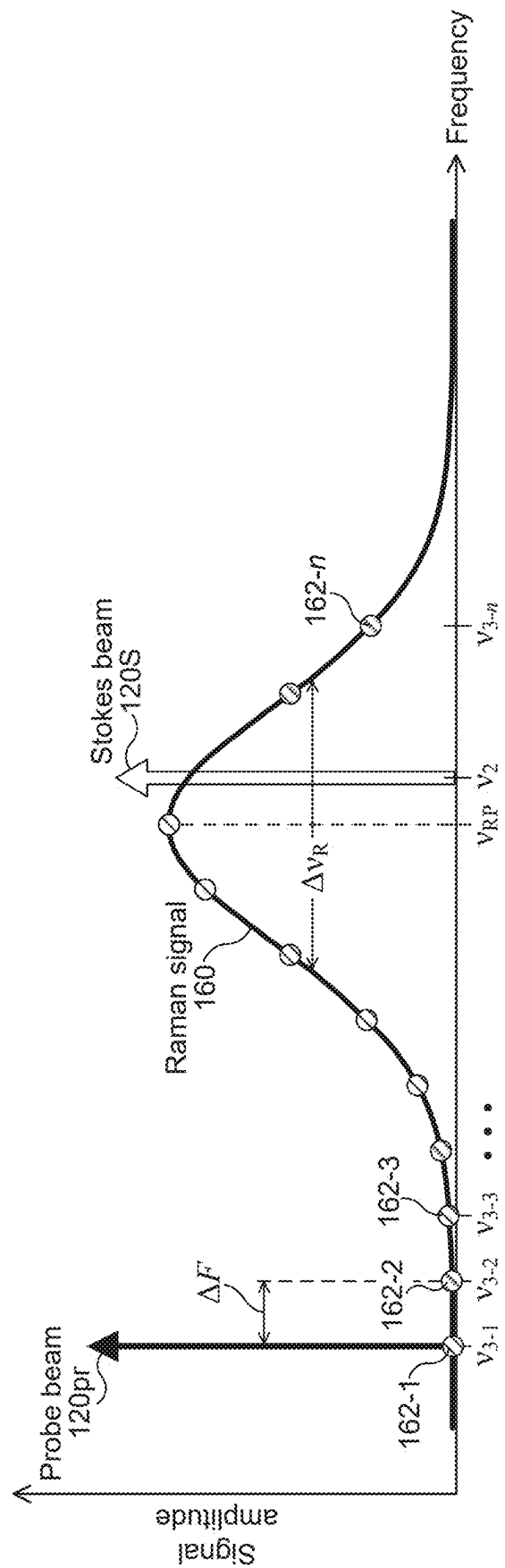


Figure 13

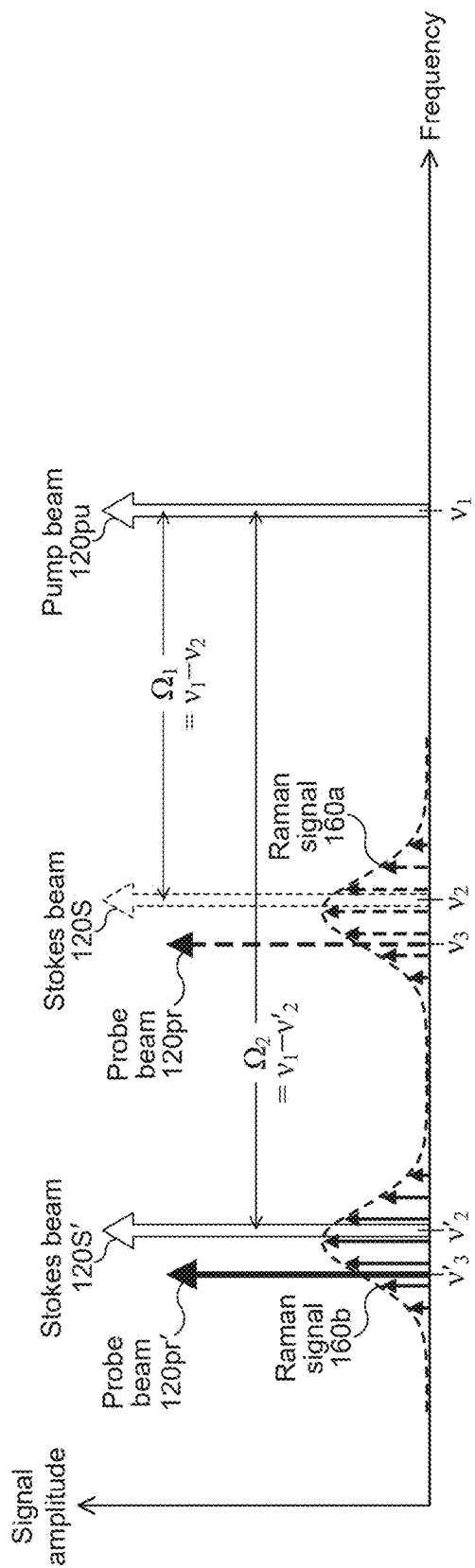


Figure 14

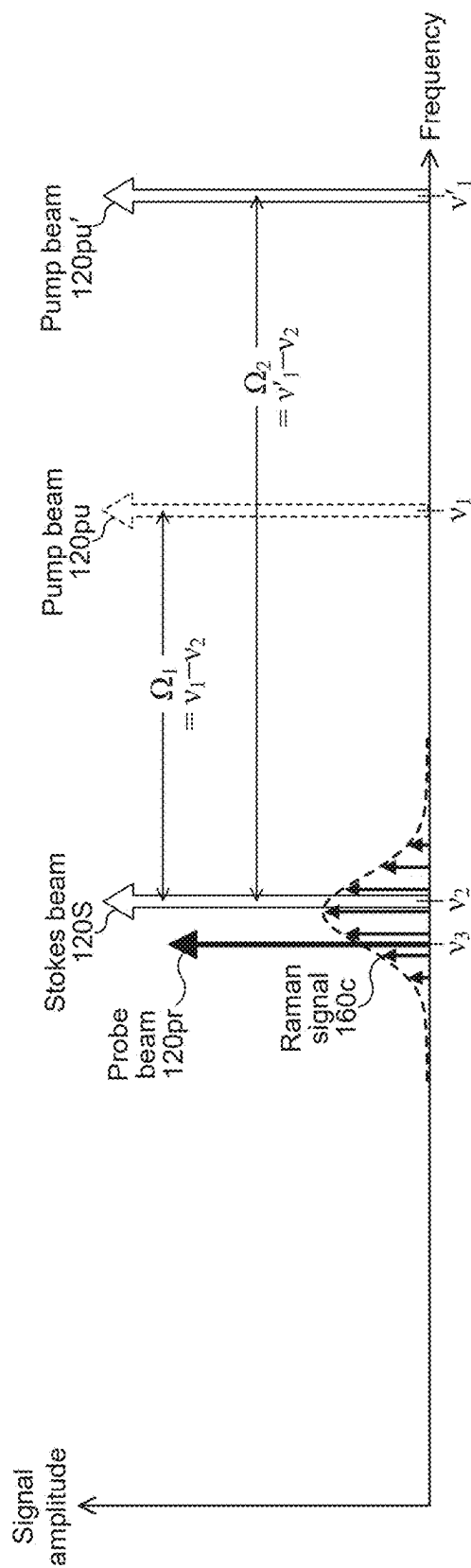


Figure 15

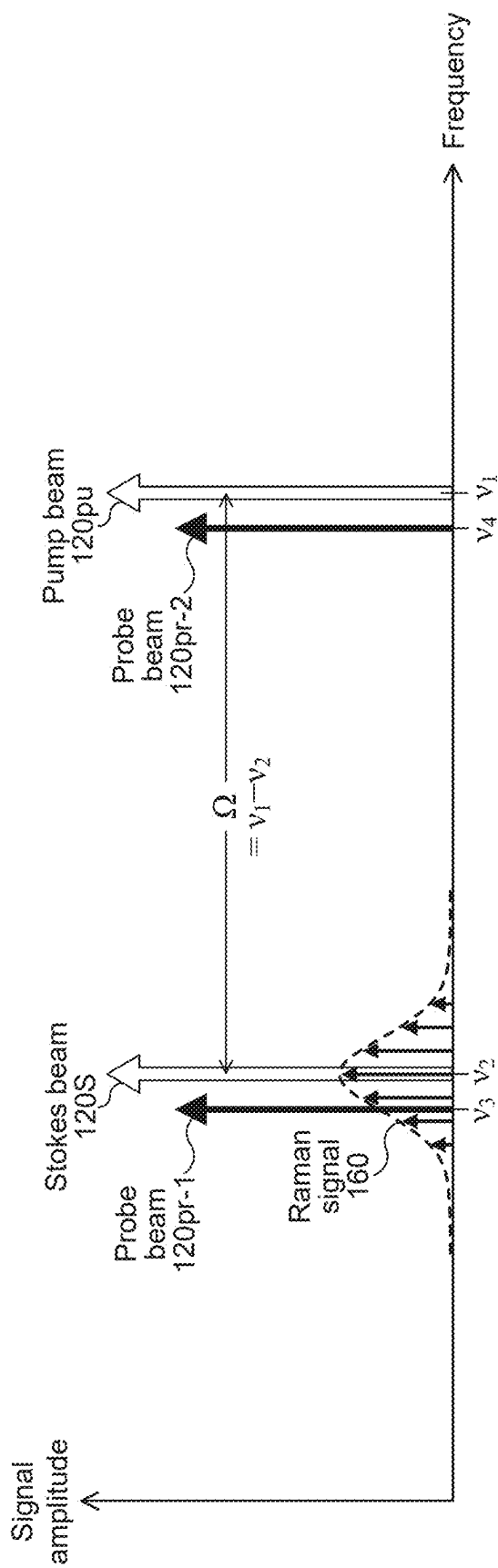


Figure 16

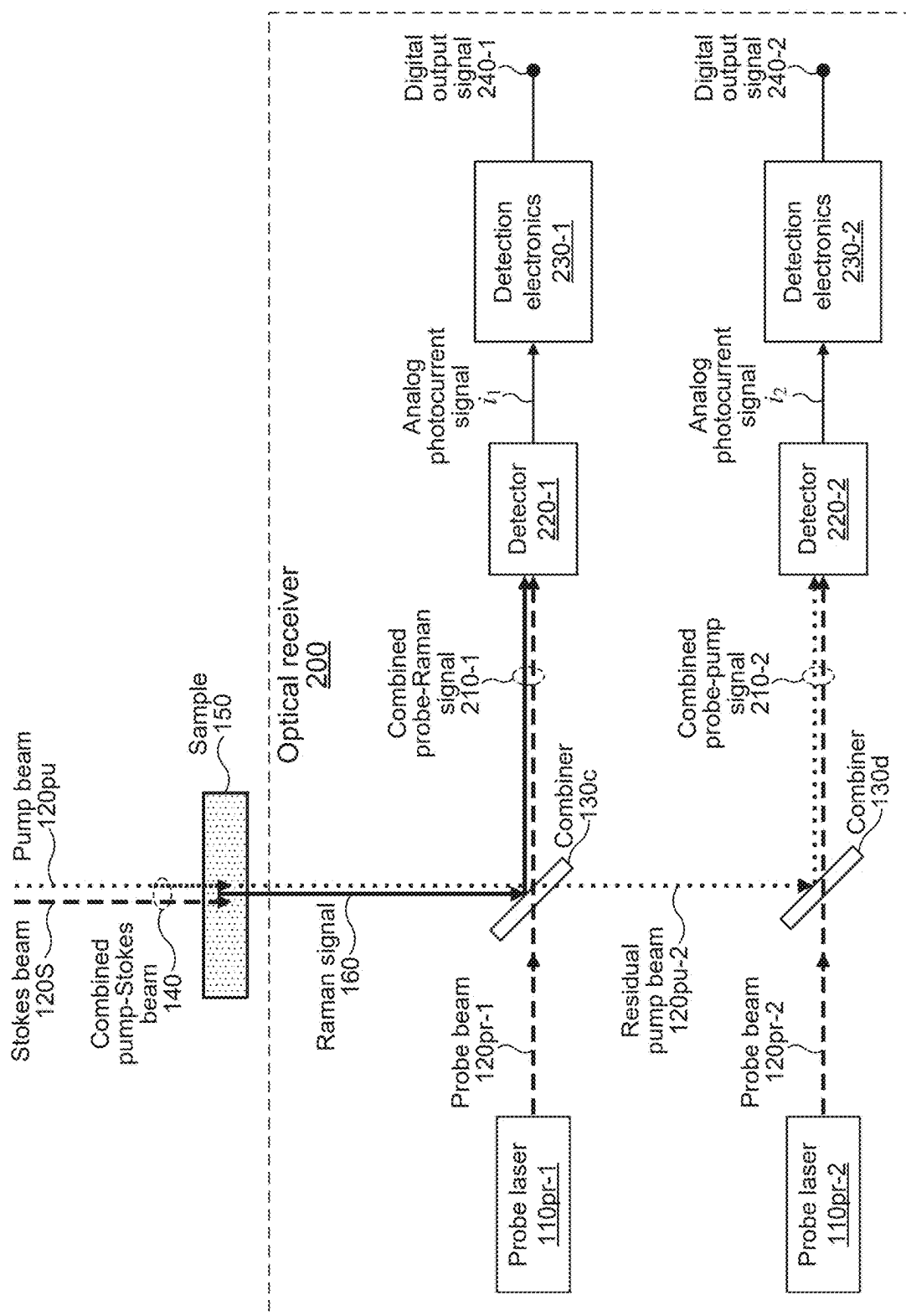


Figure 17

Raman spectroscopy system
100

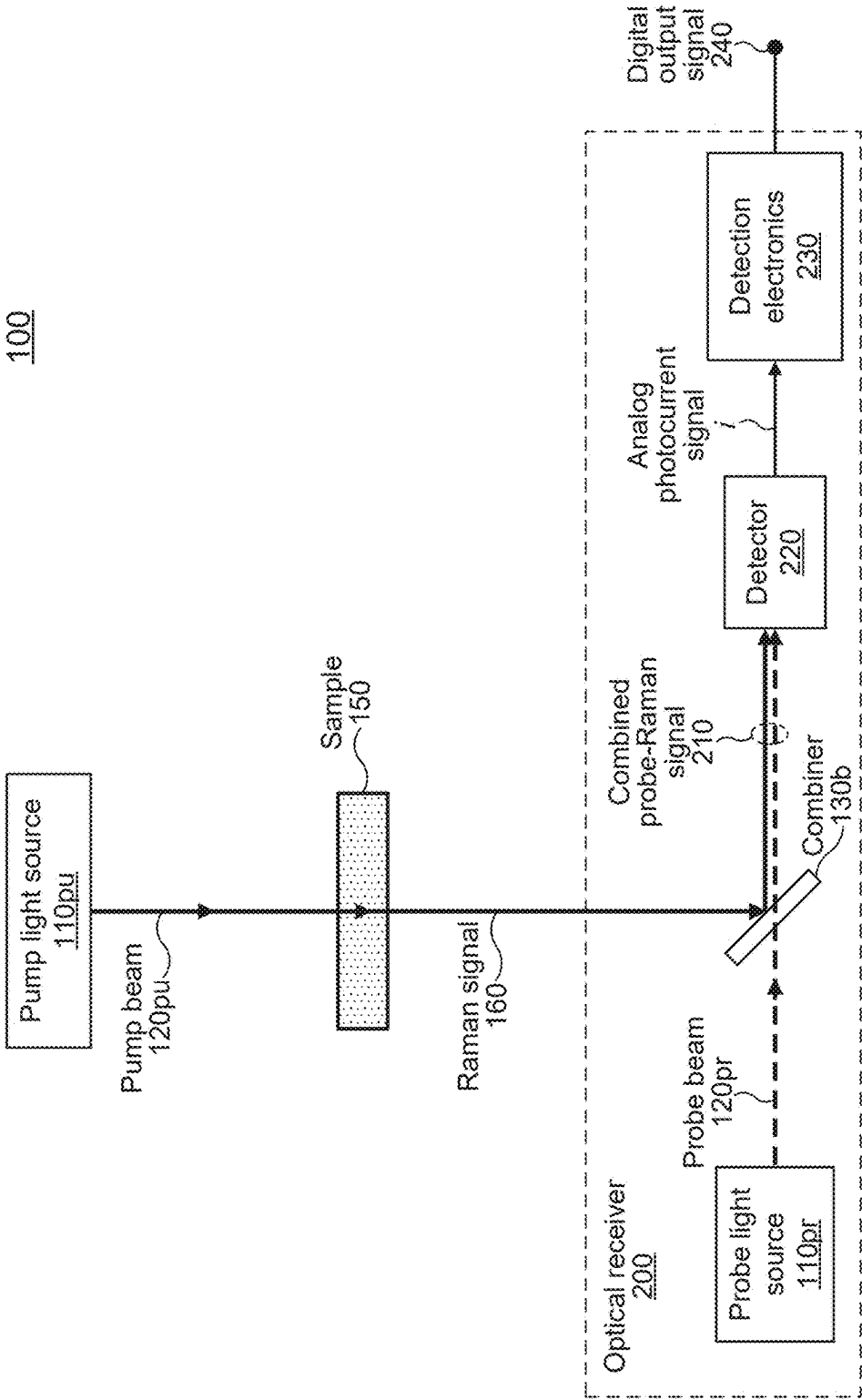


Figure 18

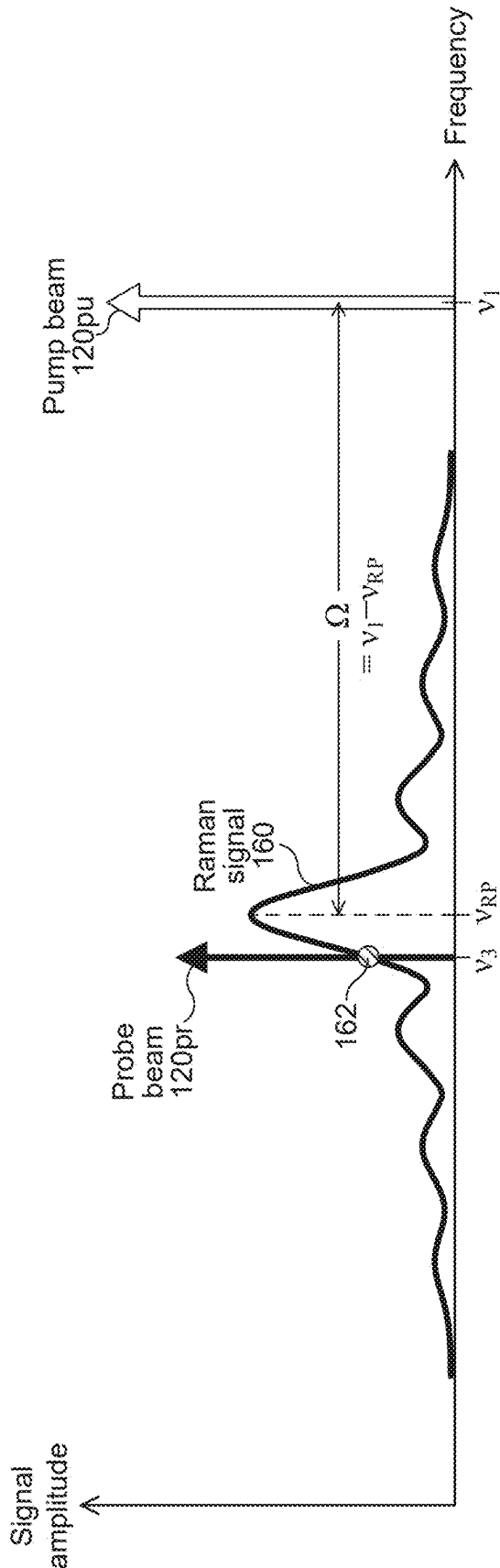


Figure 19

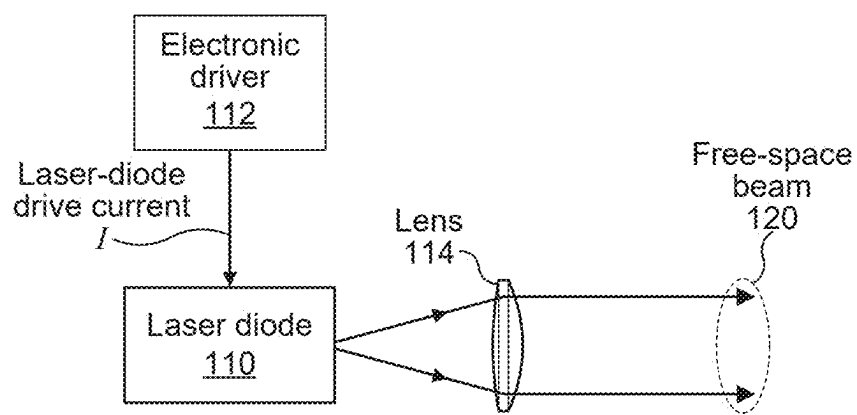


Figure 20

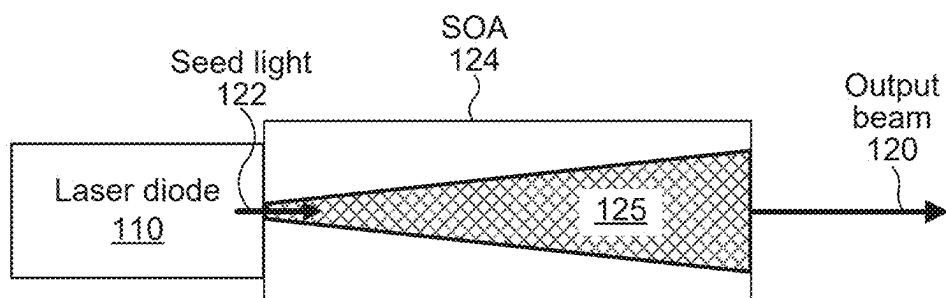


Figure 21

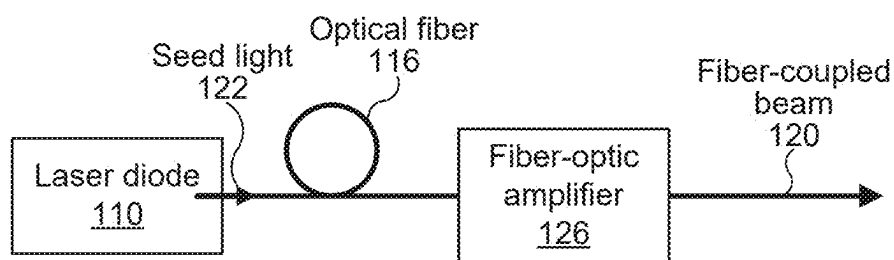


Figure 22

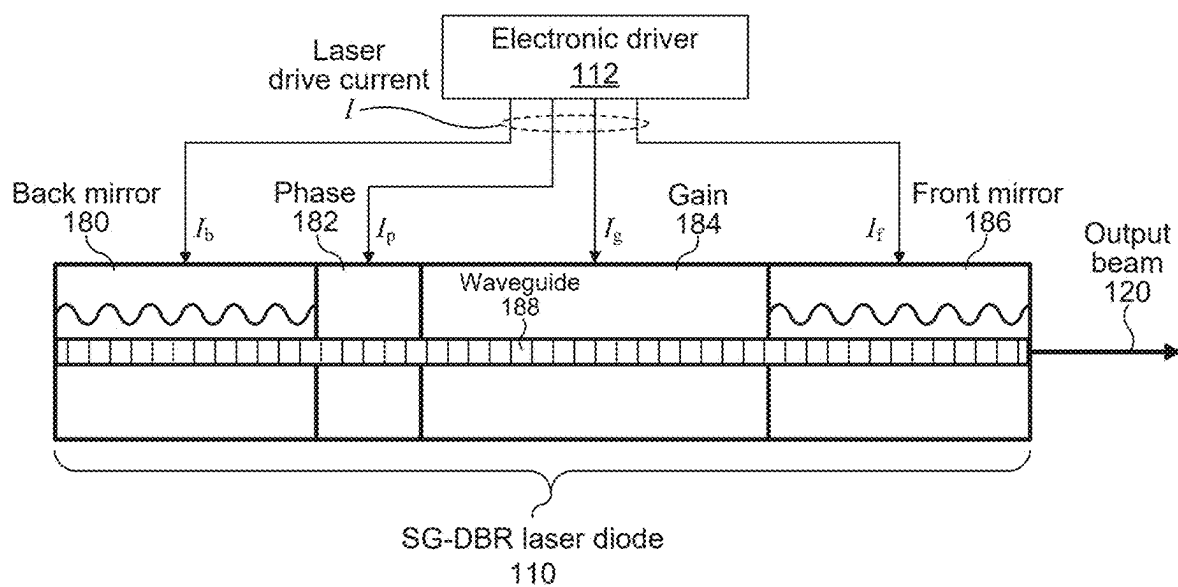


Figure 23

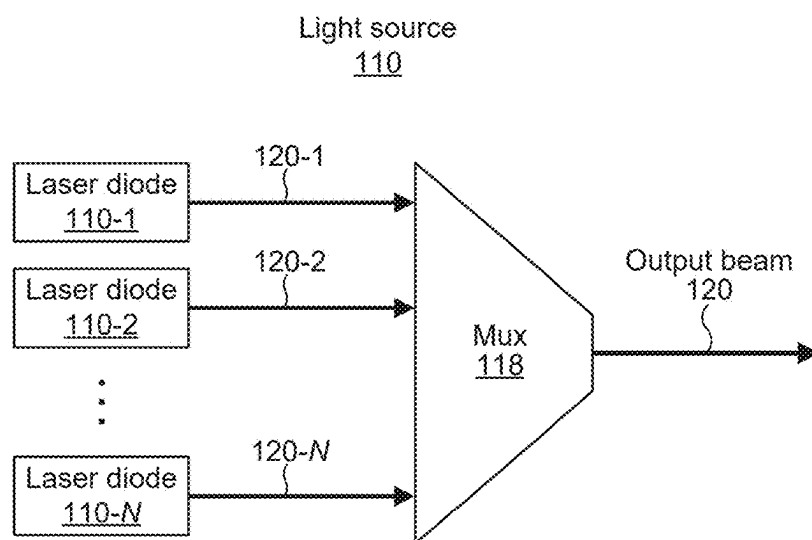


Figure 24

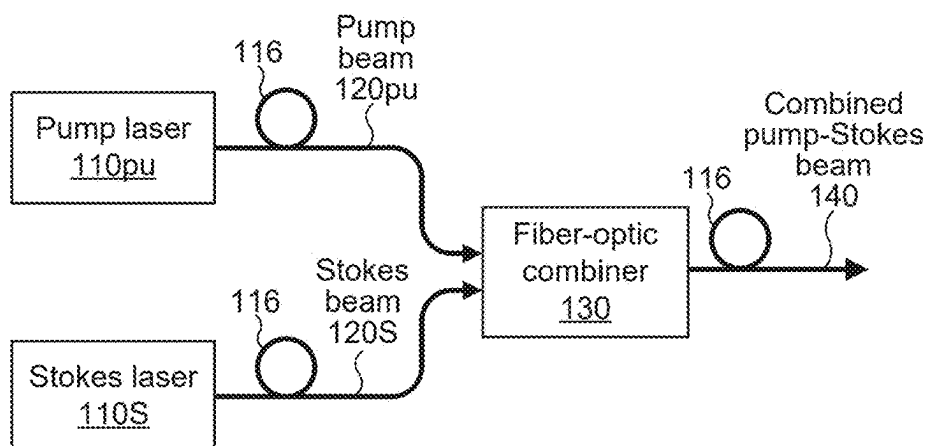


Figure 25

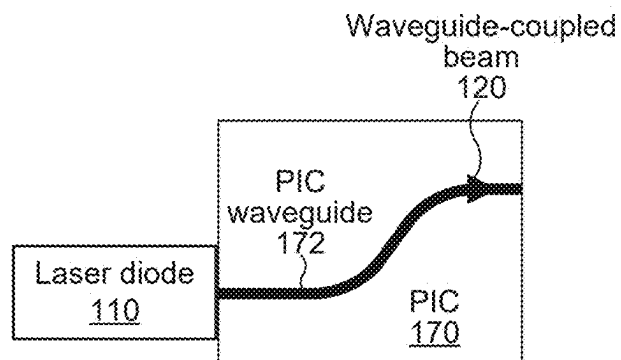


Figure 26

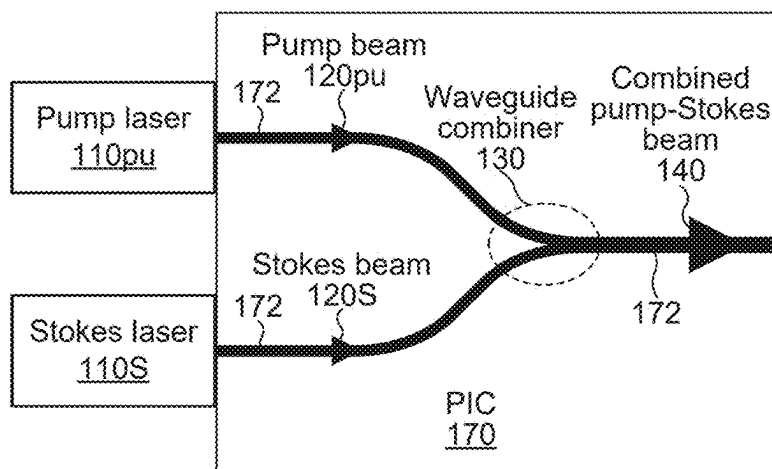


Figure 27

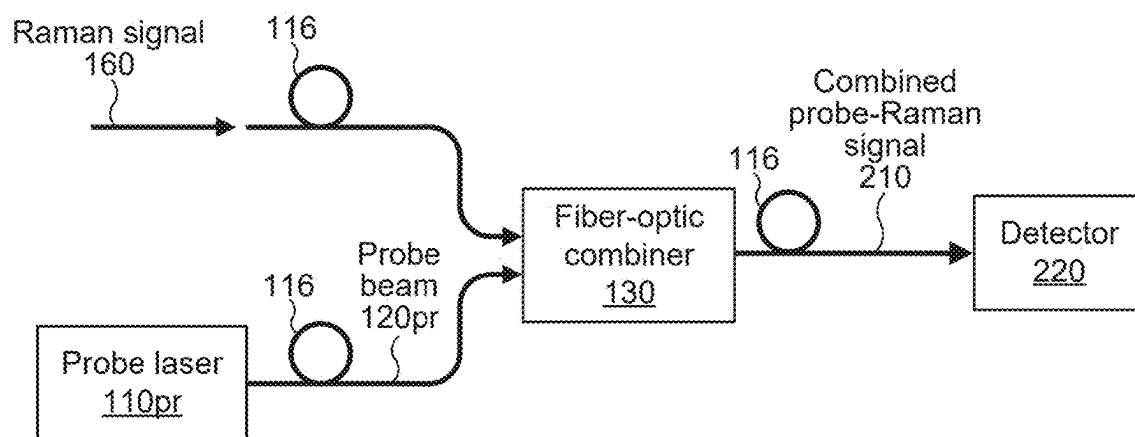


Figure 28

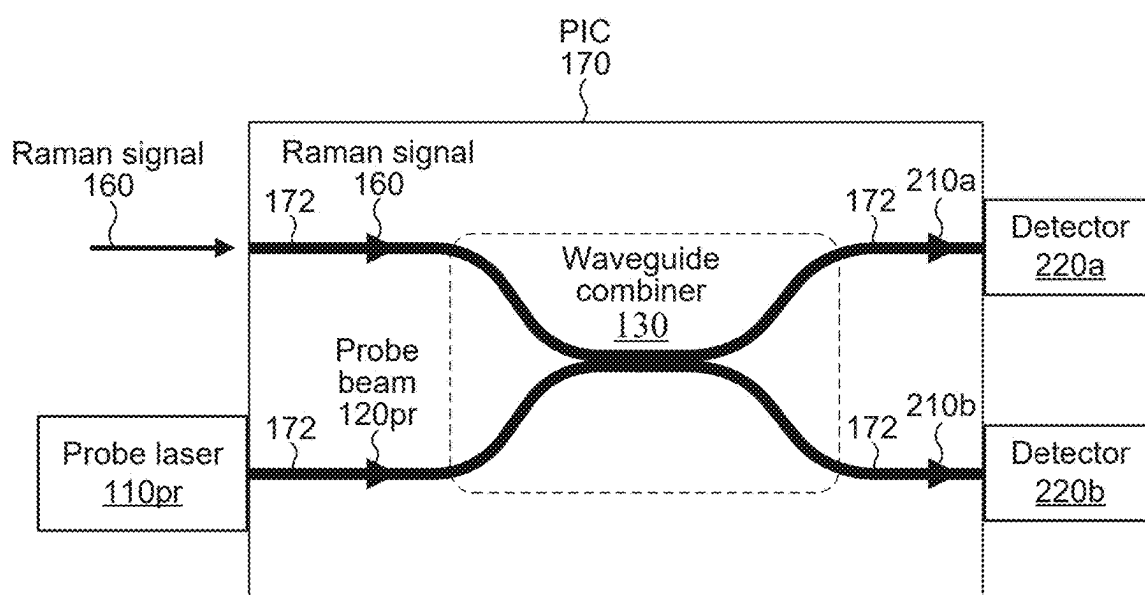


Figure 29

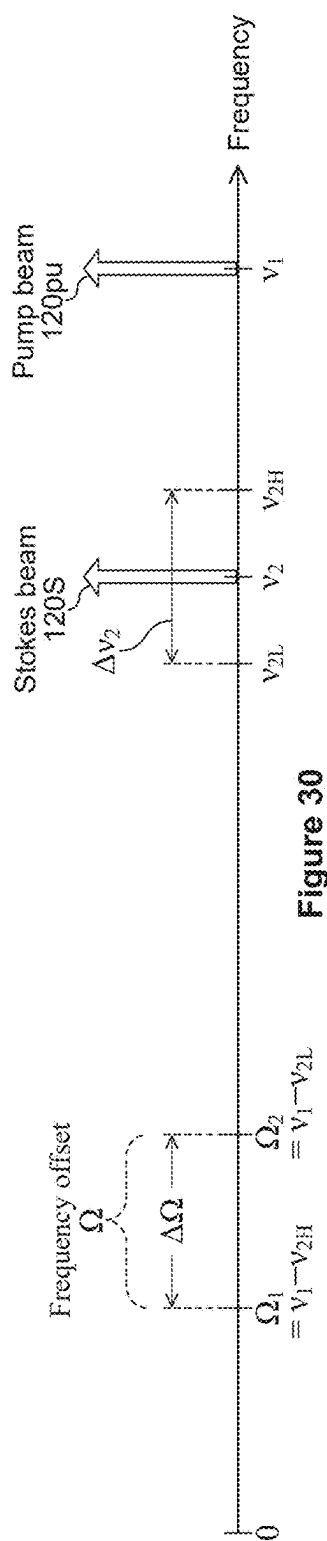


Figure 30

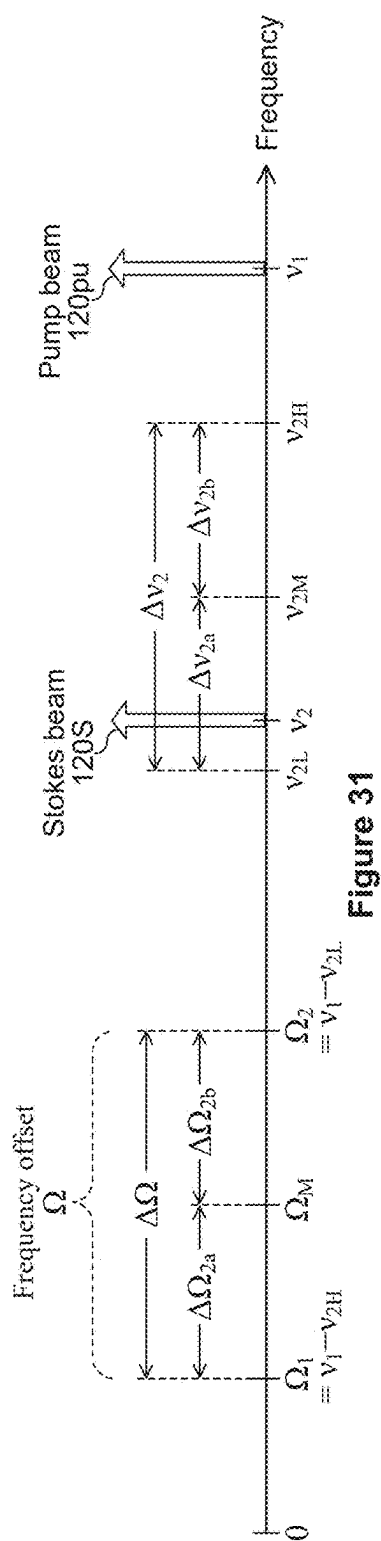


Figure 31

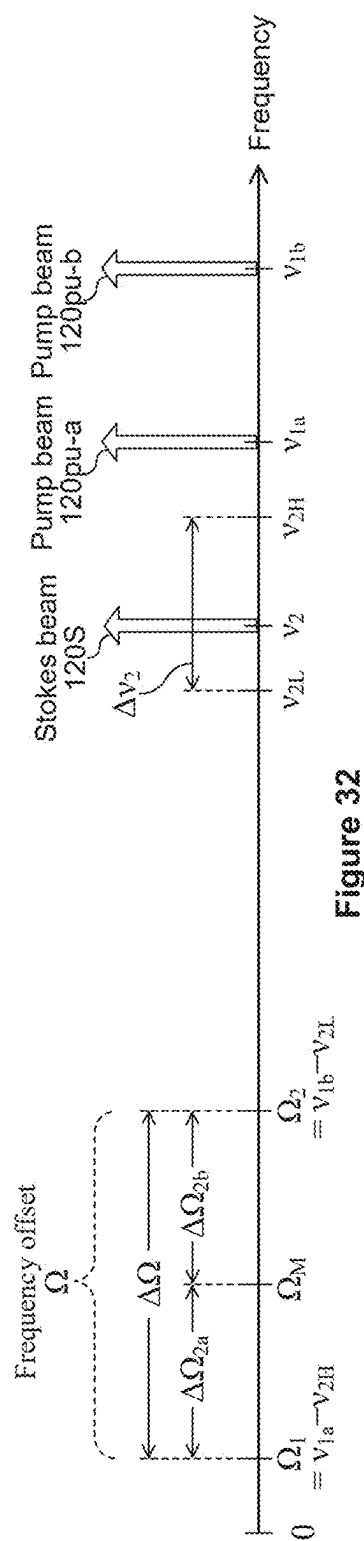


Figure 32

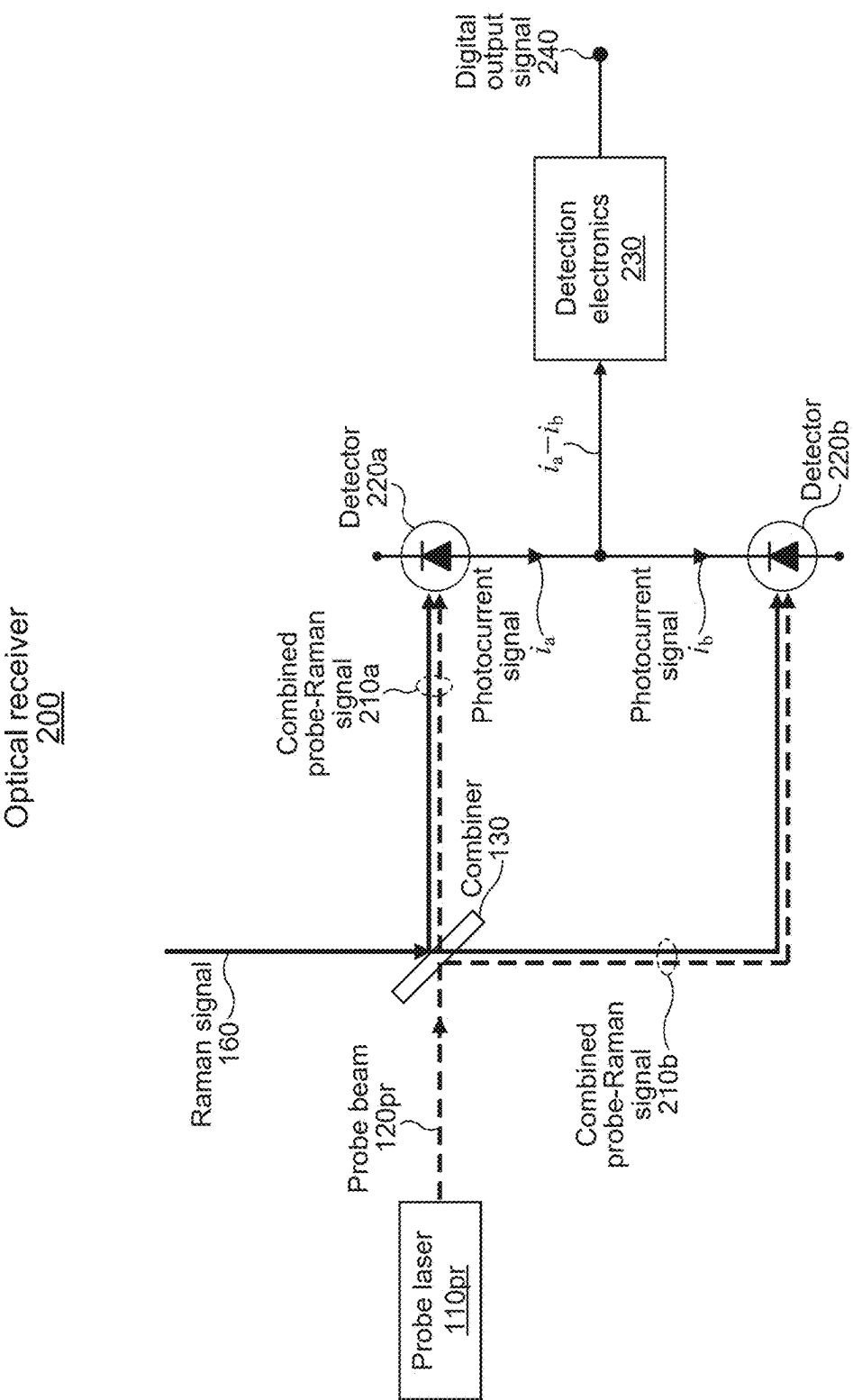


Figure 36

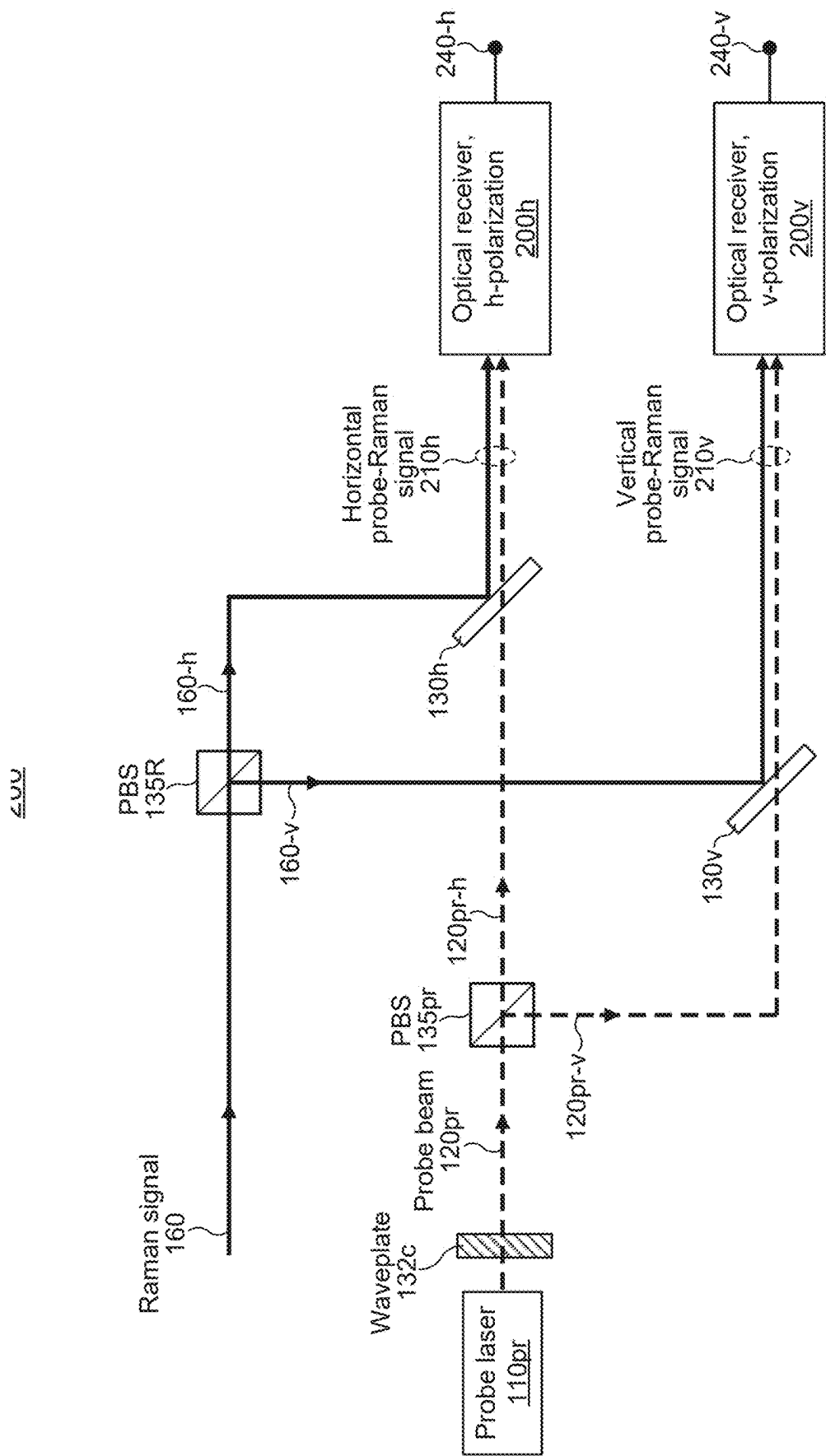


Figure 37

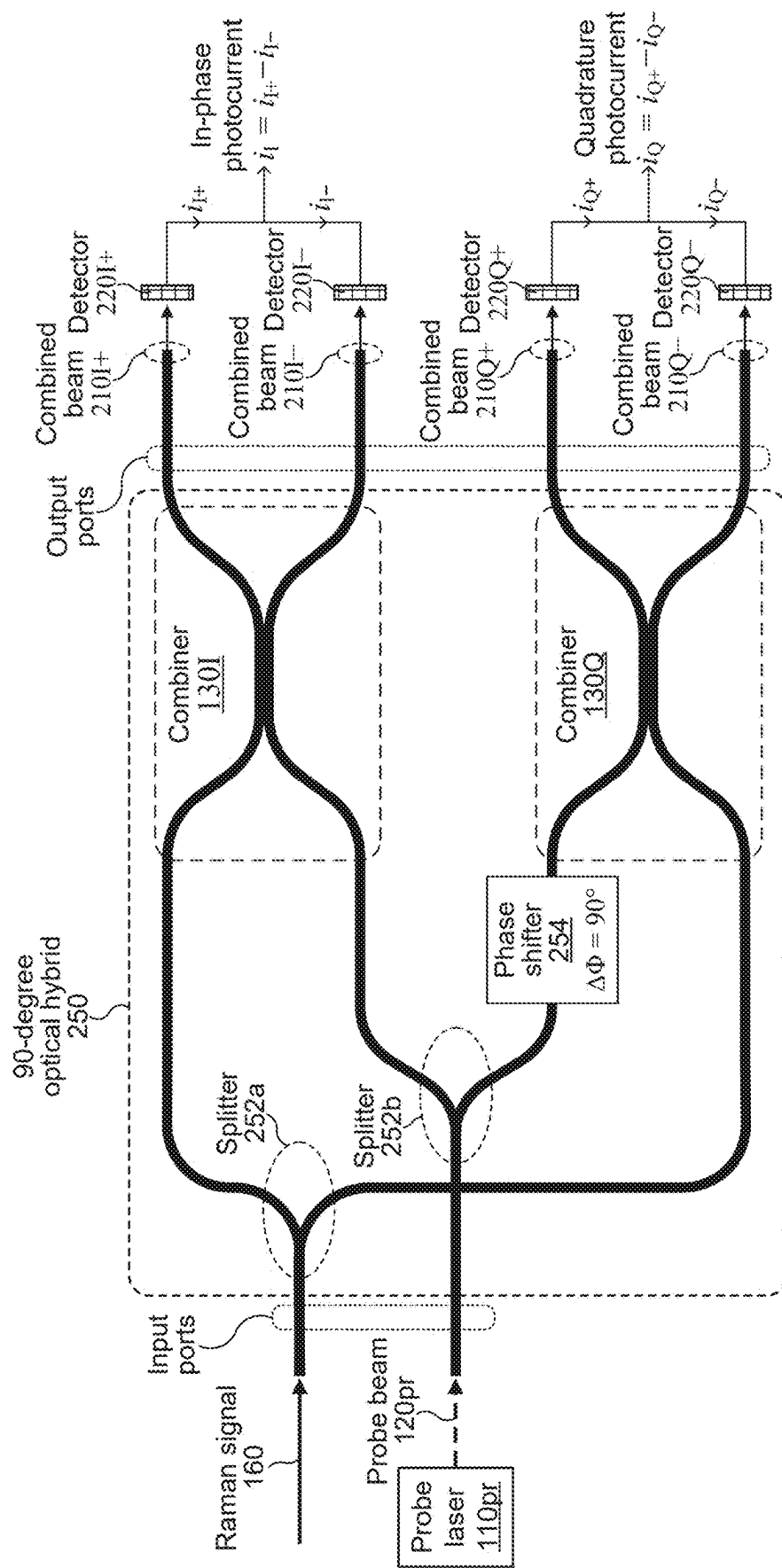


Figure 38

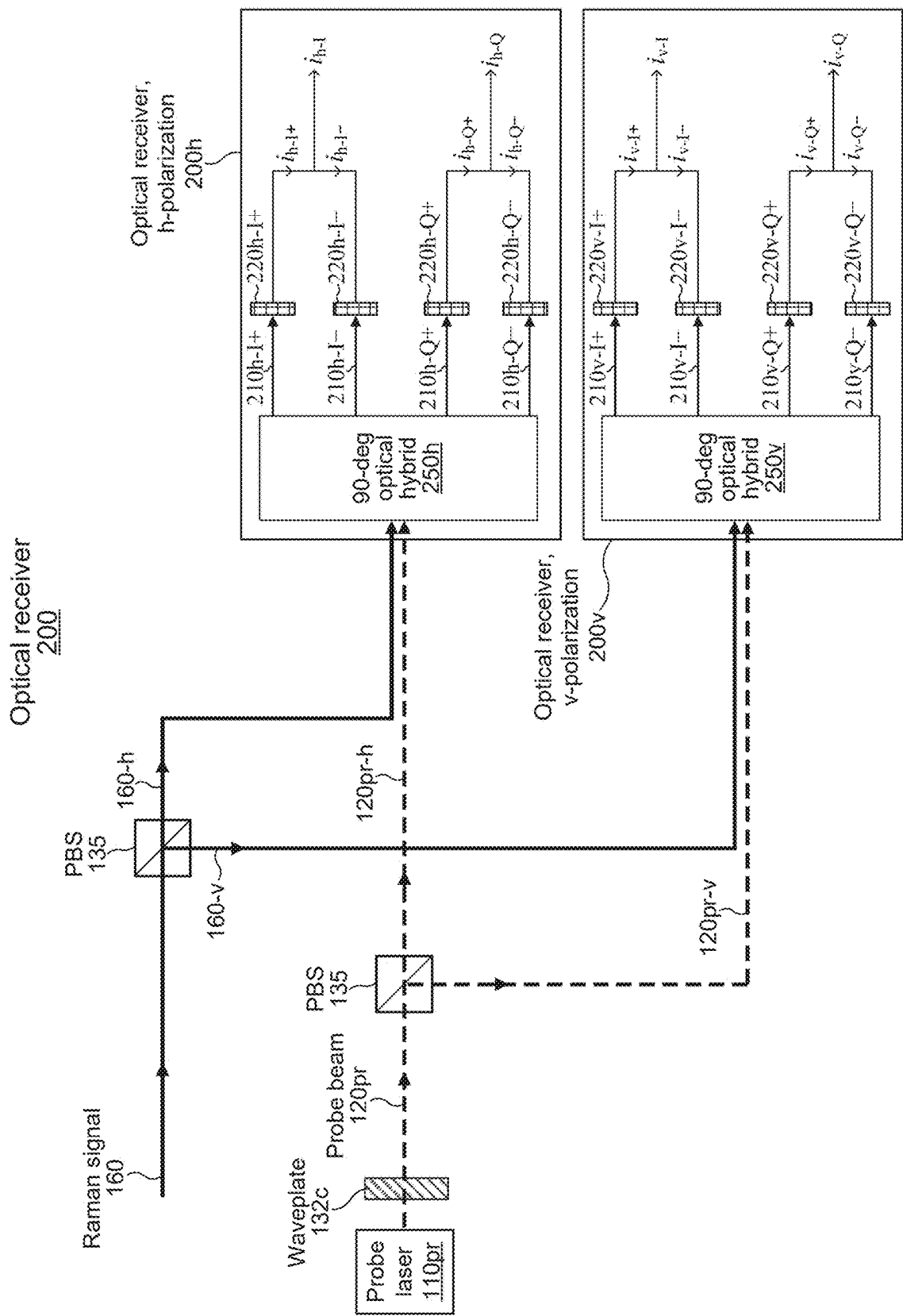


Figure 39

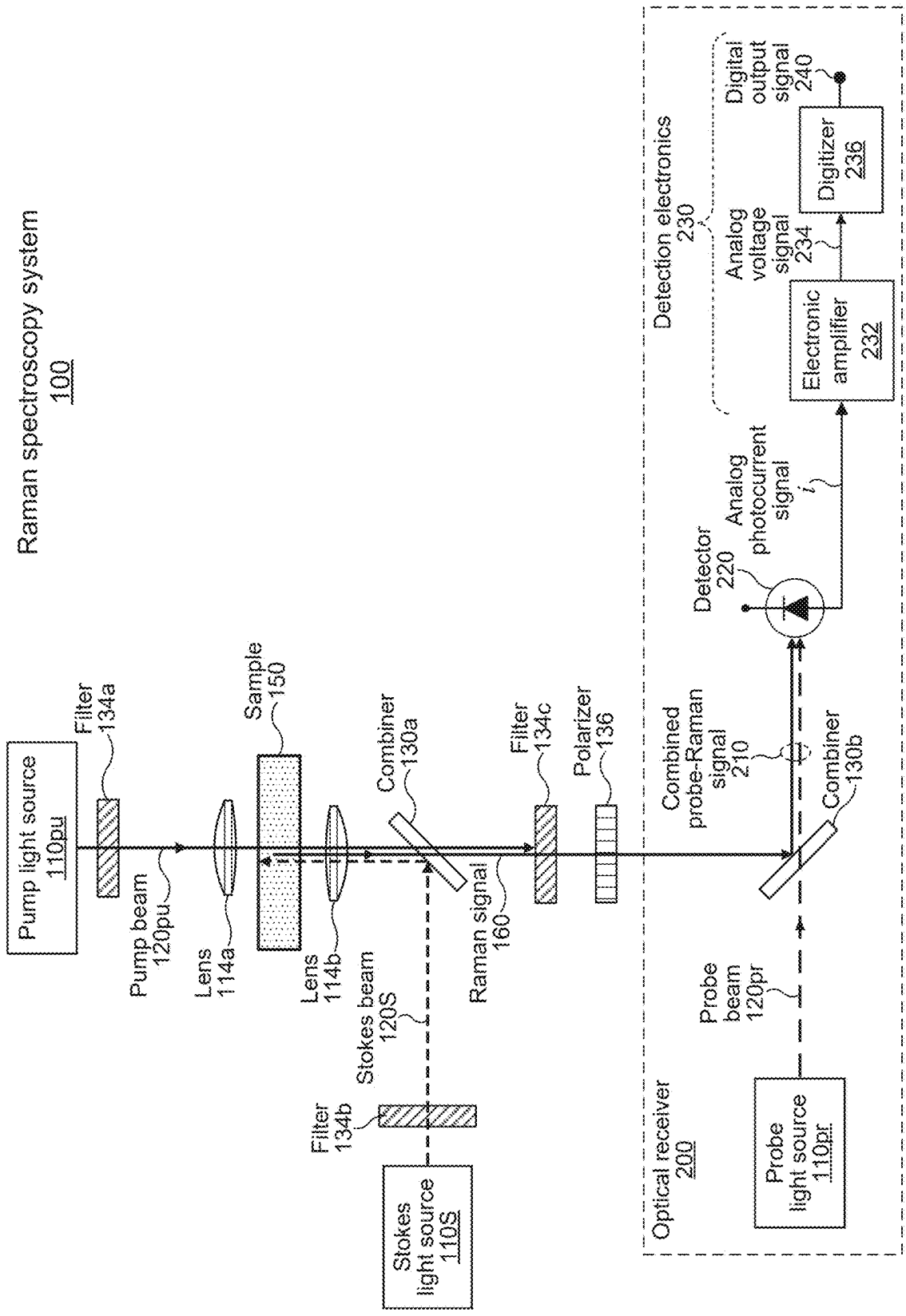


Figure 40

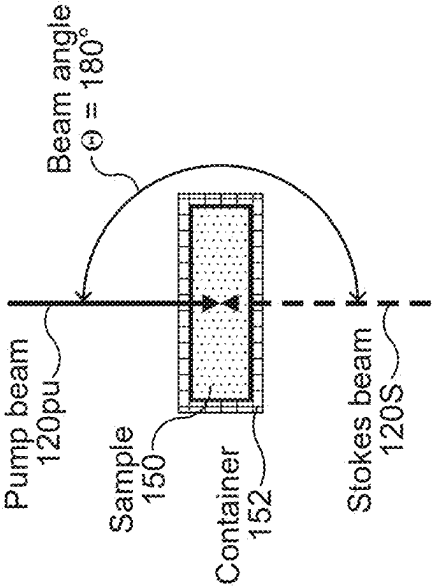


Figure 42

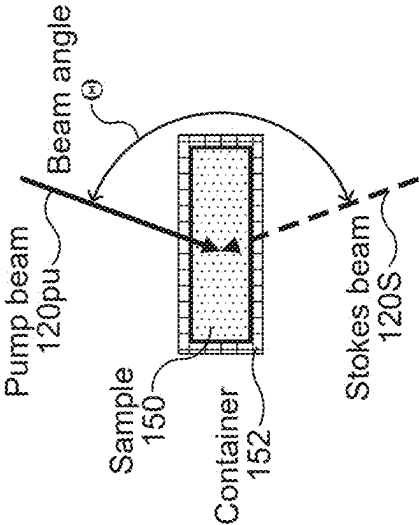


Figure 41

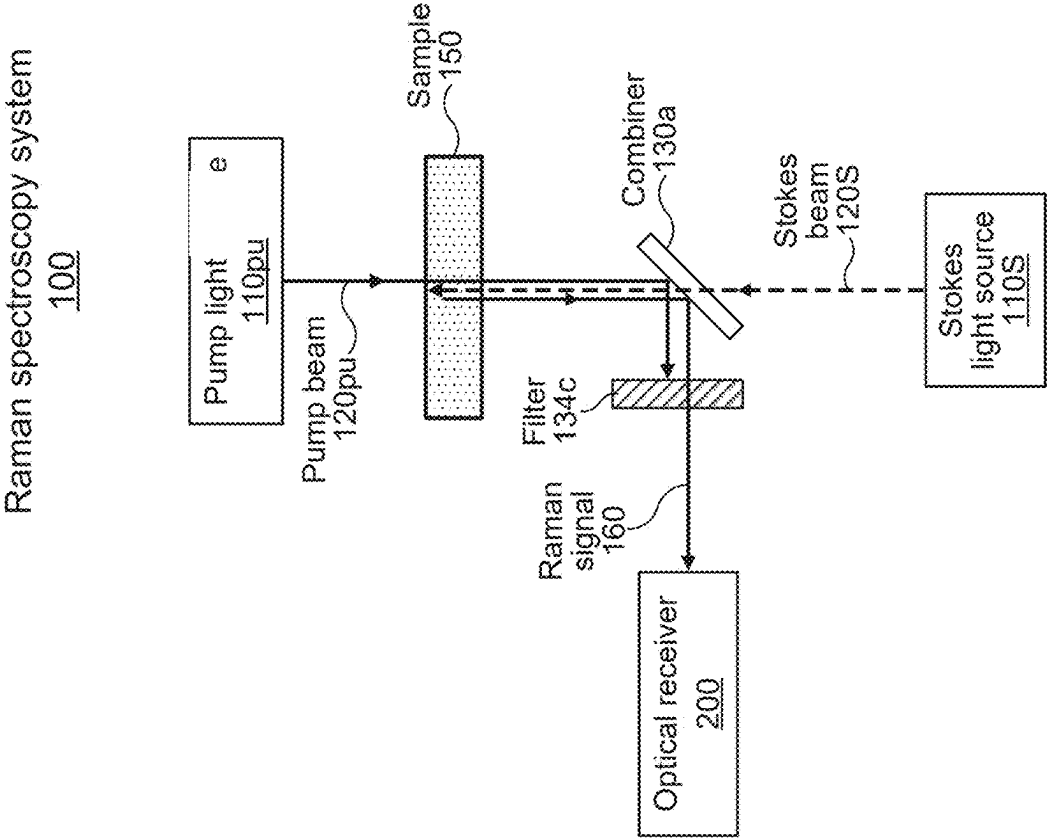


Figure 44

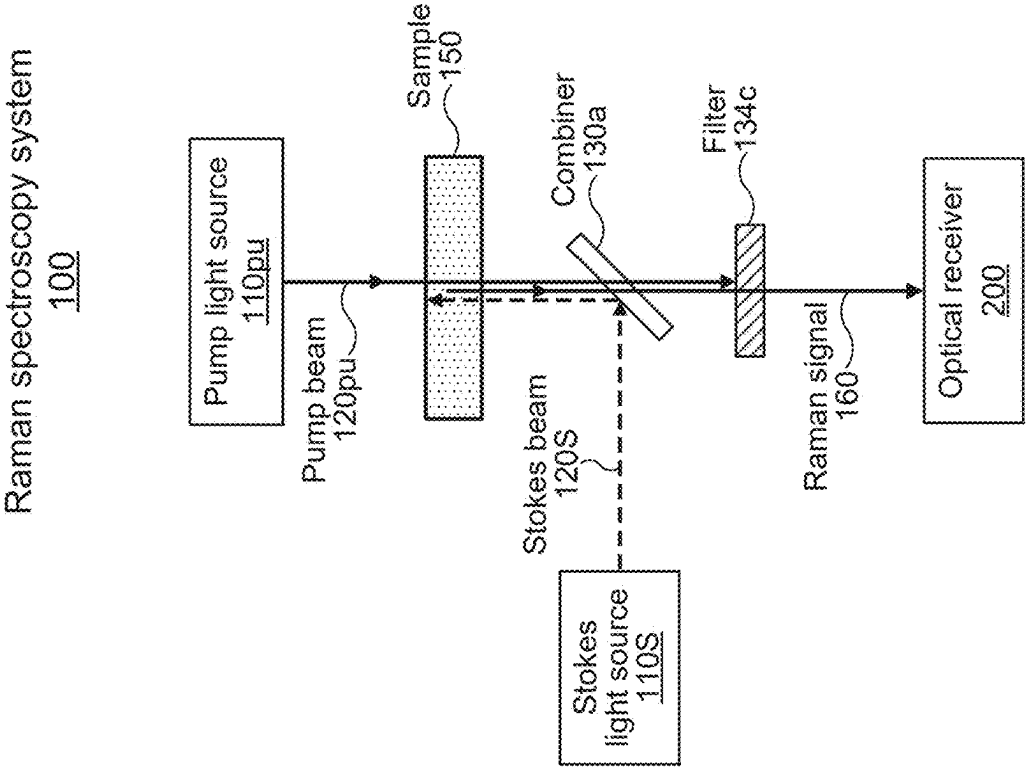


Figure 43

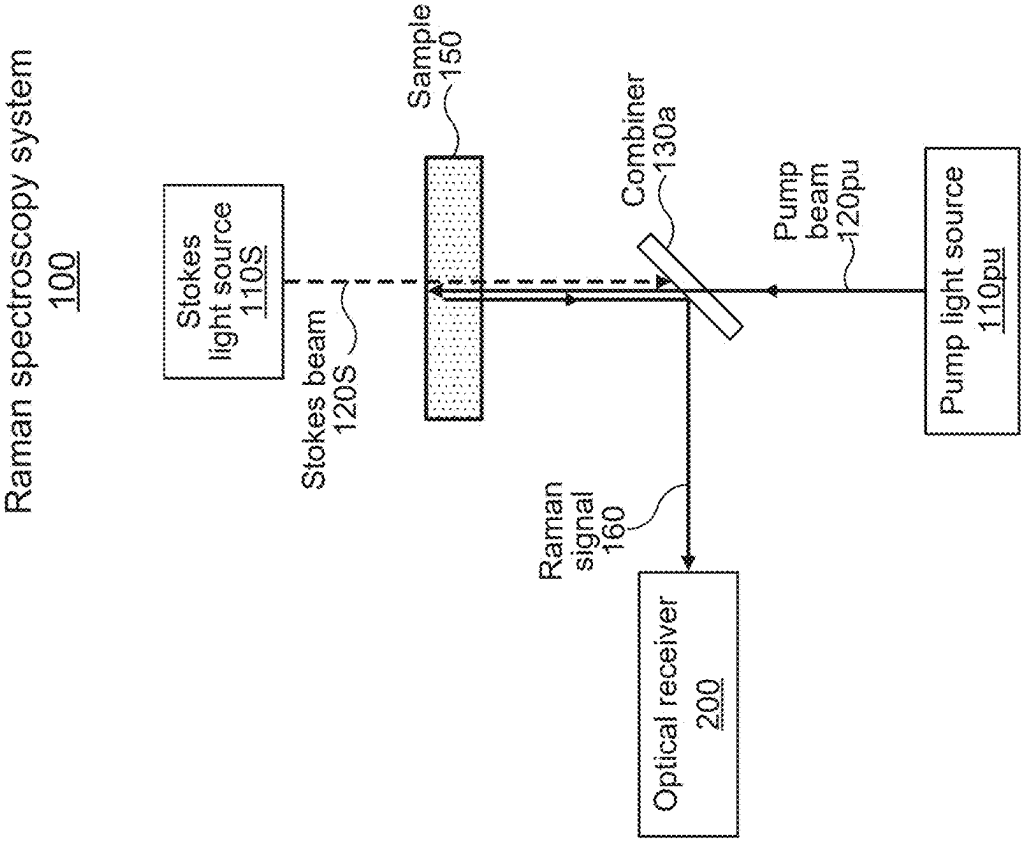


Figure 46

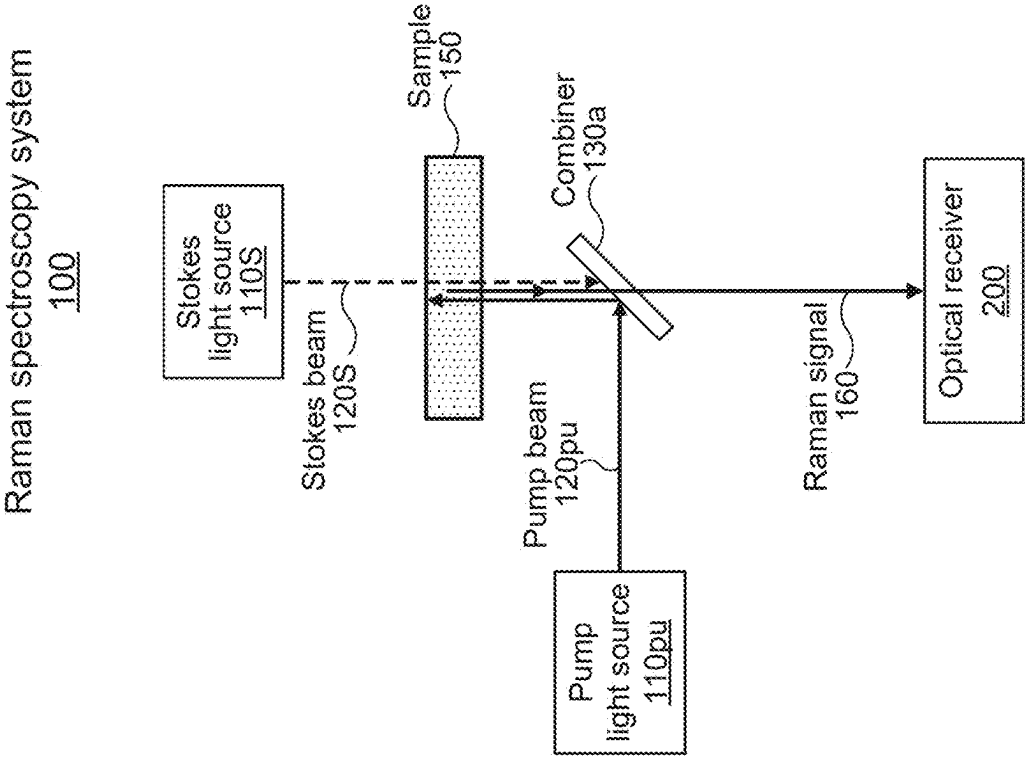


Figure 45

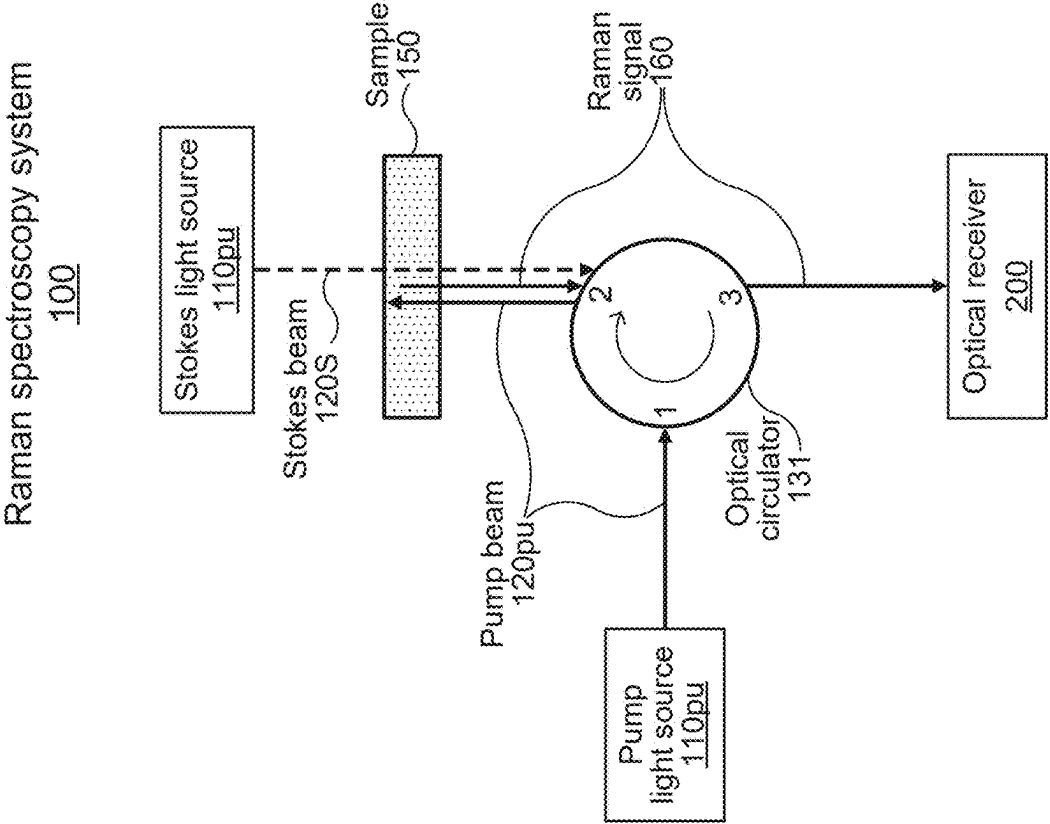


Figure 48

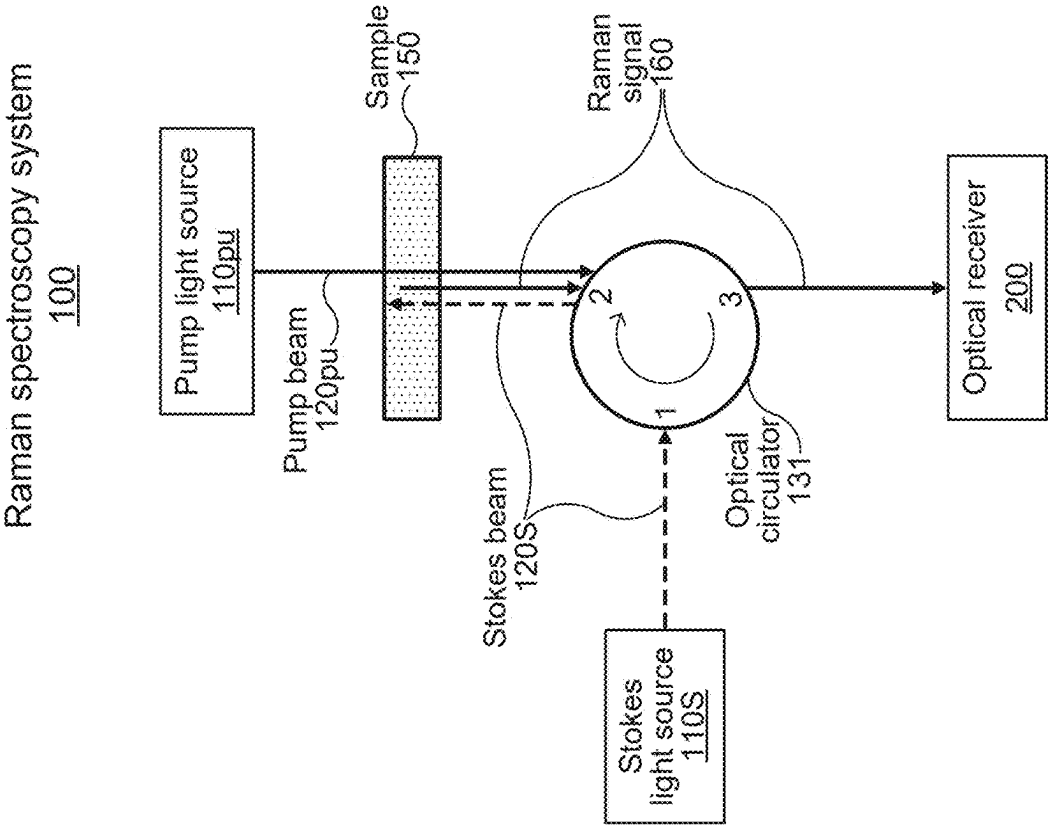


Figure 47

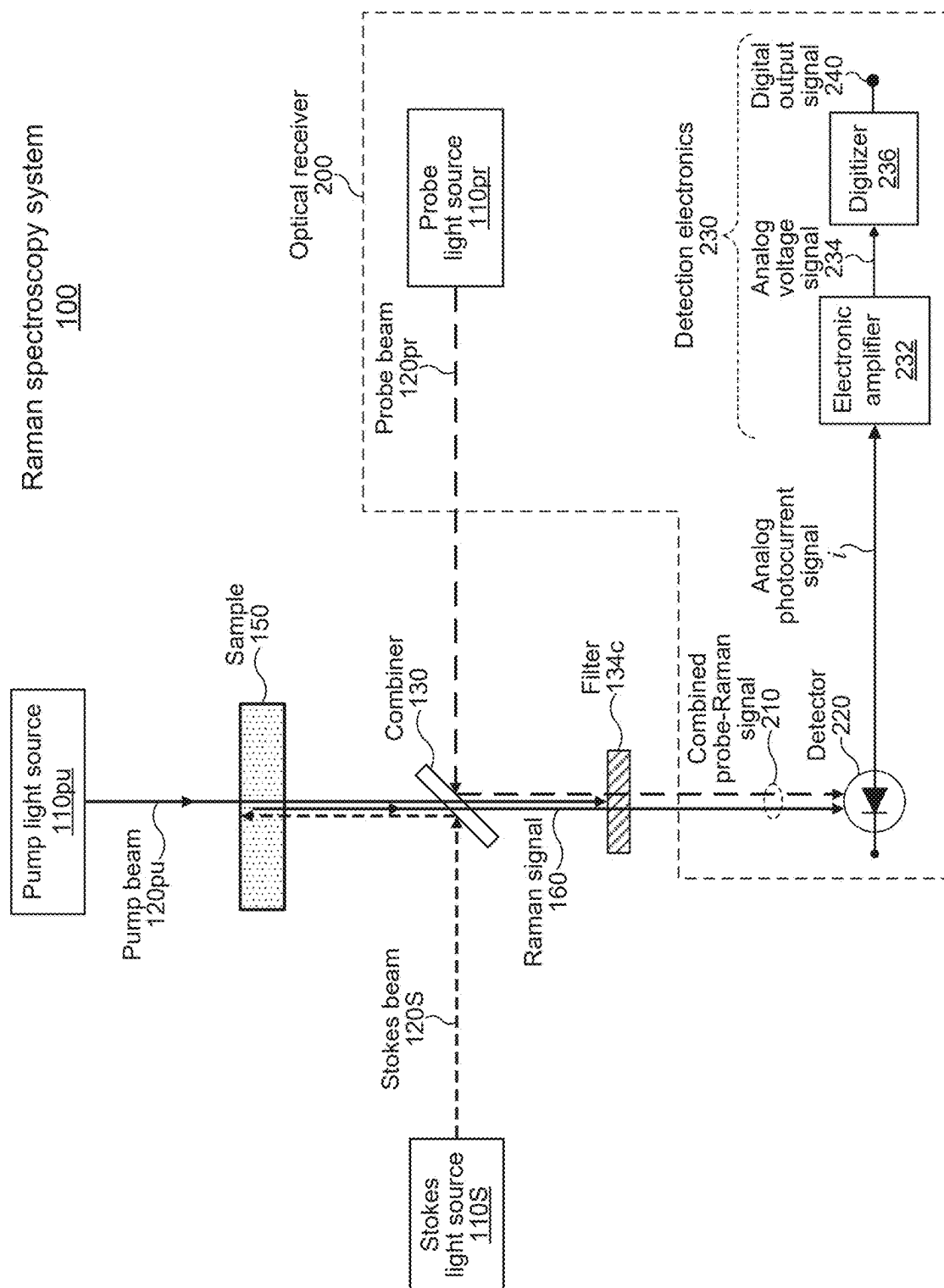


Figure 49

Raman spectroscopy system
100

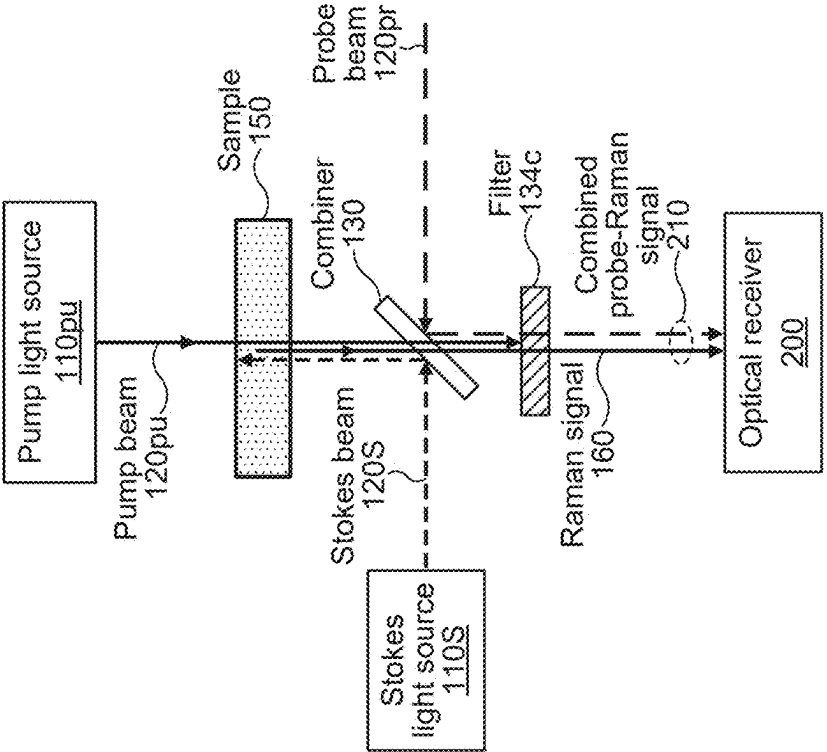


Figure 50

Raman spectroscopy system
100

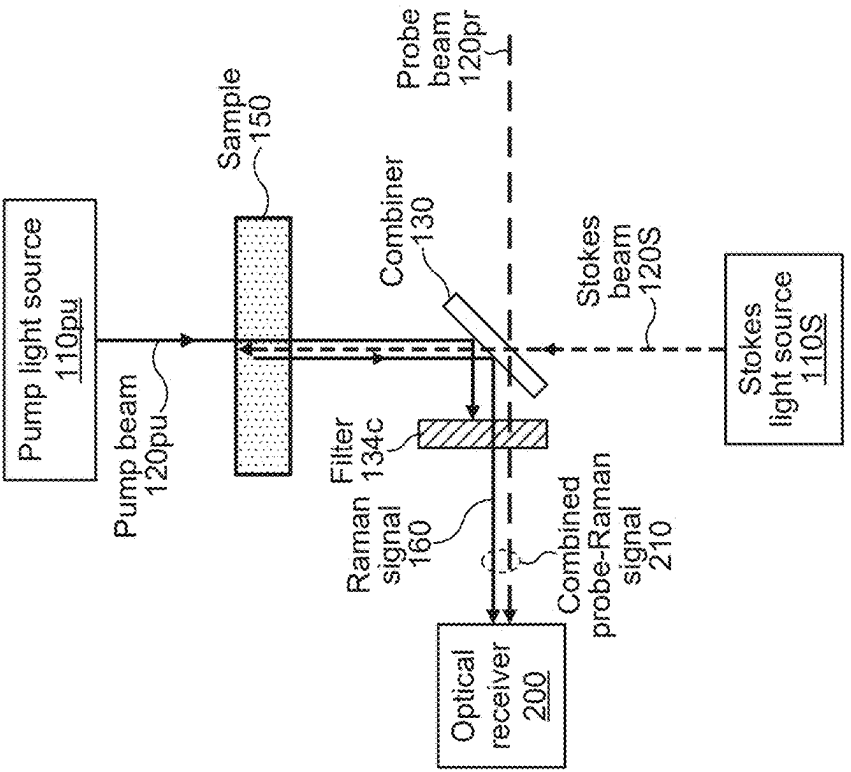


Figure 51

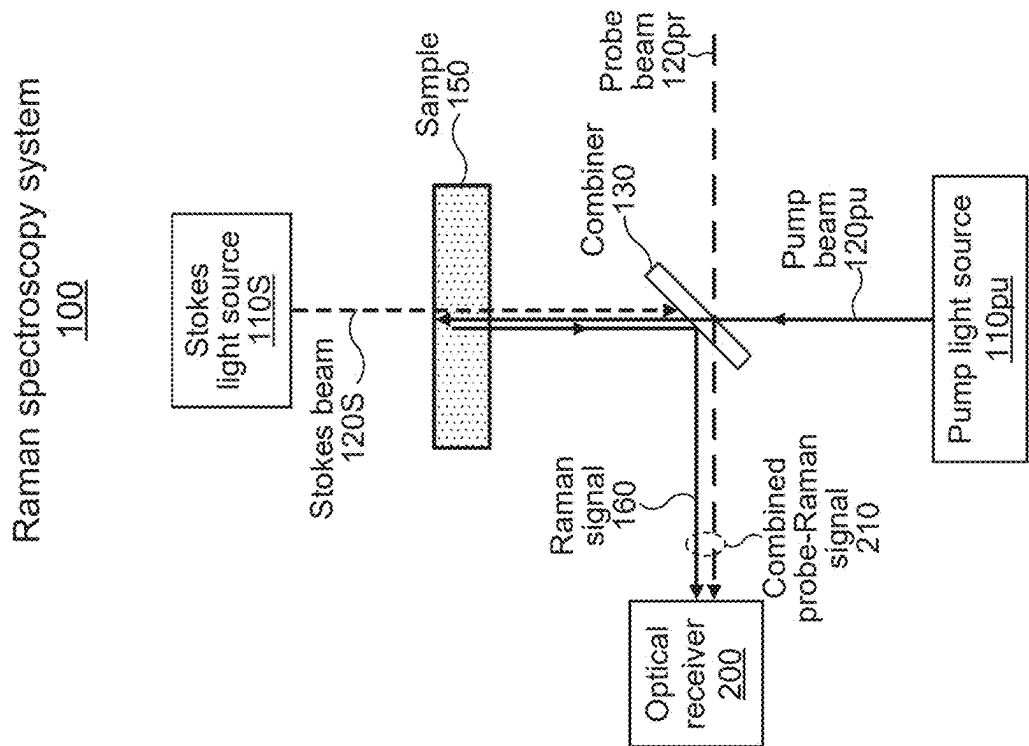


Figure 52

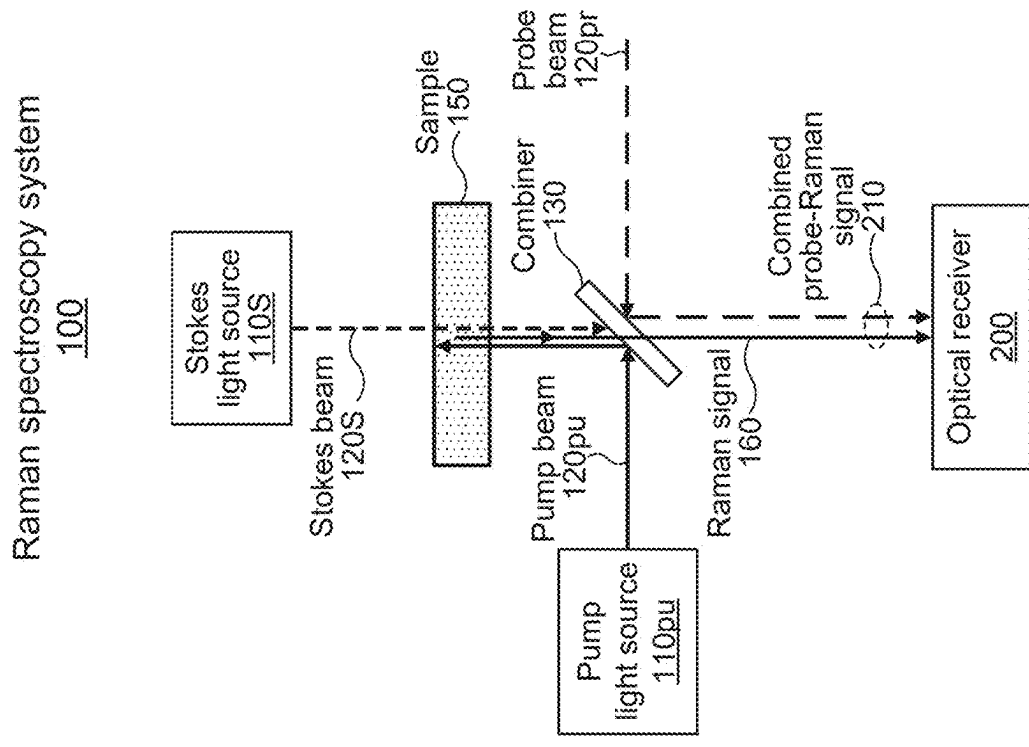


Figure 53

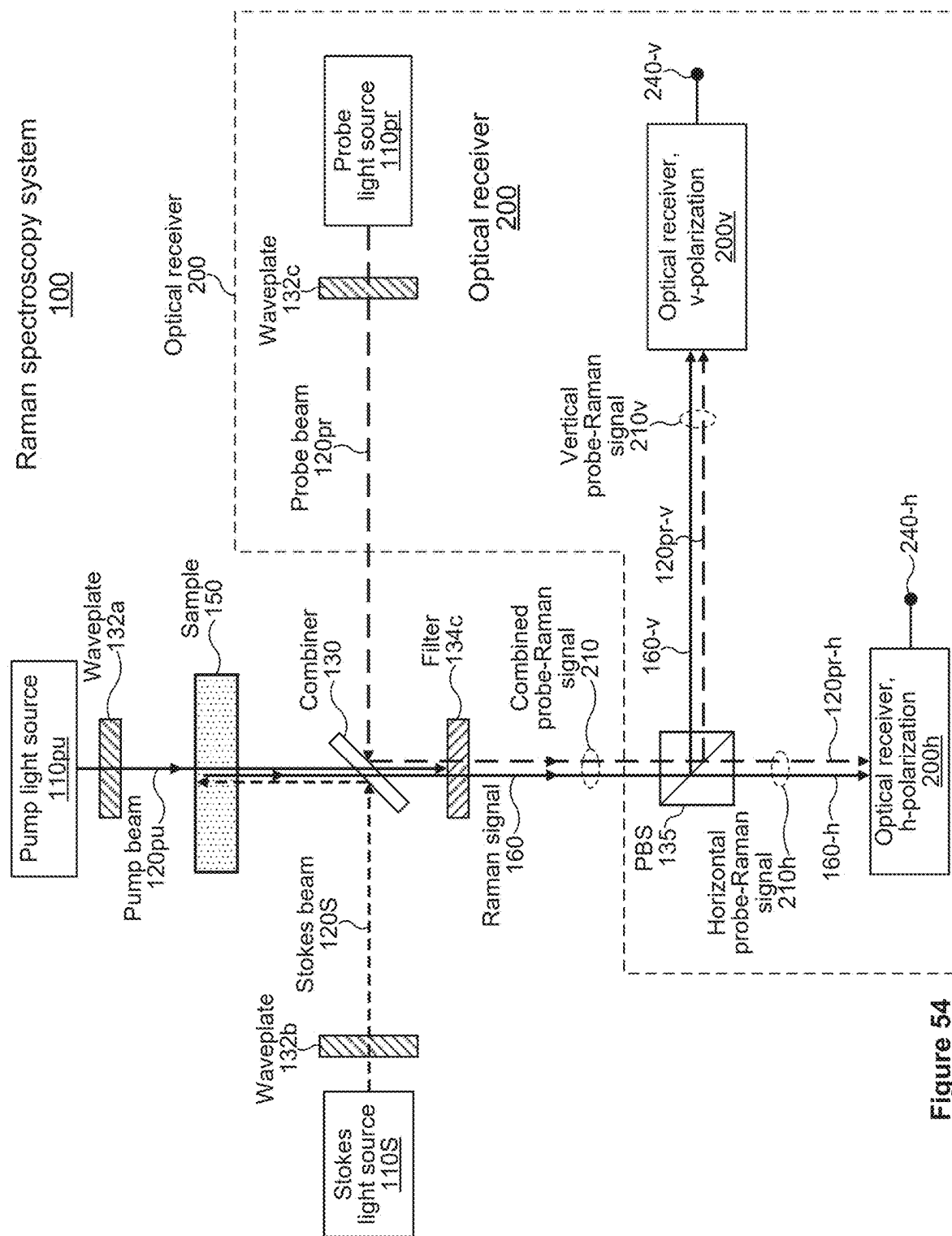
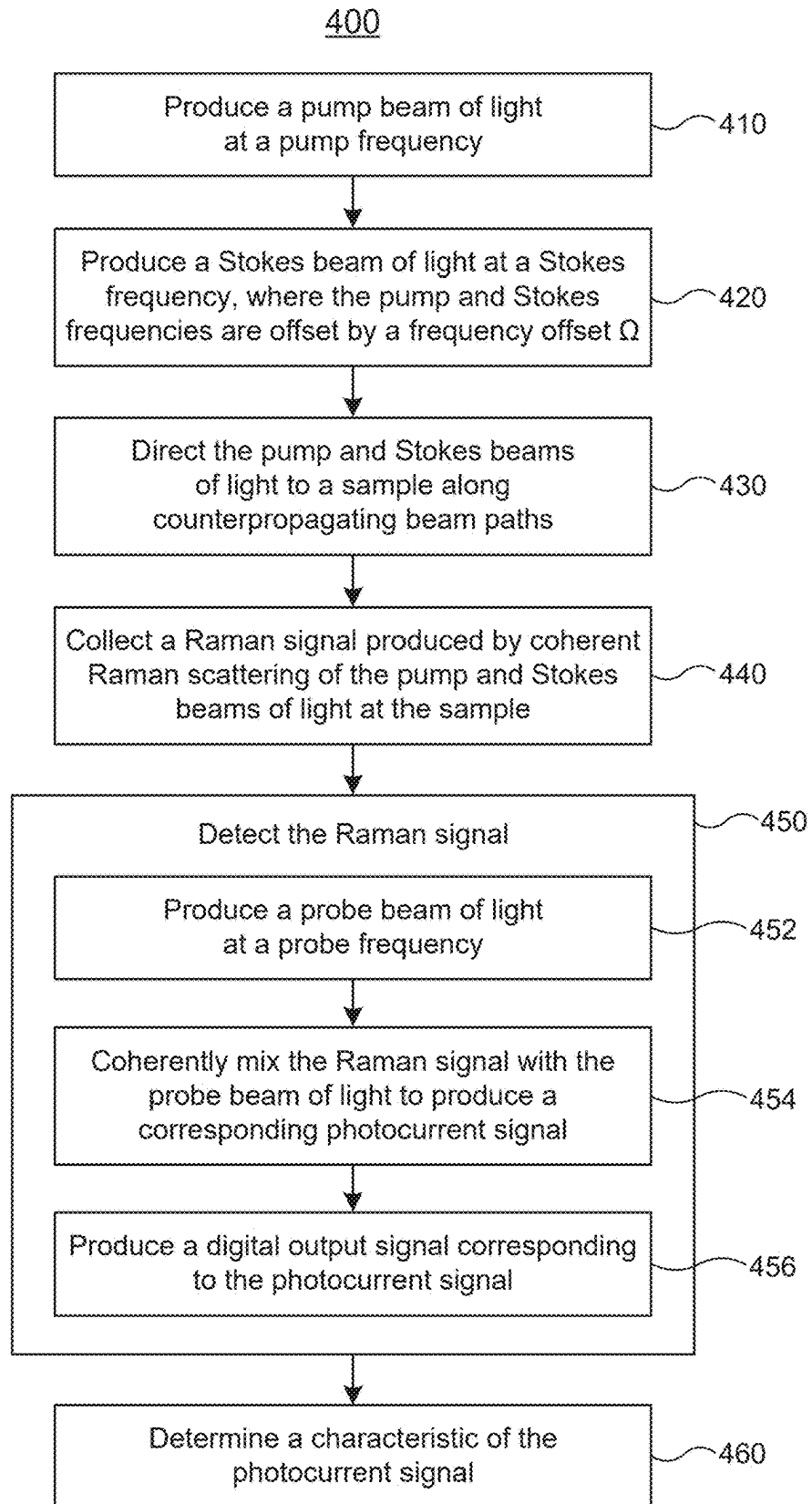


Figure 54

**Figure 55**

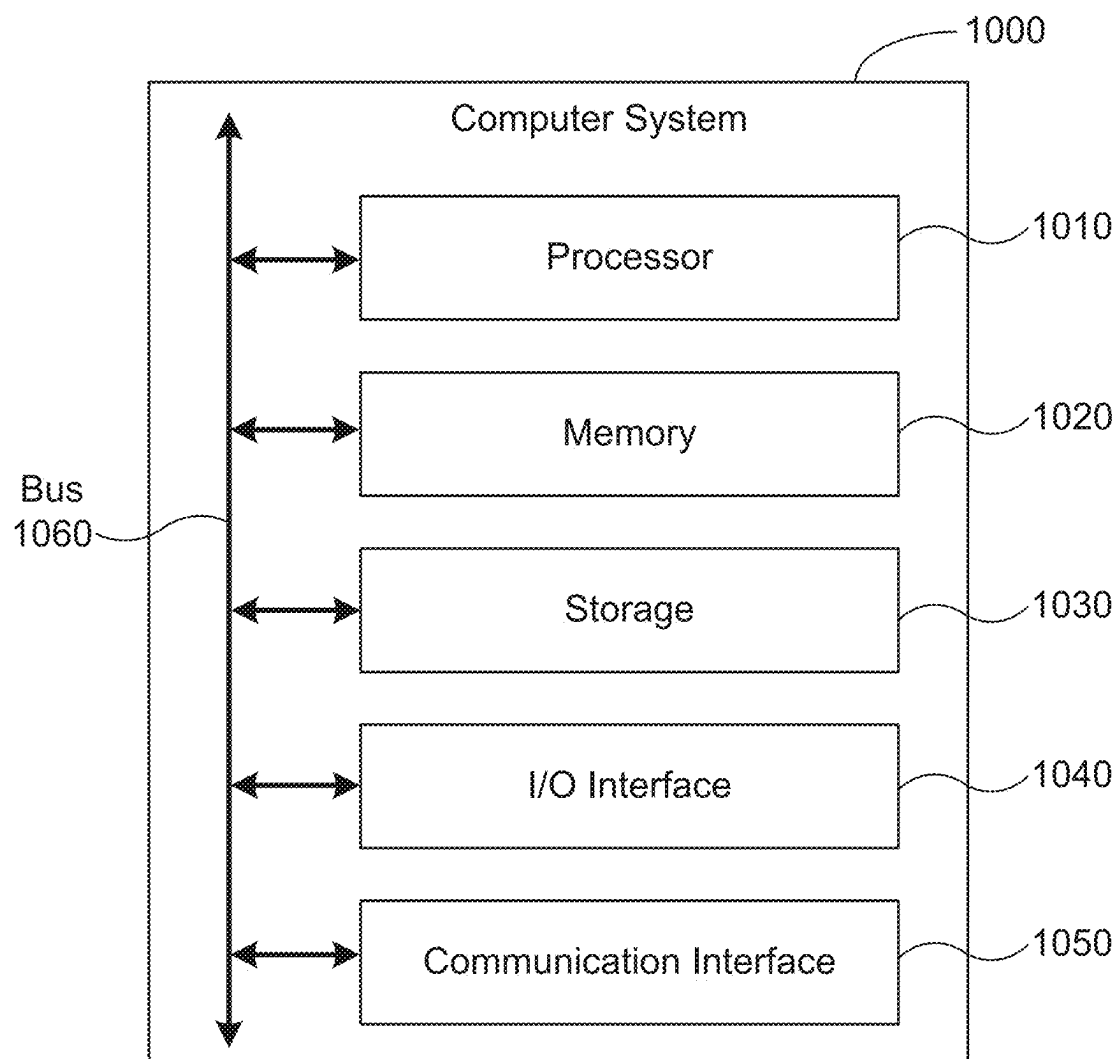


Figure 56

RAMAN SPECTROSCOPY SYSTEM WITH COUNTERPROPAGATING PUMP AND STOKES BEAMS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation-in-part of U.S. patent application Ser. No. 18/762,412, filed 2 Jul. 2024, which claims the benefit, under 35 U.S.C. § 119(e), of U.S. Provisional Patent Application No. 63/560,521, filed 1 Mar. 2024, the entireties of which are incorporated by reference herein.

TECHNICAL FIELD

[0002] This disclosure generally relates to Raman spectroscopy systems and methods that use Raman scattering.

BACKGROUND

[0003] Raman spectroscopy is an optical measurement technique that can be applied to the study of molecular dynamics (e.g., to investigate vibrational and rotational states of molecules). Molecules typically exhibit molecular vibrations with frequencies ranging from less than 10 terahertz (THz) to approximately 100 THz, which corresponds to wavenumbers of approximately 300 cm^{-1} to 3000 cm^{-1} and wavelengths of approximately 30 to 3 micrometers (μm). Raman spectroscopy is based on the inelastic scattering of photons (referred to as Raman scattering) that occurs when light interacts with molecular vibrations or phonons in a sample. Raman scattering causes the energy (or equivalently, the frequency) of scattered light to be shifted, and this shift in energy can provide information about the vibrational modes of molecules in the sample.

[0004] Raman spectroscopy can be used in various chemical sensing applications to identify molecular components in a sample. Since many molecules exhibit a unique Raman scattering spectrum, the spectrum of Raman-scattered light produced when light interacts with a sample can serve as a fingerprint to sense or identify various molecular species within the sample. A sample illuminated with light may produce Raman scattered light at different wavelengths from the illumination light, and measurement of the spectrum of the Raman scattered light is typically performed in the optical domain. For example, the spectrum of Raman scattered light can be measured using an optical spectrometer which separates the Raman scattered light into its optical frequency components using a diffractive element, such as a diffraction grating.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIGS. 1-2 each illustrate an example Raman spectroscopy system.

[0006] FIGS. 3-4 each illustrate an example Raman signal produced by coherent Raman scattering.

[0007] FIG. 5 illustrates the example Raman signal of FIG. 4 along with a probe beam.

[0008] FIG. 6 illustrates an expanded view of a portion of the Raman signal of FIG. 5.

[0009] FIG. 7 illustrates an expanded view of a portion of the Raman signal of FIG. 6.

[0010] FIGS. 8-10 illustrate time-domain and frequency-domain plots of an example electronic signal resulting from coherent mixing of the Raman signal and probe beam of FIG. 7.

[0011] FIG. 11 illustrates an example Raman signal that is measured at multiple probe frequencies.

[0012] FIG. 12 illustrates an example Raman spectrum corresponding to the Raman signal of FIG. 11.

[0013] FIG. 13 illustrates another example Raman signal that is measured at multiple probe frequencies.

[0014] FIGS. 14-15 each illustrate a second example Raman signal obtained by changing the frequency offset between a pump beam and a Stokes beam.

[0015] FIG. 16 illustrates an example Raman signal along with two probe beams of light.

[0016] FIG. 17 illustrates an example optical receiver for measuring the Raman signal and pump beam of light from FIG. 16.

[0017] FIG. 18 illustrates an example Raman spectroscopy system for measuring a Raman signal produced by spontaneous Raman scattering.

[0018] FIG. 19 illustrates an example Raman signal produced by the Raman spectroscopy system of FIG. 18.

[0019] FIG. 20 illustrates an example laser diode that produces a free-space beam of light.

[0020] FIG. 21 illustrates an example laser diode that produces seed light that is amplified by a semiconductor optical amplifier (SOA).

[0021] FIG. 22 illustrates an example laser diode that produces seed light that is amplified by a fiber-optic amplifier.

[0022] FIG. 23 illustrates an example sampled-grating distributed Bragg reflector (SG-DBR) laser.

[0023] FIG. 24 illustrates an example light source with multiple laser diodes and an optical multiplexer that combines light produced by the laser diodes into a single output beam of light.

[0024] FIG. 25 illustrates an example pump laser and Stokes laser with a fiber-optic combiner that produces a combined pump-Stokes beam coupled into an optical fiber.

[0025] FIG. 26 illustrates an example laser diode coupled to a waveguide of a photonic integrated circuit (PIC).

[0026] FIG. 27 illustrates an example pump laser and Stokes laser with a photonic integrated circuit (PIC) that produces a combined pump-Stokes beam coupled into an optical waveguide of the PIC.

[0027] FIG. 28 illustrates an example fiber-optic combiner that combines a Raman signal with a probe beam.

[0028] FIG. 29 illustrates an example photonic integrated circuit (PIC) with a waveguide combiner that combines a Raman signal with a probe beam.

[0029] FIGS. 30-35 each illustrate example frequency ranges of a pump beam and a Stokes beam.

[0030] FIG. 36 illustrates an example optical receiver with two detectors.

[0031] FIG. 37 illustrates an example optical receiver configured for polarization-sensitive detection of a Raman signal.

[0032] FIG. 38 illustrates an example optical receiver configured to detect in-phase and quadrature components of a Raman signal.

[0033] FIG. 39 illustrates an example optical receiver configured to detect polarization as well as in-phase and quadrature components of a Raman signal.

[0034] FIG. 40 illustrates an example Raman spectroscopy system with counterpropagating pump and Stokes beams.

[0035] FIGS. 41-42 each illustrate example beam angles between counterpropagating pump and Stokes beams.

[0036] FIGS. 43-46 illustrate four different example beam configurations for a Raman spectroscopy system with counterpropagating pump and Stokes beams.

[0037] FIGS. 47-48 each illustrate an example Raman spectroscopy system with counterpropagating pump and Stokes beams and with an optical circulator.

[0038] FIG. 49 illustrates an example Raman spectroscopy system with counterpropagating pump and Stokes beams and with a single optical combiner.

[0039] FIGS. 50-53 illustrate four different example beam configurations for a Raman spectroscopy system with counterpropagating pump and Stokes beams and with a single optical combiner.

[0040] FIG. 54 illustrates an example Raman spectroscopy system with counterpropagating pump and Stokes beams.

[0041] FIG. 55 illustrates an example method for measuring a Raman signal.

[0042] FIG. 56 illustrates an example computer system.

DETAILED DESCRIPTION

[0043] FIGS. 1-2 each illustrate an example Raman spectroscopy system 100. The Raman spectroscopy system 100 in each of FIGS. 1-2 may detect a Raman signal 160 by coherently mixing the Raman signal 160 with a probe beam of light 120_{pr} produced by a probe light source 110_{pr}. The Raman spectroscopy system 100 in each of FIGS. 1-2 may be referred to as a coherent Raman spectroscopy system, a coherent Raman spectroscopy system with heterodyne detection, or a high-resolution coherent Raman spectroscopy system. One or more of the systems or methods described herein may be applied to any suitable form of coherent Raman spectroscopy or coherent Raman scattering (CRS), such as for example, coherent anti-Stokes Raman scattering (CARS), stimulated Raman scattering (SRS), or Raman-induced Kerr effect (RIKE).

[0044] The Raman spectroscopy system 100 in FIG. 1 includes a pump light source 110_{pu} that produces a pump beam of light 120_{pu} and a Stokes light source 110_S that produces a Stokes beam of light 120_S. The pump light source 110_{pu} produces the pump beam of light 120_{pu} at a pump frequency, which may be referred to as a first frequency and may be represented by ν_1 , ν_{pu} , ω_1 , or ω_{pu} . The pump light source 110_{pu} may be referred to as a first light source, and the pump beam of light 120_{pu} may be referred to as a first beam of light. The Stokes light source 110_S produces the Stokes beam of light 120_S at a Stokes frequency, which may be referred to as a second frequency and may be represented by ν_2 , ν_S , ω_2 , or ω_S . The Stokes light source 110_S may be referred to as a second light source, and the Stokes beam of light 120_S may be referred to as a second beam of light. The pump and Stokes frequencies may be offset by a frequency offset Ω , where Ω equals $\nu_{pu} - \nu_S$ (or equivalently, $\Omega = \omega_1 - \omega_2$). Generally, the pump frequency ν_{pu} is greater than the Stokes frequency ν_S , and the frequency offset Ω is a positive value. The pump and Stokes light sources in FIG. 1 may each include a laser. For example, the Raman spectroscopy system 100 in FIG. 2 includes a pump laser 110_{pu} that produces a pump beam of light 120_{pu} and a Stokes laser 110_S that produces a Stokes beam of light 120_S.

[0045] In FIGS. 1-2, the pump beam 120_{pu} and the Stokes beam 120_S are directed to a sample 150, and the sample 150 produces a Raman signal 160 in response to the pump and Stokes beams. For example, the Raman signal 160 may be produced by coherent Raman scattering of the pump and Stokes beams within the sample 150. A Raman spectroscopy system may include one or more optical elements that direct the pump beam 120_{pu} and the Stokes beam 120_S to a sample 150. Additionally, the optical elements may collect the Raman signal 160 produced by the sample 150 in response to the pump beam 120_{pu} and Stokes beam 120_S and may direct the Raman signal 160 to an optical receiver 200. The optical elements may include free-space optics, fiber-optic components, waveguide-based optics, metamaterials, or any suitable combination thereof. For example, the optical elements may include a mirror, lens, optical combiner (e.g., beamsplitter), optical fiber, photonic integrated circuit (PIC), optical waveguide, or metamaterial-based optic. As another example, the optical combiner 130_a in each of FIGS. 1-2 may be a free-space dichroic beamsplitter that transmits light at the pump-beam wavelength and reflects light at the Stokes-beam wavelength. The combiner 130_a may combine the pump beam 120_{pu} and the Stokes beam 120_S to produce a combined pump-Stokes beam 140 that is directed to the sample 150. The pump and Stokes beams may be combined so that they are substantially overlapped with one another and propagate in the same direction along approximately the same optical axis. Alternatively, a Raman spectroscopy system 100 may not include a pump-Stokes optical combiner 130_a, and the pump beam 120_{pu} and the Stokes beam 120_S may be directed to a sample 150 as separate beams (e.g., the pump and Stokes beams may be overlapped or combined at the sample rather than being combined earlier). In this embodiment, the pump and Stokes beams may enter the sample 150 from opposite sides (e.g., the pump and Stokes beams may propagate to the sample in opposite directions along approximately the same optical axis). As another example, the optical elements may include a lens (not illustrated in FIGS. 1-2) that focuses the pump-Stokes beam 140 onto the sample 150. Additionally, the optical elements may include a lens (not illustrated in FIGS. 1-2) that collects the Raman signal 160 to produce a Raman-signal beam that is directed to the optical receiver 200.

[0046] In FIGS. 1-2 the combined pump-Stokes beam 140 is directed to one side of the sample 150, and the Raman signal 160 is emitted from the opposite side of the sample. A Raman signal 160 that is collected and directed to an optical receiver 200 may be emitted from a sample 150 in any suitable direction. For example, a Raman signal 160 that is sent to an optical receiver 200 may be emitted from a sample 150 in a forward-scattered direction (e.g., as illustrated in FIGS. 1-2), in a backward-scattered direction (e.g., back towards the pump or Stokes beam), or in a sideways-scattered direction (e.g., in a direction approximately orthogonal to the combined pump-Stokes beam 140 in FIGS. 1-2).

[0047] The Raman spectroscopy system 100 in each of FIGS. 1-2 includes an optical receiver 200 that detects the Raman signal 160 produced by the sample 150. The optical receiver 200 (which may be referred to as a heterodyne optical receiver or a high-resolution optical receiver) may detect the Raman signal 160 using an optical heterodyne technique in which the Raman signal 160 is coherently mixed with a probe beam of light 120_{pr}. The optical receiver

200 in FIG. 1 includes a probe light source **110_{pr}** that produces a probe beam of light **120_{pr}** at a probe frequency. The probe light source **110_{pr}** may be referred to as a third light source, and the probe beam of light **120_{pr}** may be referred to as a third beam of light. The probe frequency may be referred to as a third frequency and may be represented by ν_3 , ν_{pr} , ω_3 , or ω_{pr} . The probe light source **110_{pr}** in FIG. 1 may include a laser. For example, the optical receiver **200** in FIG. 2 includes a probe laser **110_{pr}** that produces a probe beam of light **120_{pr}**.

[0048] The optical combiner **130_b** in each of FIGS. 1-2 may be a dichroic or a non-dichroic beamsplitter that combines the Raman signal **160** and the probe beam **120_{pr}** to produce a combined probe-Raman signal **210** that is directed to a detector **220**. An optical receiver **200** may include one or more optical detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with at least a portion of a probe beam **120_{pr}** to produce an electronic signal. Each of the optical receivers **200** in FIGS. 1-2 includes one detector **220** that receives the combined probe-Raman signal **210**. The Raman signal **160** and the probe beam **120_{pr}** are coherently mixed at the detector **220**, and this heterodyne mixing process produces an electronic signal, which in FIGS. 1-2 is indicated as analog photocurrent signal *i*. In FIGS. 1-2, the detection electronics **230** receives the photocurrent signal *i* and produces a digital output signal **240** that corresponds to the photocurrent signal. Detection electronics **230** may include or may be referred to as an electronic circuit. The digital output signal **240** may be sent to a processor, and the processor may determine a characteristic of the analog photocurrent signal *i* based on the digital output signal **240**. For example, the characteristic of the analog photocurrent signal (or the characteristic of an analog voltage signal that corresponds to the photocurrent signal) determined by the processor may include one or more of: a peak amplitude (e.g., a peak amplitude of the photocurrent signal or a peak amplitude of the corresponding voltage signal), an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization (e.g., a polarization of a Raman signal). An analog photocurrent signal may be referred to as a photocurrent, a photocurrent signal, or a current signal, and an analog voltage signal may be referred to as a voltage signal. A processor of a Raman spectroscopy system **100** may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. A processor may be similar to the computer system **1000** illustrated in FIG. 56 and described herein. In some embodiments, a processor or a portion of a processor may be located in the detection electronics **230**. For example, a processor located in the detection electronics **230** may receive a digital output signal **240** and perform some preprocessing of the digital signal or determine a characteristic of a photocurrent signal based on the digital signal.

[0049] In FIGS. 1-2, the probe beam **120_{pr}** does not travel through the sample **150**, and the probe beam is combined with the Raman signal **160** after the Raman signal has exited the sample **150**. In other embodiments, a Raman spectroscopy system **100** may include a probe laser **110_{pr}** that produces a probe beam **120_{pr}** that is directed through the sample **150**. For example, the probe beam **120_{pr}** may be

combined with the pump beam **120_{pu}** and the Stokes beam **120_S**, and all three beams may be directed to the sample **150**. The probe beam **120_{pr}** may travel through the sample **150** and may exit the sample along with the Raman signal **160**. The optical receiver **200** may not include a combiner **130_b**, since the probe beam **120_{pr}** and the Raman signal **160** may already be combined after they exit the sample **150**. The probe beam **120_{pr}** and the Raman signal **160** may be directed to a detector **220** without being transmitted or reflected by an optical combiner **130_b**.

[0050] The detection electronics **230** in FIG. 2 includes an electronic amplifier **232** and a digitizer **236**. The electronic amplifier **232** may include a transimpedance amplifier that amplifies the photocurrent signal *i* to produce an analog voltage signal **234** that corresponds to the photocurrent signal *i* (e.g., the photocurrent signal *i* and the analog voltage signal **234** may have similar temporal shapes or may include similar electronic frequency components). The electronic amplifier **232** may include an additional gain stage that further amplifies an intermediate voltage signal produced by the transimpedance amplifier to produce the voltage signal **234**. Additionally, the electronic amplifier **232** may include an electronic filter (e.g., a low-pass, high-pass, or band-pass filter) that filters the photocurrent signal or voltage signal. For example, the electronic amplifier **232** may include (i) a high-pass filter that removes a DC offset and low-frequency components (e.g., frequency components below 10 MHz) from the photocurrent signal or (ii) a band-pass filter that removes the DC and low-frequency components as well as high-frequency components (e.g., frequency components above 5 GHz). Herein, an electronic signal produced in response to coherent mixing may refer to a current signal (e.g., a photocurrent *i* produced by a detector **220**) or may refer to a corresponding voltage signal (e.g., a voltage signal **234** produced by an electronic amplifier that amplifies a photocurrent produced by a detector to produce the voltage signal). In FIG. 2, the digitizer **236** receives the voltage signal **234** and produces a digital output signal **240** that includes a digital representation of the voltage signal **234**. The digital output signal **240** may be a time-domain digital representation of the voltage signal **234**. The digital output signal **240** may be referred to as corresponding to or representing the voltage signal **234** or the photocurrent signal *i*. The digitizer **236** may include an analog-to-digital converter (ADC) that produces a digital version of the voltage signal **234**. Additionally or alternatively, the digitizer **236** may include a peak detector that determines a peak value of the voltage signal **234**.

[0051] A Raman spectroscopy system **100** may include one or more optical detectors **220**. An optical detector **220** (which may be referred to as a detector, photodetector, or photodiode) may include a PN photodiode, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), silicon photomultiplier (SiPM), or photomultiplier tube (PMT). A PN photodiode refers to a photodiode structure formed by a p-doped semiconductor and an n-doped semiconductor, where the PN acronym refers to the structure having p-doped and n-doped regions. A PIN photodiode refers to a photodiode structure formed by an undoped intrinsic semiconductor region located between p-doped and n-doped regions, where the PIN acronym refers to the structure having p-doped, intrinsic, and n-doped regions.

[0052] A PN photodiode, PIN photodiode, APD, or SPAD may include any suitable semiconductor material, such as for example: silicon, germanium, gallium arsenide (GaAs), indium phosphide (InP), indium arsenide (InAs), aluminum arsenide (AlAs), indium antimonide (InSb), aluminum antimonide (AlSb), gallium antimonide (GaSb), aluminum gallium arsenide (AlGaAs), indium gallium arsenide (InGaAs), indium aluminum arsenide (InAlAs), indium arsenide antimonide (InAsSb), aluminum arsenide antimonide (AlAsSb), aluminum gallium antimonide (AlGaSb), gallium arsenide antimonide (GaAsSb), aluminum indium arsenide antimonide (AlInAsSb), indium gallium arsenide antimonide (InGaAsSb), indium gallium aluminum arsenide (InGaAlAs), aluminum gallium arsenide antimonide (AlGaAsSb), or silicon germanium (SiGe). For example, the Raman signal **160** and the probe beam **120_{pr}** in FIGS. 1-2 may each have a wavelength in the 400-1100 nanometer (nm) range, and the detector **220** may include a silicon PIN photodiode. As another example, the Raman and probe beams in FIGS. 1-2 may each have a wavelength in the 1000-1700 nm range, and the detector **220** may include an InGaAs PIN photodiode. As another example, a detector **220** may include an APD that includes a semiconductor material with antimonide (e.g., InSb, AlSb, GaSb, InAsSb, AlAsSb, AlGaSb, GaAsSb, AlInAsSb, InGaAsSb, or AlGaAsSb).

[0053] In order for a detector **220** to detect an optical signal, the wavelength of the optical signal must be within the detector's wavelength range of responsivity (e.g., approximately 400-1100 nm for a silicon detector, and approximately 1000-1700 nm for an InGaAs detector) and a frequency of an amplitude modulation of the optical signal must be within the electronic bandwidth of the detector. The electronic bandwidth (Δf) of a detector **220** refers to the range of electronic modulation frequencies over which a detector may detect an optical signal, where detection of the optical signal refers to (i) the detector producing a photocurrent signal **i** that corresponds to the optical signal and (ii) an electronic amplifier **232** producing a voltage signal **234** that corresponds to the photocurrent signal. If a silicon detector **220** has an electronic bandwidth that extends from 100 MHz to 5 GHz, then the detector may detect optical signals with (i) wavelengths between approximately 400 nm and 1100 nm and (ii) amplitude modulation between 100 MHz and 5 GHz. For example, a 900-nm optical signal with an amplitude modulation at a frequency between 100 MHz and 5 GHz may be detected by the silicon detector **220**. The silicon detector **220** may not detect a continuous-wave or substantially constant portion of a 900-nm optical signal (e.g., the substantially constant portion of the optical signal may produce a DC current in the detector that may be electronically filtered out), and the silicon detector **220** may not detect a portion of the optical signal with an amplitude modulation greater than approximately 5 GHz. Herein, reference to the electronic bandwidth (Δf) of a detector **220** may refer to (i) the electronic bandwidth of just the detector or (ii) the overall bandwidth of the detector in combination with an electronic amplifier **232**.

[0054] A detector **220** may have an electronic bandwidth Δf between approximately 100 MHz and approximately 50 GHz. For example, the detector **220** in each of FIGS. 1-2 may have an electronic bandwidth between 100 MHz and 10 GHz. As another example, the detector **220** in each of FIGS. 1-2 may have an electronic bandwidth that extends from a low-frequency cutoff to a high-frequency cutoff. The low-

frequency cutoff may be approximately DC (i.e., zero hertz), 1 MHz, 10 MHz, 50 MHz, or 100 MHz, and the high-frequency cutoff may be approximately 500 MHz, 1 GHz, 2 GHz, 5 GHz, 10 GHz, 20 GHz, or 50 GHz. The electronic bandwidth of a detector **220** may refer to the bandwidth of only the detector **220**. For example, a detector **220** may have an electronic bandwidth that extends from DC to 10 GHz, and the detector may be referred to as having a 10-GHz bandwidth that extends from DC to 10 GHz. Alternatively, the electronic bandwidth of a detector **220** may refer to the overall bandwidth of the detector in combination with an electronic amplifier **232** that amplifies the photocurrent signal **i** produced by the detector. An electronic amplifier **232** may have a low-frequency cutoff (e.g., DC, 1 MHz, 10 MHz, 50 MHz, or 100 MHz) and a high-frequency cutoff (e.g., 500 MHz, 1 GHz, 2 GHz, 5 GHz, 10 GHz, 20 GHz, or 50 GHz), and a detector-amplifier combination may be referred to as having an electronic bandwidth that extends from the low-frequency cutoff to the high-frequency cutoff. For example, if the detector bandwidth extends from DC to 10 GHz, and the electronic amplifier bandwidth extends from DC to 5 GHz, then the detector (or, the detector-amplifier combination) may be referred to as having an electronic bandwidth of 5 GHz that extends from DC to 5 GHz. As another example, if the detector bandwidth extends from DC to 10 GHz, and the electronic amplifier bandwidth extends from 100 MHz to 5 GHz, then the detector may be referred to as having an electronic bandwidth of 4.9 GHz that extends from 100 MHz to 5 GHz.

[0055] A Raman spectroscopy system **100** may include one or more optical waveplates that change or rotate the polarization of a beam of light. For example, a half-wave plate may be used to rotate a linearly polarized beam of light to a different polarization orientation (e.g., from vertically polarized to horizontally polarized), and a quarter-wave plate may be used to convert a linearly polarized beam of light to a circular or elliptical polarization. The pump laser **110_{pu}** in FIG. 2 may produce linearly polarized light, and the waveplate **132a** may be a half-wave plate that rotates the polarization of the pump beam **120_{pu}** prior to the pump beam being directed to the sample **150**. Alternatively, the waveplate **132a** may be a quarter-wave plate that converts the linearly polarized pump beam **120_{pu}** to a circular or elliptical polarization prior to the pump beam being directed to the sample **150**. Similarly, the Stokes laser **110_s** in FIG. 2 may produce linearly polarized light, and the waveplate **132b** may be (i) a half-wave plate that rotates the polarization of the Stokes beam **120_s** or (ii) a quarter-wave plate that converts the Stokes beam **120_s** to a circular or elliptical polarization.

[0056] The probe laser **110_{pr}** in FIG. 2 may produce linearly polarized light, and the waveplate **132c** may be (i) a half-wave plate that rotates the polarization of the probe beam **120_{pr}** or (ii) a quarter-wave plate that converts the probe beam **120_{pr}** to a circular or elliptical polarization prior to the probe beam being combined with the Raman signal **160**. Changing the polarization of the probe beam **120_{pr}** may allow the Raman signal **160** and the probe beam to be coherently mixed. The polarization of the probe beam **120_{pr}** can be changed so that it has both horizontal and vertical polarization components, which ensures that at least a portion of the probe beam **120_{pr}** and the Raman signal **160** have polarizations that are oriented in the same direction so that their electric fields may be added together.

[0057] Each of the optical waveplates 132 in FIG. 2 may be a free-space optical element, a fiber-optic component, a waveguide-based optical element, or a metamaterial-based optic. Additionally, each of the optical waveplates 132 in FIG. 2 may be a fixed waveplate or an adjustable waveplate. A fixed waveplate may have a fixed optical phase difference between the two axes of the waveplate (e.g., a quarter-wave plate may have a one-quarter wavelength phase difference, and a half-wave plate may have a one-half wavelength phase difference). An adjustable waveplate may allow for the phase difference between the two axes of the waveplate to be dynamically changed. For example, an electronically adjustable waveplate may include a Pockels cell, a liquid crystal device, or a photoelastic modulator that allows the phase difference to be adjusted electronically so that the waveplate can be dynamically configured to act as a waveplate having any suitable phase difference (e.g., a phase difference between zero wavelengths and one-half wavelength). An adjustable waveplate may switch between (i) applying no phase difference to incident light so that the transmitted light has the same polarization as the incident light and (ii) applying a one-quarter wavelength or one-half wavelength phase difference so that linearly polarized incident light is converted to circularly polarized light or is rotated to a different polarization. For example, the pump laser 110_{pu} in FIG. 2 may produce vertically polarized light, and the waveplate 132_a may be an adjustable waveplate that switches between (i) applying no polarization rotation to the pump beam 120_{pu} so that the pump beam remains vertically polarized and (ii) applying a 90-degree rotation to the pump-beam polarization so that the pump beam 120_{pu} after the waveplate 132_a is horizontally polarized. Dynamically changing the polarization of the pump beam 120_{pu} may allow the Raman spectroscopy system 100 in FIG. 2 to perform measurements at two different pump-beam polarizations, which may produce additional data for characterization of the sample 150.

[0058] In some embodiments, an optical waveplate 132 may be a metamaterial-based waveplate. A metamaterial refers to an engineered material having features or repeating patterns at scales smaller than the wavelength of light interacting with the metamaterial. A metamaterial may be configured to act as a mirror, lens, waveplate, diffractive optical element, optical combiner, or optical waveguide. A metamaterial-based waveplate may affect the polarization of a beam of light based on wavelength. For example, the Raman spectroscopy system 100 in FIG. 2 may include a metamaterial-based waveplate (not illustrated in FIG. 2) located after the combiner 130_a and before the sample 150. The metamaterial waveplate may change the polarization of the pump beam 120_{pu} from linear to circular while preserving the polarization of the Stokes beam 120_S (e.g., the Stokes beam may remain linearly polarized and may not be significantly changed by the waveplate).

[0059] A Raman spectroscopy system 100 may include an optical filter that transmits light at one or more wavelengths and blocks light at one or more other wavelengths. The optical filter 134 in FIG. 2 is located between the sample 150 and the optical receiver 200 and may be configured to substantially transmit one or more optical wavelengths associated with the Raman signal 160 and substantially block one or more wavelengths associated with the pump beam 120_{pu} or the Stokes beam 120_S. For example, the optical filter 134 may transmit greater than 90% of the Raman signal

160 and may block greater than 90% of both the pump and Stokes beams. As another example, the optical filter 134 may transmit greater than 90% of the Raman signal 160 and the Stokes beam 120_S, and the optical filter 134 may block greater than 98% of the pump beam 120_{pu}.

[0060] A Raman spectroscopy system 100 may include an optical polarizer that transmits light having a particular polarization (e.g., horizontal) and blocks light having an orthogonal polarization (e.g., vertical). The optical polarizer 136 in FIG. 2 is located between the sample 150 and the optical receiver 200 and may be oriented to transmit light with a polarization associated with the Raman signal. Additionally, the polarizer 136 may block a polarization associated with the pump or Stokes beams. For example, the Stokes beam 120_S incident on the sample 150 may be vertically polarized, and the Raman signal 160 may be at least partially horizontally polarized. The polarizer 136 may be oriented to block vertically polarized light and transmit horizontally polarized light so that the Stokes beam 120_S is blocked and the Raman signal 160 is at least partially transmitted by the polarizer. A Raman spectroscopy system 100 may include both an optical filter 134 and an optical polarizer 136. For example, the optical filter 136 in FIG. 2 may be configured to transmit the Raman signal 160 and the Stokes beam 120_S and block the pump beam 120_{pu}. Additionally, the Raman signal 160 and the Stokes beam 120_S may be orthogonally polarized, and the polarizer 136 may be oriented to transmit the Raman signal 160 and block the Stokes beam 120_S. Using a filter or polarizer located after the sample 150 to block light from the pump beam 120_{pu} or the Stokes beam 120_S may reduce noise in the system by reducing the amount of unwanted background light that reaches the detector 220.

[0061] The sample 150 in FIGS. 1-2 may be a solid, liquid, or gas, or any combination thereof. The sample 150 may include a biological material, an organic material, an inorganic material, a crystalline material, an amorphous solid material, or any other suitable material or combination of suitable materials. For example, the sample 150 may include a drug, mineral, food, contaminant, or explosive material that produces a Raman signal 160 in response to excitation by the pump and Stokes beams. As another example, the sample 150 may be a biological material (e.g., blood, urine, saliva, sweat, or cerebrospinal fluid) and a component or molecule (e.g., glucose or cortisol) that is part of the biological material may produce a Raman signal 160. As another example, the sample 150 may be water or wastewater that may include a contaminant, virus, bacteria, or an indicator of an infectious disease. The Raman signal 160 may be produced by coherent Raman scattering of the pump and Stokes beams within the sample 150, and the frequency offset Ω between the pump and Stokes beams may be approximately equal to a vibrational frequency or an electronic-transition frequency of a particular material that is part of the sample. The vibrational frequency of the particular material may correspond to a molecular vibration of a molecule, or in the case of a crystalline material, may correspond to a lattice vibration of a crystal. For example, the sample 150 may include glucose, and the frequency offset Ω may be approximately equal to a frequency of a molecular vibration of glucose.

[0062] The Raman spectroscopy systems 100 in FIGS. 1-2 may perform one or more measurements of a sample 150, and each measurement may include determining a charac-

teristic of an electronic signal that results from the coherent mixing of the Raman signal **160** and the probe beam **120_{pr}**. The electronic signal may include a photocurrent *i* or a corresponding voltage signal **234**. The frequency offset Ω may be approximately equal to a vibrational frequency or electronic-transition frequency of a particular material, and based on the one or more measurements, a processor may determine whether the particular material is present in the sample **150**. Additionally or alternatively, the processor may determine an amount or a concentration of the particular material in the sample based on the measurements. For example, the frequency offset Ω may be approximately equal to a vibrational frequency of glucose, and, based on one or more optical heterodyne measurements of a Raman signal **160** produced by a sample **150**, the processor may determine (i) whether glucose is present in the sample or (ii) an amount or a concentration of glucose in the sample. The amount of glucose that is present in the sample **150** may be proportional to the amplitude of a photocurrent signal *i* produced by coherent mixing of the Raman signal **160** and the probe beam **120_{pr}**. Based on the amplitude of the photocurrent signal *i*, the processor may determine the amount or concentration of glucose in the sample **150**.

[0063] A technical advantage of a coherent Raman spectroscopy system **100** as described herein is a higher spectral resolution or a better chemical sensitivity than a conventional Raman spectroscopy system. As such, a Raman spectroscopy system **100** as described herein may be referred to as a high-resolution Raman spectroscopy system or as a high-resolution coherent Raman spectroscopy system. In a conventional Raman spectroscopy system, a Raman signal produced by a sample may be measured in the optical domain using an optical spectrometer. A spectrometer typically uses a dispersive optical element (e.g., a diffraction grating) to separate the Raman signal into its various spectral components. However, this type of measurement performed in the optical domain typically has a spectral resolution on the order of 1 cm^{-1} (or, about 30 GHz). In contrast, the spectral resolution of a coherent Raman spectroscopy system with heterodyne detection, as described herein, is determined primarily by the spectral linewidth $\Delta\nu_{pr}$ of the probe beam **120_{pr}** that is coherently mixed with the Raman signal **160**. The probe laser **110_{pr}** may include a wavelength-tunable laser diode with a linewidth of 200 MHz or less, which corresponds to a spectral resolution of the Raman spectroscopy system **100** of less than 200 MHz (or, less than 0.007 cm^{-1}). This 200-MHz spectral resolution is more than 100 times better than the 30-GHz spectral resolution of a conventional Raman spectroscopy system. In some embodiments, the probe laser **110_{pr}** may have a linewidth of 1 MHz or less, which corresponds to a spectral resolution of the Raman spectroscopy system **100** of less than 1 MHz (or, less than $33\times 10^{-6}\text{ cm}^{-1}$). A related advantage of a coherent Raman spectroscopy system **100** is that the signal capture and analysis are performed in the electronic domain (e.g., at electronic frequencies between DC and 50 GHz) rather than in the optical domain (e.g., at optical frequencies between 60 THz and 1,000 THz). The coherent mixing of two optical signals (Raman signal **160** and probe beam **120_{pr}**) produces an electronic signal which can be analyzed with relatively high resolution compared to an optical signal. This electronic signal analysis, along with the relatively narrow spectral linewidth of the probe laser **110_{pr}**, provides a coherent Raman spectroscopy system **100**

with a high spectral resolution. Additionally, the wavelength tunability of the probe laser **110_{pr}** allows a Raman spectrum of a material to be determined at multiple frequencies with high spectral resolution.

[0064] The higher spectral resolution of a coherent Raman spectroscopy system **100** may provide a corresponding improvement in the ability of the coherent Raman spectroscopy system to sense various chemical species. For example, a high-resolution coherent Raman spectroscopy system **100** may be able to distinguish between different chemical species that have Raman peaks located relatively close together, whereas a conventional Raman spectroscopy system may not be able to resolve spectral features below about 1 cm^{-1} . Additionally, the higher spectral resolution of a coherent Raman spectroscopy system **100** may allow for lower concentrations of materials to be detected, as compared to a conventional Raman spectroscopy system. For example, a coherent Raman spectroscopy system **100** may be able to detect small deviations in the chemical signature of a biological sample, which may indicate the presence of damage or a mutation, which in turn may be correlated with a disease or pathogen.

[0065] Another technical advantage of a coherent Raman spectroscopy system **100** as described herein is its relatively compact size. A coherent Raman spectroscopy system may be packaged in a relatively compact enclosure as compared to a conventional Raman spectroscopy system. Since the spectral resolution of an optical spectrometer scales inversely with the optical path length of the spectrometer (e.g., a longer path length provides better spectral resolution), an optical spectrometer with a spectral resolution around 1 cm^{-1} can be quite large or bulky. In contrast, since the spectral resolution of a coherent Raman spectroscopy system is determined primarily by the spectral linewidth of the probe laser **110_{pr}**, a coherent Raman spectroscopy system does not require a long optical path length to provide high spectral resolution. Thus, an enclosure for a coherent Raman spectroscopy system may be significantly smaller than that for a conventional Raman spectroscopy system. In some embodiments, a coherent Raman spectroscopy system may be packaged as a compact device that may be referred to as a lab-on-a-chip or a spectrometer on a chip. For example, a coherent Raman spectroscopy system may be packaged as a wearable device that provides ongoing, continual monitoring for a person or an animal.

[0066] FIGS. 3-4 each illustrate an example Raman signal **160** produced by coherent Raman scattering. In a coherent Raman spectroscopy system **100**, the pump beam **120_{pu}** and the Stokes beam **120_S** are directed to a sample **150**, which produces a Raman signal **160** by coherent Raman scattering of the pump and Stokes beams within the sample. The frequency of the pump beam **120_{pu}** is ν_1 , and the frequency of the Stokes beam **120_S** is ν_2 . The frequency offset Ω between the pump and Stokes beams equals $\nu_1 - \nu_2$. The frequencies of the pump and Stokes beams may be set so that the frequency offset Ω is approximately equal to a vibrational frequency or an electronic-transition frequency of a particular material. A coherent Raman spectroscopy system **100** may measure a Raman signal **160** produced by a sample **150** to determine (i) whether the particular material is present in the sample or (ii) an amount or a concentration of the particular material within the sample.

[0067] The frequency offset Ω between the pump and Stokes beams may be any suitable fixed or adjustable value

between approximately 5 terahertz (THz) and approximately 100 THz. Expressed in wavenumbers, this corresponds to the frequency offset Ω being between approximately 167 cm^{-1} and approximately 3336 cm^{-1} . For example, in a coherent Raman spectroscopy system **100**, the pump beam **120_{pu}** may have a wavelength between approximately 1220 nanometers (nm) and approximately 1450 nm (which corresponds to a pump-beam frequency ν_1 between approximately 246 THz and approximately 207 THz), and the Stokes beam **120_S** may have a wavelength between approximately 1490 nm and approximately 1570 nm (which corresponds to a Stokes-beam frequency ν_2 between approximately 201 THz and approximately 191 THz). This system may produce pump and Stokes beams having a frequency offset Ω between approximately 5.6 THz and approximately 54.8 THz (or, in wavenumbers, between approximately 185 cm^{-1} and 1827 cm^{-1}). For example, if the pump beam **120_{pu}** has a wavelength of 1330 nm (or equivalently, a frequency ν_1 of 225.4 THz) and the Stokes beam **120_S** has a wavelength of 1550 nm (or equivalently, a frequency ν_2 of 193.4 THz), then the frequency offset Ω between the pump and Stokes beams is approximately 32 THz (or, in wavenumbers, 1067 cm^{-1}). The pump and Stokes beams in a Raman spectroscopy system **100** may each have any suitable wavelength between approximately 300 nm and approximately 5,000 nm. For example, if the pump beam **120_{pu}** has a wavelength of approximately 785 nm (or equivalently, a frequency ν_1 of 381.9 THz) and the Stokes beam **120_S** has a wavelength of approximately 840 nm (or equivalently, a frequency ν_2 of 356.9 THz), then the frequency offset Ω between the pump and Stokes beams is approximately 25 THz (or, in wavenumbers, 834 cm^{-1}). As another example, the pump beam **120_{pu}** may have a wavelength between approximately 700 nm and approximately 850 nm, between approximately 890 nm and approximately 920 nm, or between approximately 1000 nm and approximately 1100 nm.

[0068] The Raman signal **160** in each of FIGS. 3-4 is an optical signal with a spectral linewidth of $\Delta\nu_R$. In FIG. 3, the Raman signal **160** has a center frequency approximately equal to $2\nu_1 - \nu_2$ (which is equal to $\nu_1 + \Omega$, since $\Omega = \nu_1 - \nu_2$). The center frequency of a Raman signal **160** may refer to the frequency of a central peak or the frequency of an approximate center of the Raman signal. Additionally or alternatively, the center frequency of a Raman signal **160** may correspond to a Raman peak of an associated Raman spectrum. The Raman signal **160** in FIG. 3 may be produced by coherent anti-Stokes Raman scattering (CARS) in which the pump and Stokes beams interact with a sample **150** to produce a Raman signal **160** at or around the frequency $2\nu_1 - \nu_2$. For example, if the pump and Stokes beams have respective wavelengths of 1064 nm and 1550 nm (which corresponds to frequencies of approximately 281.8 THz and 193.4 THz), then the frequency offset Ω is 88.3 THz (or, 2947 cm^{-1}), and the anti-Stokes Raman signal **160** has a center wavelength of approximately 810 nm (which corresponds to a frequency of approximately 370 THz). The Raman signal **160** in FIG. 4, which overlaps the frequency of the Stokes beam **120_S**, may be produced by stimulated Raman scattering. The Raman signal **160** in FIG. 4 may be centered at or near the frequency ν_2 of the Stokes beam **120_S**. For example, the Raman signal **160** in FIG. 4 may have a center frequency that is within approximately 200 GHz (or, 6.7 cm^{-1}) of the Stokes-beam frequency ν_2 .

[0069] FIG. 5 illustrates the example Raman signal **160** of FIG. 4 along with a probe beam **120_{pr}**. The Raman signal **160** may be produced by stimulated Raman scattering of the pump beam **120_{pu}** and Stokes beam **120_S**, and the probe beam **120_{pr}** may be used to measure the Raman signal. The Raman signal **160** is centered at or near the frequency ν_2 of the Stokes beam **120_S**, and the frequency ν_3 of the probe beam of light **120_{pr}** overlaps the Raman signal **160** and is relatively close to the frequency ν_2 of the Stokes beam of light **120_S** (e.g., the probe-beam frequency ν_3 may be within 200 GHz of the Stokes-beam frequency ν_2). In a coherent Raman spectroscopy system **100**, the pump frequency ν_1 , Stokes frequency ν_2 , and probe frequency ν_3 may each be between approximately 60 THz and approximately 1,000 THz (which corresponds to a wavelength between approximately 5,000 nm and approximately 300 nm). For example, in FIG. 5 the pump frequency ν_1 may be 291 THz (which corresponds to a wavelength of approximately 1030 nm), the Stokes frequency ν_2 may be 240.03 THz (which corresponds to a wavelength of approximately 1249.00 nm), and the probe frequency ν_3 may be 240.00 THz (which corresponds to a wavelength of approximately 1249.14 nm).

[0070] Each of the light sources **110** of a coherent Raman spectroscopy system **100** may include a wavelength-tunable light source. A wavelength-tunable light source refers to a light source **110** that can produce light at multiple different wavelengths within a range of wavelengths (or equivalently, at multiple different frequencies within a range of frequencies). For example, the probe light source **110_{pr}** in FIG. 1 may be a wavelength-tunable light source where the wavelength of the probe beam **120_{pr}** is adjustable over an 80-nm wavelength range from 1490 nm to 1570 nm (which corresponds to a 10.3 THz frequency range from approximately 201.2 THz to approximately 191.0 THz). At any given time, a wavelength-tunable light source **110** may operate at any one of the different wavelengths within its wavelength-tuning range. For example, during a first period of time, the probe light source **110_{pr}** with a 1490-1570 nm wavelength-tuning range may produce a probe beam **120_{pr}** at 1500 nm, and during a subsequent second period of time, the probe light source may be tuned to operate at 1560 nm. A wavelength-tunable light source may be adjustable over a frequency range that corresponds to a wavelength range having a width or span between approximately 10 nm and approximately 100 nm. For example, the width of the wavelength-tuning range of a wavelength-tunable light source may be approximately 10 nm, 20 nm, 30 nm, 40 nm, 50 nm, 60 nm, 80 nm, or 100 nm. A wavelength-tuning range from 1490 nm to 1570 nm may be referred to as having an 80-nm width or an 80-nm span. Around 1550 nm, a 10-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 1.25 THz (or equivalently, 41.6 cm^{-1}), and a 100-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 12.5 THz (or equivalently, 416 cm^{-1}). Around 1050 nm, a 10-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 2.72 THz (or equivalently, 90.7 cm^{-1}), and a 100-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 27.3 THz (or equivalently, 909 cm^{-1}). A wavelength-tunable light source may be referred to as a frequency-tunable light source or a tunable light source.

[0071] Each of the light sources **110** of a coherent Raman spectroscopy system **100** may include one or more laser

diodes, and each of the laser diodes may be a fixed-wavelength laser diode or a wavelength-tunable laser diode. A fixed-wavelength laser diode may operate at a single wavelength or within a relatively narrow wavelength range (e.g., within 0.1 nm of a particular wavelength). A fixed-wavelength laser diode may include a distributed feedback (DFB) laser diode, a distributed Bragg reflector (DBR) laser diode, a fiber-Bragg-grating (FBG) stabilized laser diode, a temperature-stabilized laser diode, or any other suitable fixed-wavelength laser diode. A wavelength-tunable laser diode may produce light at multiple different wavelengths within a range of wavelengths. For example, a wavelength-tunable laser diode may be configured to produce light at any wavelength within a wavelength range having a width between approximately 10 nm and approximately 100 nm. At any given time, a wavelength-tunable laser diode may operate at any one wavelength of multiple different wavelengths within a range of wavelengths. A wavelength-tunable laser diode may include an external-cavity laser diode, a thermally tuned laser diode, or a sampled-grating distributed Bragg reflector (SG-DBR) laser. For example, a wavelength-tunable SG-DBR laser may have a 40-nm wavelength-tuning range that extends from 1530 nm to 1570 nm, and the SG-DBR laser may be adjustable to operate at any single wavelength within the 40-nm wavelength range. A wavelength-tunable laser diode may be referred to as a frequency-tunable laser diode or a tunable laser diode. For example, a tunable laser with a 40-nm wavelength-tuning range that extends from 1530 nm to 1570 nm may also be referred to as a frequency-tunable laser with a 5.0-THz frequency-tuning range that extends from approximately 191 THz to approximately 196 THz.

[0072] Each of the light sources **110** of a coherent Raman spectroscopy system **100** may include one or more of the following: light-emitting diode (LED), super-luminescent light source, short-pulse laser, broadband light source, fiber laser, solid-state laser, quantum-cascade laser. For example, a light source that produces light over a relatively broad range of wavelengths (e.g., a super-luminescent light source, short-pulse laser, or broadband light source) may be used to investigate a sample over the broad range of wavelengths without having to use a wavelength-tunable light source.

[0073] FIG. 6 illustrates an expanded view of a portion of the Raman signal **160** of FIG. 5. The portion of FIG. 5 enclosed by the dashed-line box is expanded in FIG. 6. The peak of the Raman signal **160** in FIG. 6 is approximately coincident with the frequency ν_2 of the Stokes beam **120S**, and the frequency ν_3 of the probe beam **120pr** overlaps the Raman signal.

[0074] The Raman signal **160** in FIG. 6 is an optical signal with a spectral linewidth of $\Delta\nu_R$. The spectral linewidth of a Raman signal **160** may have a value between approximately 30 GHz and 300 GHz (or, in wavenumbers, between approximately 1 cm^{-1} and approximately 10 cm^{-1}). The spectral linewidth $\Delta\nu_R$ of a Raman signal **160** may represent the spectral width of a peak of the Raman signal (e.g., a full-width-at-half-maximum of the peak) or may represent an approximate extent or width of the full Raman signal. For example, the spectral linewidth $\Delta\nu_R$ of the Raman signal **160** in FIG. 13 corresponds to a full-width-at-half-maximum of a peak of the Raman signal. In FIG. 6, the spectral linewidth $\Delta\nu_R$ of the Raman signal **160** corresponds to an extent or width of the full Raman signal. For example, the spectral linewidth of a Raman signal **160** may equal a spectral width

at which an envelope of the Raman signal has decreased to a particular level (e.g., to 50%, 20%, or 10% of a peak value). The envelope may be a curve that decreases monotonically away from a peak of the Raman signal and approximately follows an overall shape of the Raman signal. In FIG. 6, the dashed-line curve that traces the peaks of the Raman signal **160** represents an envelope of the Raman signal, and the spectral linewidth $\Delta\nu_R$ corresponds to a full-width-at-10% of the Raman-signal envelope (i.e., the points at which the envelope has decreased to 10% of its maximum value).

[0075] In a coherent Raman spectroscopy system **100**, the difference between the frequency ν_2 of the Stokes beam **120S** and the frequency ν_3 of the probe beam **120pr** may be greater than a low-frequency limit F_1 and less than a high-frequency limit F_2 (i.e., $F_1 < |\nu_2 - \nu_3| < F_2$). For example, the low-frequency limit F_1 may be approximately 1 MHz, 10 MHz, 50 MHz, 100 MHz, 200 MHz, 500 MHz, or 1 GHz, and the high-frequency limit F_2 may be approximately 10 GHz, 20 GHz, 50 GHz, 100 GHz, 200 GHz, 500 GHz, or 1 THz. As another example, the low-frequency limit F_1 may be related to the spectral linewidth $\Delta\nu_{pr}$ of the probe beam **120pr** and the spectral linewidth $\Delta\nu_S$ of the Stokes beam **120S** (e.g., F_1 may be greater than $\Delta\nu_{pr} + \Delta\nu_S$). As another example, the high-frequency limit F_2 may be related to the spectral linewidth $\Delta\nu_R$ of a Raman signal **160** (e.g., F_2 may be approximately equal $(0.5)\Delta\nu_R$, $\Delta\nu_R$, or $2\Delta\nu_R$). As another example, F_1 may be 100 MHz and F_2 may be 200 GHz, which indicates that the frequency ν_2 of the Stokes beam **120S** and the frequency ν_3 of the probe beam **120pr** may differ by greater than 100 MHz and less than 200 GHz (i.e., $100\text{ MHz} < |\nu_2 - \nu_3| < 200\text{ GHz}$). The frequencies F_1 and F_2 represent the frequency range with respect to the Stokes-beam frequency ν_2 over which the probe-beam frequency ν_3 may be scanned. The two hatched rectangles along the frequency axis in FIG. 6 represent the allowed frequency ranges for the probe beam **120pr**. The frequency of the probe beam **120pr** may be (i) between $\nu_2 - F_2$ and $\nu_2 - F_1$ or (ii) between $\nu_2 + F_1$ and $\nu_2 + F_2$. In an optical receiver **200**, the probe beam **120pr** and the Raman signal **160** may be coherently mixed together at an optical detector **220**. The frequency ν_3 of the probe beam **120pr** may be kept away from the frequency ν_2 of the Stokes beam **120S** by at least the low-frequency limit F_1 (e.g., $|\nu_2 - \nu_3| > F_1$) to avoid mixing between the probe and Stokes beams, which could cause the detector to produce an unwanted electronic signal not related to the mixing between the probe beam and Raman signal. Additionally, the frequency ν_3 of the probe beam **120pr** may be kept within the high-frequency limit F_2 of the frequency ν_2 of the Stokes beam **120S** (e.g., $|\nu_2 - \nu_3| < F_2$), since measurements outside the high-frequency limit may not produce a significant electronic signal.

[0076] FIG. 7 illustrates an expanded view of a portion of the Raman signal **160** of FIG. 6. The portion of FIG. 6 enclosed by the dashed-line box is expanded in FIG. 7. The peak of the Raman signal **160** in FIG. 7 is located at frequency ν_{RP} , and the Stokes beam **120S** is located at the Stokes frequency ν_2 . The Raman-signal peak frequency ν_{RP} may correspond to a Raman peak of a material that is part of a sample **150** being measured by a Raman spectroscopy system **100**. For example, the difference between the pump-beam frequency ν_1 and the Raman-signal peak frequency ν_{RP} may equal a vibrational frequency Ω of the material (e.g., $\nu_1 - \nu_{RP} = \Omega$). In some embodiments, a Stokes light source **110S** may be operated so that the Stokes-beam

frequency ν_2 is approximately equal to the Raman-signal peak frequency ν_{RP} , and in other embodiments (e.g., as illustrated in FIG. 7), the Stokes-beam frequency ν_2 may be slightly off-resonance or detuned with respect to the Raman-signal peak frequency ν_{RP} . For example, in FIG. 7, the Stokes-beam frequency ν_2 may differ from the Raman-signal peak frequency ν_{RP} by less than or equal to 30 GHz, 10 GHz, 5 GHz, or 1 GHz.

[0077] The beams of light **120** produced by the pump, Stokes, and probe light sources **110** in a Raman spectroscopy system **100** may each have a spectral linewidth of less than 200 MHz. Additionally, one or more of the beams of light **120** may have a spectral linewidth of less than 1 MHz. For example, the spectral linewidth of a beam of light **120** may be less than 200 MHz, 100 MHz, 50 MHz, 10 MHz, 1 MHz, or 100 kHz. In FIG. 7, the spectral linewidth $\Delta\nu_{pr}$ of the probe beam **120pr** and the spectral linewidth $\Delta\nu_S$ of the Stokes beam **120S** may each be less than 200 MHz. In some embodiments, the probe light source **110pr** may be configured to produce a probe beam **120pr** having a relatively narrow spectral linewidth, which may allow a Raman spectroscopy system **100** to measure a Raman spectrum with a high degree of spectral resolution. For example, the spectral linewidth $\Delta\nu_{pr}$ of the probe beam **120pr** in FIG. 7 may be less than 1 MHz.

[0078] An optical receiver **200** of a coherent Raman spectroscopy system **100** may include one or more detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with at least a portion of a probe beam of light **120pr** to produce an electronic signal. For an optical receiver **200** with a single detector **220**, all or most of the probe beam of light **120pr** may be mixed with the Raman signal **160**. For an optical receiver **200** with multiple detectors **220**, the probe beam of light **120pr** may be split so that a portion of the probe beam of light is sent to each of the detectors. For example, in an optical receiver **200** with four detectors **220**, the probe beam **120pr** may be split into four portions, and each detector may receive one of the four portions of the probe beam. Similarly, for an optical receiver **200** with a single detector **220**, all or most of the Raman signal **160** may be sent to the single detector, and for an optical receiver with multiple detectors **220**, the Raman signal **160** may be split so that a portion of the Raman signal is sent to each of the detectors.

[0079] Coherent mixing of a probe beam **120pr** and a Raman signal **160**, which may be referred to as heterodyne detection, may occur when the two optical signals are optically combined and then detected by a detector **220**. Optically combining the probe beam **120pr** and the Raman signal **160** may refer to combining the two optical signals so that their electric fields are summed together. For example, the probe beam and Raman signal may be combined (e.g., with an optical combiner **130**) so that the two signals are substantially coaxial and travel together in the same direction and along approximately the same optical path. Additionally, the probe beam and Raman signal may be combined so that at least a portion of their polarizations have the same orientation to allow at least a portion of their electric fields to be summed together. Once the probe beam and Raman signal are optically combined to produce a combined probe-Raman signal **210**, the probe beam and Raman signal may be coherently mixed at a detector **220**. The detector **220** may

produce a photocurrent signal i corresponding to the coherent mixing of the probe beam **120pr** and a portion of the Raman signal **160**.

[0080] The portion of a Raman signal **160** that is coherently mixed with a probe beam of light **120pr** at a detector **220** to produce an electronic signal may refer to a spectral portion of the Raman signal. The spectral portion of a Raman signal **160** that is coherently mixed with a probe beam **120pr** may include optical frequency components of the Raman signal that are within a particular frequency range of the frequency ν_3 of the probe beam of light, where the particular frequency range is based on or depends on the electronic bandwidth Δf of the optical detector **220**. In FIG. 7, the hatched region around the probe frequency ν_3 represents the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120pr**. The particular frequency range illustrated by the hatched region extends from $\nu_3 - \Delta f$ to $\nu_3 + \Delta f$, where Δf is the electronic bandwidth of the detector. The particular frequency range includes the optical frequency components of the Raman signal **160** that are coherently mixed with the probe beam **120pr** to produce an electronic signal.

[0081] The electronic signal produced by the detector **220** in response to the coherent mixing of the portion of the Raman signal **160** and probe beam **120pr** may include one or more electronic frequency components, where each electronic frequency component has a frequency less than or equal to approximately Δf . For example, the electronic bandwidth Δf of the optical detector may be 10 GHz, and the electronic signal produced by the detector may include one or more electronic frequency components having frequencies less than or equal to approximately 10 GHz. Other optical frequency components of the Raman signal **160** that are outside the hatched region (e.g., optical frequencies less than $\nu_3 - \Delta f$ and greater than $\nu_3 + \Delta f$) may produce a coherent-mixing response in the detector **220**. However, since these Raman-signal optical frequency components would produce an electronic response at frequencies greater than Δf (which is outside of the electronic bandwidth of the detector), these optical frequency components will not result in any significant contribution to the electronic signal. The electronic bandwidth of the detector **220** effectively limits or filters the optical frequency components of the Raman signal **160** that are measured by the optical receiver **200** to optical frequency components that are within a particular frequency range of the probe frequency. Accordingly, the electronic bandwidth of the detector **220**, in combination with the relatively narrow spectral linewidth of the probe beam **120pr**, may allow a Raman spectroscopy system **100** to measure a Raman signal **160** with a high degree of spectral resolution. Herein, an optical frequency or an optical frequency component refers to a signal in the optical domain between approximately 60 THz and approximately 1,000 THz, and an electronic frequency or an electronic frequency component refers to a signal in the electronic domain between 0 Hz and approximately 50 GHz.

[0082] The electronic signal that results from coherent mixing of a Raman signal **160** and a probe beam **120pr** may include a coherent-mixing term that is proportional to a product of (i) ϵ_R , the amplitude of the electric field of the Raman signal and (ii) ϵ_{pr} , the amplitude of the electric field of the probe beam. The photocurrent signal i produced by a detector **220** in response to the coherent mixing of a Raman signal **160** and a probe beam **120pr** may be proportional to

the square of the summed electric fields of the probe beam **120pr** and a spectral portion of the Raman signal **160**. This type of detector **220** that produces a photocurrent signal i that is proportional to the square of a received electric field may be referred to as a square-law detector. The photocurrent signal i may be expressed as $i(t) = k|\epsilon_R(t) + \epsilon_{pr}(t)|^2$, where k is a constant (e.g., k may account for the responsivity of the detector **220** as well as other constant parameters or conversion factors). For clarity, the constant k or other constants (e.g., conversion constants or factors of 2 or 4) may be excluded from expressions herein related to the photocurrent i or the voltage signal **234**. In the above expression for $i(t)$, $\epsilon_R(t)$ is the electric field of the Raman signal **160**, and $\epsilon_{pr}(t)$ is the electric field of the probe beam **120pr**. The electric field of the Raman signal **160** may be expressed as $E_R \cos[2\pi\nu_R t + \phi_R]$, where E_R is the amplitude of the electric field of the Raman signal. The electric field of the probe beam **120pr** may be expressed as $E_{pr} \cos[2\pi\nu_3 t + \phi_{pr}]$, where E_{pr} is the amplitude of the electric field of probe beam. The frequency ν_R is the optical frequency of the electric field of the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120pr**. The frequency ν_R may include the optical frequency components of the Raman signal **160** from $\nu_3 - \Delta f$ to $\nu_3 + \Delta f$, where Δf is the electronic bandwidth of the detector. The frequency ν_3 is the optical frequency of the electric field of the probe beam **120pr**. The term ϕ_R is the phase of the electric field of the Raman signal **160**, and the term ϕ_{pr} is the phase of the electric field of the probe beam **120pr**.

[0083] The above expression for the photocurrent signal i may be expanded and written as $i(t) = E_R^2 + E_{pr}^2 + 2E_R E_{pr} \cos[2\pi(\nu_R - \nu_3)t + \Delta\phi]$, where, for clarity, the constant k is not included. In this expanded expression for the photocurrent signal $i(t)$, the first term E_R^2 corresponds to the optical power (P_R) of the Raman signal **160**, and the second term E_{pr}^2 corresponds to the optical power (P_{pr}) of the probe beam **120pr**. The third term in the above expression is $2E_R E_{pr} \cos[2\pi(\nu_R - \nu_3)t + \Delta\phi]$ and may be referred to as a coherent-mixing term that represents coherent mixing between the electric fields of the Raman signal **160** and probe beam **120pr**. The phase difference $\Delta\phi$ is the phase difference between the electric fields of the Raman signal and the probe beam (e.g., $\Delta\phi = \phi_R - \phi_{pr}$). The coherent-mixing term is proportional to $E_R \times E_{pr}$, which is the product of the electric-field amplitudes of the Raman signal **160** and the probe beam **120pr**. Additionally, the coherent-mixing term includes a cosine function that varies in time based on the frequency difference ($\nu_R - \nu_3$) between the Raman signal **160** and the probe beam **120pr**. Since the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120pr** includes the optical frequency components of the Raman signal from $\nu_3 - \Delta f$ to $\nu_3 + \Delta f$, the frequency-difference term ($\nu_R - \nu_3$) may include frequency components from zero to Δf . Accordingly, the coherent-mixing term may be referred to as including multiple electronic frequency components, where each electronic frequency component is proportional to $E_R E_{pr} \cos[2\pi f t + \Delta\phi]$. The frequency f , which may be referred to as an electronic frequency, is equal to the frequency difference ($\nu_R - \nu_3$) and has a value between zero and Δf . The coherent-mixing term may also be expressed as $2\sqrt{P_R P_{pr}} \cos[2\pi f t + \Delta\phi]$, where P_R is the optical power of the Raman signal **160** and P_{pr} is the optical power of the probe beam **120pr**.

[0084] FIGS. **8-10** illustrate time-domain and frequency-domain plots of an example electronic signal resulting from coherent mixing of the Raman signal **160** and probe beam **120pr** of FIG. **7**. The electronic signal produced by a detector **220** may be a photocurrent signal i , and an electronic amplifier **232** may produce a voltage signal **234** that corresponds to the photocurrent signal i . The voltage signal **234** may be approximately proportional to the photocurrent signal i , and both the voltage and photocurrent signals may be proportional to $P_R + P_{pr} + 2\sqrt{P_R P_{pr}} \cos[2\pi f t + \Delta\phi]$, where the electronic frequency f is equal to the frequency difference ($\nu_R - \nu_3$). The electronic frequency f may take on multiple values (or a continuous range of values) between zero and Δf , and so, the voltage signal **234** in the time domain may be viewed as a summation over these multiple frequency components. In the frequency domain, the voltage signal **234** may have frequency components that extend from DC to Δf (or, for an amplifier **232** with a low-frequency cutoff filter, from the low-frequency cutoff to Δf). The two terms P_R and P_{pr} represent the optical powers of the Raman signal and probe beam. Over the time duration of a measurement, the powers of the Raman signal and probe beam may be approximately constant, and so the contribution of $P_R + P_{pr}$ in the above expression may correspond to a DC offset with little or no time variation. An electronic amplifier **232** may include a high-pass or band-pass filter that removes or attenuates the DC or low-frequency components, resulting in an electronic signal with little or no DC offset (e.g., as illustrated by the time-domain voltage signal **234** in FIG. **8**).

[0085] The signal characteristic **162** in each of FIGS. **7-10** represents a characteristic of an electronic signal (e.g., a characteristic of a photocurrent signal i or a characteristic of a corresponding voltage signal **234**) associated with a Raman signal **160**. A signal characteristic **162** may be determined from a digital signal **240**, where the digital signal is a digital representation of a photocurrent signal i or a voltage signal **234**. A characteristic **162** of an electronic signal may be determined by a processor of a Raman spectroscopy system **100** and may include one or more data points **163**, where each data point represents: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency (e.g., an electronic frequency or an optical frequency), a phase, or a polarization. A Raman spectroscopy system **100** may measure one or more signal characteristics **162** associated with one or more Raman signals **160**, and based on the measured characteristics, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0086] FIG. **8** illustrates an example time-domain plot of a voltage signal **234** resulting from coherent mixing of the Raman signal **160** and probe beam **120pr** of FIG. **7**. A digitizer **236** (e.g., an ADC) may produce a digital signal **240** that represents the voltage signal **234** in the time domain, and a processor may determine a characteristic **162** of the electronic signal based on the digital signal. The signal characteristic **162** in FIG. **8** includes a single data point **163** which represents a peak amplitude A_p of the time-domain voltage signal **234**. The signal characteristic **162** may also include the optical frequency ν_3 that the probe beam **120pr** was set to when the electronic signal was obtained. Other

time-domain-type data points **163** associated with the signal characteristic **162** in FIG. **8** may include (i) an amplitude of the time-domain voltage signal **234** at a particular time (e.g., at a temporal center) or (ii) an area associated with the voltage signal. For an area associated with the voltage signal **234**, a processor may first convert the voltage signal into a non-negative signal (e.g., by taking the absolute value or by squaring the values of the voltage signal) and then integrate the non-negative signal to determine an area under the curve.

[0087] A processor of a Raman spectroscopy system **100** may determine a Fourier transform of a digital signal **240**. A voltage signal **234** may be a time-domain signal, and the digital signal **240** may be a time-domain digital representation of the voltage signal **234**. The Fourier transform of the digital signal **240** may produce a frequency-domain representation of the voltage signal **234**. From the Fourier transform, the processor may determine one or more electronic frequency components of the voltage signal **234**. Determining a frequency component of a voltage signal **234** may include determining an amplitude of the frequency-domain voltage signal at a particular frequency (e.g., at 4 GHz). FIGS. **9-10** each illustrate an example frequency-domain plot of a voltage signal **234** resulting from coherent mixing of the Raman signal **160** and probe beam **120pr** of FIG. **7**. The frequency-domain plots may be determined by taking a Fourier transform (e.g., a discrete Fourier transform or a fast Fourier transform) of a time-domain digital signal **240** that represents a time-domain voltage signal **234**. The x-axis of each of the frequency-domain plots in FIGS. **9-10** is labeled as “electronic frequency” to clarify and distinguish it from the x-axis of the frequency-domain plots in FIGS. **3-7**. The x-axis of each of the frequency-domain plots in FIGS. **3-7** may be referred to as an “optical frequency,” where the pump beam **120pu**, Stokes beam **120S**, Raman signal **160**, and probe beam **120pr** each have frequencies in the optical domain (e.g., in the range of 60-1,000 THz). In contrast, the x-axis of each of the frequency-domain plots in FIGS. **9-10** includes frequencies in the electronic domain (e.g., in the range of DC to 50 GHz). Coherent mixing of a Raman signal **160** and probe beam **120pr** (which are signals in the optical domain) produces an electronic signal that can be detected and analyzed using electronic techniques. The y-axis of each of the plots in FIGS. **8-10** is labeled “electronic signal amplitude” and may have units of voltage, current, or electrical power (e.g., watts).

[0088] In FIG. **9**, the frequency-domain voltage signal **234** has non-zero frequency components that extend from DC (i.e., zero hertz) to the detector-bandwidth frequency Δf . In FIG. **10**, the frequency-domain voltage signal drops off before reaching zero hertz, indicating that the corresponding voltage signal **234** may have been AC-coupled or high-pass filtered to remove the DC or low-frequency components. The signal characteristic **162** in FIG. **9** includes a single data point **163f** which represents a peak amplitude A_p of the frequency-domain voltage signal **234**. The data point **163f** may include the peak amplitude A_p or the frequency f_p at which the peak amplitude is located. The signal characteristic **162** in FIG. **10** includes five data points: data point **163a** at frequency f_a , data point **163b** at frequency f_b , data point **163c** at frequency f_c , data point **163d** at frequency f_d , and data point **163e** at frequency f_e . Each data point **163** in FIG. **10** may include an amplitude value (e.g., amplitude A_a may be associated with data point **163a**) or a frequency (e.g., frequency f_a may be associated with data point **163a**). The

frequencies associated with the data points **163** in FIG. **10** may have particular values. For example, the detector bandwidth Δf may be 5 GHz, and the frequencies f_a , f_b , f_c , f_d , and f_e may have respective values of approximately 1 GHz, 2 GHz, 3 GHz, 4 GHz, and 5 GHz. Other frequency-domain-type data points **163** associated with the signal characteristic **162** in FIG. **9** or **10** may include (i) an amplitude of the frequency-domain voltage signal **234** at a particular frequency or at a center frequency or (ii) an area or an average amplitude associated with the frequency-domain voltage signal **234**. Additionally, the signal characteristic **162** in FIG. **9** or **10** may include the optical frequency ν_3 that the probe beam **120pr** was set to when the voltage signal **234** was obtained.

[0089] A processor of a coherent Raman spectroscopy system **100** may associate a determined signal characteristic **162** with a Raman frequency shift. For example, the signal characteristic **162** in FIG. **7** may be associated with a Raman frequency shift having a frequency $\nu_1 - \nu_3$, which is the difference between the pump-beam frequency ν_1 and the probe-beam frequency ν_3 . A sample **150** under investigation may include a material with a Raman spectrum having a peak at a frequency Ω , which is approximately equal to $\nu_1 - \nu_{RP}$, the difference between the pump-beam frequency ν_1 and the Raman-signal peak frequency ν_{RP} . The peak frequency Ω may be approximately equal to a vibrational frequency of the material and may correspond to a Raman frequency shift of Ω (e.g., when excited with light at a frequency ν , the material may produce Raman-shifted light at the frequencies $\nu + \Omega$ and $\nu - \Omega$). In FIG. **7**, the probe beam **120pr** is being used to measure the Raman signal **160** at the probe-frequency ν_3 , and the resulting signal characteristic **162** that is determined may be associated with a Raman spectrum of the material at the frequency $\nu_1 - \nu_3$, which may be referred to as a Raman frequency shift of $\nu_1 - \nu_3$.

[0090] FIG. **11** illustrates an example Raman signal **160** that is measured at multiple probe frequencies ν_3 . The probe beam **120pr** is tuned to multiple different probe frequencies (ν_{3-1} , ν_{3-2} , ν_{3-3} , . . . ν_{3-n}) to measure multiple respective signal characteristics (**162-1**, **162-2**, **162-3**, . . . **162-n**) of the Raman signal **160**. The Raman signal **160** is produced by coherent Raman scattering of pump and Stokes beams within a sample **150**. Throughout the measurements of the multiple signal characteristics **162** in FIG. **11**, the pump-beam frequency ν_1 and the Stokes-beam frequency ν_2 may remain substantially constant so that the resulting Raman signal **160** also remains substantially constant (e.g., the Raman signal may exhibit substantially the same shape and amplitude throughout the measurements). The number n of signal characteristics **162** that are measured may be 1, 5, 10, 50, 100, 500, 1,000, or any other suitable number of signal characteristics. At each of the n probe frequencies, the probe beam **120pr** may be coherently mixed with a spectral portion of the Raman signal **160** that is within a particular frequency range of the probe frequency ν_3 (e.g., within Δf of the probe frequency, as illustrated in FIG. **7**) to measure one signal characteristic **162**. Each signal characteristic **162** may provide information about the Raman signal **160** in the frequency region from $\nu_3 - \Delta f$ to $\nu_3 + \Delta f$. Based on one or more measured signal characteristics **162**, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0091] At each of the n probe frequencies, a single measurement may be performed, or multiple measurements may be performed. For example, with the probe beam **120pr** in FIG. 11 set to the probe-beam frequency ν_{3-1} , an optical receiver **200** may measure a single photocurrent signal i and produce a corresponding single digital output signal **240**. From that one output signal **240**, a processor may determine a signal characteristic **162-1**. Alternatively, with the probe beam **120pr** in FIG. 11 set to the probe-beam frequency ν_{3-1} , an optical receiver **200** may measure a series of multiple photocurrent signals i (e.g., 2, 4, 10, 20, 50, 100, or any other suitable number of signals) and produce multiple corresponding digital output signals **240**. A processor may determine one signal characteristic **162-1** from the multiple digital output signals. For example, the processor may average or otherwise combine the multiple digital output signals **240** to produce one signal characteristic **162-1**. Measurement of a series of multiple photocurrent signals i may improve the measurement accuracy by averaging out noise or removing outliers from the measurements.

[0092] By tuning the probe-beam frequency ν_3 to multiple frequencies across at least a portion of a Raman signal **160**, a coherent Raman spectroscopy system **100** may measure the Raman signal at multiple points **162**. For example, the probe-beam frequency ν_3 in FIG. 6 may be tuned across at least a portion of the frequency range between ν_2-F_2 and ν_2-F_1 or between ν_2+F_1 and ν_2+F_2 . Each of the different probe-beam frequencies to which the probe beam **120pr** is changed may be offset from an adjacent probe-beam frequency by a frequency increment Δf between approximately 10 MHz and approximately 10 GHz. For example, the probe-beam frequency in FIG. 11 may be changed in frequency increments Δf of approximately 5 GHz as it is tuned across at least a portion of the Raman signal **160**. Each signal characteristic **162** in FIG. 11 is plotted along the x-axis at the probe frequency ν_3 at which the signal characteristic was obtained. The y-axis in FIG. 11 may represent an amplitude or area associated with the signal characteristics **162**. For example, each signal characteristic **162** in FIG. 11 may be plotted along the y-axis at a value corresponding to an amplitude or an area of an electronic signal from which the signal characteristic was obtained.

[0093] A probe light source **110pr** of a coherent Raman spectroscopy system **100** may include a wavelength-tunable laser, where the frequency ν_3 of the probe beam **120pr** is adjustable by changing the wavelength of light produced by the wavelength-tunable laser. For example, a wavelength-tunable laser may be adjustable over a wavelength range having a width between approximately 10 nm and approximately 100 nm. As another example, the wavelength-tuning range of a wavelength-tunable laser may be between approximately 1000 nm and approximately 1100 nm, between approximately 1490 nm and approximately 1570 nm, or between approximately 1600 nm and approximately 1690 nm. A wavelength-tunable laser may be continuously tunable over a wavelength-tuning range or may be tunable to multiple discrete wavelengths within a wavelength-tuning range. For example, a wavelength-tunable laser may be continuously tunable to any wavelength between 1530 nm and 1570 nm. Alternatively, a wavelength-tunable laser may be tunable to a set of approximately 10, 100, or 1,000 discrete wavelengths between 1530 nm and 1570 nm (e.g., the wavelengths may be separated from one another by approximately 4 nm, 0.4 nm, or 0.04 nm, respectively). A

probe light source **110pr** may include a wavelength-tunable laser that sequentially changes the probe-beam frequency ν_3 to multiple different frequencies. For example, the probe-beam frequency in FIG. 6 may be tuned to approximately 100 different frequencies between the frequencies ν_2-F_2 and ν_2-F_1 . At each of the different probe-beam frequencies, the probe beam **120pr** and a spectral portion of the Raman signal **160** may be coherently mixed at a detector **220** to produce a corresponding electronic signal, and a processor may determine a signal characteristic **162** based on the electronic signal.

[0094] In FIG. 11, the probe beam **120pr** may initially be set to the probe-beam frequency ν_{3-1} and coherently mixed at a detector **220** with a spectral portion of the Raman signal **160** around the probe frequency ν_{3-1} . For example, the spectral portion of the Raman signal **160** may include optical frequency components of the Raman signal from $\nu_{3-1}-\Delta f$ to $\nu_{3-1}+\Delta f$, where Δf is the electronic bandwidth of the optical detector **220**. A processor may determine a signal characteristic **162-1** based on the electronic signal resulting from the coherent mixing of the probe beam **120pr** at frequency ν_{3-1} and the associated spectral portion of the Raman signal **160**. After measuring the Raman signal **160** around the probe frequency ν_{3-1} , the probe light source **110pr** may change the probe-beam frequency by a frequency change Δf to the frequency ν_{3-2} . The probe frequency ν_{3-2} is equal to $\nu_{3-1}+\Delta f$, and the frequency change Δf between adjacent frequencies may be between approximately 10 MHz and approximately 10 GHz. The frequency change Δf may be a fixed value or may be dynamically adjusted during a measurement of a Raman signal **160**. After the probe-beam frequency is changed to ν_{3-2} , the probe beam **120pr** may be coherently mixed with the spectral portion of the Raman signal **160** around the probe frequency ν_{3-2} (e.g., the spectral portion may include optical frequency components of the Raman signal from $\nu_{3-2}-\Delta f$ to $\nu_{3-2}+\Delta f$). A processor may determine a signal characteristic **162-2** based on the electronic signal resulting from the coherent mixing of the probe beam **120pr** at frequency ν_{3-2} and the associated spectral portion of the Raman signal **160**. After measuring the Raman signal **160** around the probe frequency ν_{3-2} , the probe light source **110pr** may change the probe-beam frequency by the frequency change Δf to the frequency ν_{3-3} , and a measurement of the Raman signal **160** around the frequency ν_{3-3} may be performed. The Raman spectroscopy system **100** may sequentially change the frequency ν_3 of the probe light source and measure a spectral portion of the Raman signal **160** at each frequency until reaching the final probe frequency ν_{3-n} . During the measurements, the probe-beam frequency ν_3 may be tuned so that it avoids overlapping with the frequency ν_2 of the Stokes beam to prevent mixing between the probe and Stokes beams.

[0095] FIG. 12 illustrates an example Raman spectrum corresponding to the Raman signal of FIG. 11. A Raman spectroscopy system **100** may measure a sample **150** and determine the signal characteristics **162** of the Raman signal **160** in FIG. 11, and the corresponding Raman spectrum in FIG. 12 may represent the Raman spectrum of one or more materials that are part of the sample **150**. For example, the sample **150** may include glucose and the peak frequency $\nu_1-\nu_{RP}$ of the Raman spectrum may be approximately equal to a frequency Ω of a molecular vibration of glucose. A processor of the Raman spectroscopy system **100** may determine the Raman spectrum in FIG. 12 based on the

signal characteristics **162** (and the associated probe-beam frequencies ν_3) of the Raman signal **160** in FIG. **11**. Each signal characteristic **162** of a Raman signal **160** measured at a probe frequency ν_3 may be associated with a Raman frequency shift having a frequency $\nu_1 - \nu_3$, where ν_1 is the pump-beam frequency. To determine the Raman spectrum, each signal characteristic **162** may be transformed from its Raman-signal frequency ν_3 to a corresponding Raman-shift frequency $\nu_1 - \nu_3$. For example, the processor may associate the signal characteristic **162-1** at frequency ν_{3-1} in FIG. **11** with a Raman shift having the frequency $\nu_1 - \nu_{3-1}$ in FIG. **12**. Similarly, the signal characteristic **162-2** at frequency ν_{3-2} in FIG. **11** may be associated with a Raman shift having the frequency $\nu_1 - \nu_{3-2}$ in FIG. **12**, and the signal characteristic **162-3** at frequency ν_{3-3} in FIG. **11** may be associated with a Raman shift having the frequency $\nu_1 - \nu_{3-3}$ in FIG. **12**. Additionally, the peak frequency ν_{RP} of the Raman signal **160** in FIG. **11** may correspond to the Raman-signal peak in FIG. **12** with a frequency of $\nu_1 - \nu_{RP}$, which in turn may correspond to the vibrational frequency Ω of a particular material.

[0096] Based on the Raman spectrum in FIG. **12**, a processor may determine (i) whether a particular material is present in a sample **150** or (ii) an amount or a concentration of the particular material in the sample **150**. For example, the processor may compare one or more peaks, signal characteristics **162**, or other features of the Raman spectrum in FIG. **12** to a previously determined Raman spectrum for glucose. If one or more peaks or characteristics **162** of the Raman spectrum in FIG. **12** match or line up with peaks or characteristics of the Raman spectrum for glucose, then the processor may determine that glucose is present in the sample **150**. Alternatively, if the Raman spectrum in FIG. **12** is missing one or more peaks or characteristics of the Raman spectrum for glucose, then the processor may determine that little or no glucose is present in the sample **150**. As another example, the processor may determine the amount or concentration of glucose in the sample **150** based on the Raman spectrum in FIG. **12**. The concentration of glucose in the sample **150** may be related to the amplitude or height of one or more peaks of the Raman spectrum in FIG. **12** (e.g., the glucose concentration may be approximately proportional to the height of one or more Raman peaks). The concentration of glucose may then be determined based at least in part on the height of the Raman peak located at the Raman-shift frequency $\nu_1 - \nu_{RP}$.

[0097] FIG. **13** illustrates another example Raman signal **160** that is measured at multiple probe frequencies. The probe beam **120pr** is tuned to multiple different probe frequencies (ν_{3-1} , ν_{3-2} , ν_{3-3} , . . . ν_{3-n}) to measure multiple respective signal characteristics (**162-1**, **162-2**, **162-3**, . . . **162-n**) of the Raman signal **160**. The Raman signal **160** in FIG. **13** has a single peak centered at frequency ν_{RP} with a spectral linewidth of $\Delta\nu_R$. The Raman signal **160** in FIG. **11** has one main peak located at frequency ν_{RP} along with multiple smaller peaks located on either side of the main peak. A Raman signal **160** may include a single peak (e.g., as illustrated in FIG. **13**) or may include multiple peaks (e.g., as illustrated in FIG. **11**).

[0098] FIGS. **14-15** each illustrate a second example Raman signal obtained by changing the frequency offset Ω between a pump beam **120pu** and a Stokes beam **120S**. A coherent Raman spectroscopy system **100** may include a pump light source **110pu** or a Stokes light source **110S** with

a wavelength-tunable laser, and the frequency offset Ω may be adjustable by changing the wavelength of the wavelength-tunable laser. A wavelength-tunable pump or Stokes laser may be continuously tunable over a wavelength-tuning range or may be tunable to multiple discrete wavelengths within a wavelength-tuning range. For example, Stokes beam **120** in FIG. **14** may be produced by a continuously tunable laser diode that can produce light at the frequencies ν_2 and ν'_2 . As another example, a Stokes laser **110S** may include two fixed-wavelength laser diodes that operate at the respective frequencies ν_2 and ν'_2 . Similarly, pump beam **120pu** in FIG. **15** may be produced by a continuously tunable laser diode that can produce light at the frequencies ν_1 and ν'_1 , or a pump laser **110pu** may include two fixed-wavelength laser diodes that operate at the respective frequencies ν_1 and ν'_1 .

[0099] In FIG. **14**, the pump frequency ν_1 is fixed, and the Stokes frequency is changed from ν_2 to ν'_2 to change the frequency offset from Ω_1 to Ω_2 . Frequency offset Ω_1 equals $\nu_1 - \nu_2$, and frequency offset Ω_2 equals $\nu_1 - \nu'_2$. In FIG. **15**, the Stokes frequency ν_2 is fixed, and the pump frequency is changed from ν_1 to ν'_1 to change the frequency offset from Ω_1 to Ω_2 . Frequency offset Ω_1 equals $\nu_1 - \nu_2$, and frequency offset Ω_2 equals $\nu'_1 - \nu_2$. In other embodiments, both the pump and Stokes frequencies may be changed to change the frequency offset Ω from one value to another.

[0100] FIG. **14** includes two Raman signals **160a** and **160b**. Initially, a Stokes light source **110S** may produce a Stokes beam **120S** at the frequency ν_2 to produce the frequency offset Ω_1 between the Stokes beam and the pump beam **120pu**, where $\Omega_1 = \nu_1 - \nu_2$. For example, the Stokes beam **120S** may have a frequency ν_2 of 193 THz, and the pump beam **120pu** may have a frequency ν_1 of 207 THz, which corresponds to a frequency offset Ω_1 of 14 THz (or, 467 cm^{-1}). The Raman signal **160a** is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu** within a sample **150**. The Raman signal **160a** is centered at or near the frequency ν_2 of the Stokes beam **120S**, and the frequency ν_3 of the probe beam **120pr** overlaps the Raman signal **160a** and is relatively close to the frequency ν_2 of the Stokes beam **120S** (e.g., the probe-beam frequency ν_3 may be within 200 GHz of the Stokes-beam frequency ν_2). The probe beam **120pr** and a spectral portion of the Raman signal **160a** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency ν_3 of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160a** to measure multiple signal characteristics **162** associated with the Raman signal.

[0101] After measuring the first Raman signal **160a**, the Stokes light source **110S** may change the frequency of the Stokes beam to produce a Stokes beam **120S'** at the frequency ν'_2 , resulting in a frequency offset of Ω_2 between the Stokes beam **120S'** and the pump beam **120pu**, where $\Omega_2 = \nu_1 - \nu'_2$. For example, the Stokes beam **120S** may be changed to a frequency ν'_2 of 182 THz, and the pump-beam frequency ν_1 may remain at 207 THz, which corresponds to a frequency offset Ω_2 of 25 THz (or, 834 cm^{-1}). The second Raman signal **160b** is produced by coherent Raman scattering of the Stokes beam **120S'** and pump beam **120pu** within the sample **150**. The Raman signal **160b** is centered at or near the frequency ν'_2 of the Stokes beam **120S'**. Additionally, a probe light source **110pr** may change the frequency of

the probe beam to produce a probe beam **120pr'** at a frequency ν_3 that overlaps the Raman signal **160b** and is relatively close to the frequency ν_2 of the Stokes beam **120S'** (e.g., the probe-beam frequency ν_3 may be within 200 GHz of the Stokes-beam frequency ν_2).

[0102] The probe beam **120pr'** and a spectral portion of the Raman signal **160b** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency ν_3 of the probe beam **120pr'** may be tuned across at least a portion of the second Raman signal **160b** to measure multiple signal characteristics **162** associated with the Raman signal.

[0103] The two Raman signals **160a** and **160b** in FIG. 14 may correspond to two respective peaks of a Raman spectrum of a material that is in the sample **150**. One Raman peak, associated with Raman signal **160a**, is located at a frequency of approximately Ω_1 , and the frequency Ω_1 may correspond to a vibrational frequency of the material. The other Raman peak, associated with Raman signal **160b**, is located at a frequency of approximately Ω_2 , and the frequency Ω_2 may correspond to another vibrational frequency of the material. A Raman spectroscopy system **100** may perform measurements of signal characteristics **162** at 1, 2, 4, 10, 20, or 50 different values of the frequency offset Ω . The frequency offset Ω between the Stokes beam **120S** and the pump beam **120pu** may be set to each of the different frequency-offset values to produce an associated Raman signal **160**. At each value of Ω , the frequency ν_3 of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal. Then, the frequency of the Stokes beam or pump beam may be adjusted to the next value of Ω , where another measurement of an associated Raman signal **160** is performed. Based on the signal characteristics **162** associated with each of the values of the frequency offset Ω , a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0104] FIG. 15 includes the Raman signal **160c**. A first Raman signal (similar to Raman signal **160a** in FIG. 14) may be approximately overlapped with the second Raman signal **160c** in FIG. 15, and for clarity, the first Raman signal is not included in FIG. 15. Initially, a Stokes light source **110S** may produce a Stokes beam **120S** at the frequency ν_2 , and a pump light source **110pu** may produce a pump beam **120pu** at the frequency ν_1 , which results in the frequency offset Ω_1 between the Stokes and pump beams in FIG. 15. For example, the Stokes beam **120S** may have a frequency ν_2 of 193 THz, and the pump beam **120pu** may have a frequency ν_1 of 207 THz, which corresponds to a frequency offset Ω_1 of 14 THz (or, 467 cm^{-1}). A first Raman signal (similar to Raman signal **160a** in FIG. 14) is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu** within a sample **150**. The first Raman signal may be centered at or near the frequency ν_2 of the Stokes beam **120S**, and the frequency ν_3 of the probe beam **120pr** may overlap the first Raman signal and may be relatively close to the frequency ν_2 of the Stokes beam of light **120S**. The probe beam **120pr** and a spectral portion of the first Raman signal may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency ν_3 of the probe beam **120pr** may be tuned across

at least a portion of the first Raman signal to measure multiple signal characteristics **162** associated with the first Raman signal.

[0105] In FIG. 15, after measuring the first Raman signal, the pump light source **110pu** may change the frequency of the pump beam **120pu** from ν_1 to ν_1' . This results in a frequency offset of Ω_2 between the Stokes beam **120S** and the pump beam **120pu'**, where $\Omega_2 = \nu_1' - \nu_2$. For example, the pump beam may be changed to a frequency ν_1' of 218 THz, and the Stokes-beam frequency ν_2 may remain at 193 THz, which corresponds to a frequency offset Ω_1 of 25 THz (or, 834 cm^{-1}). The second Raman signal **160c** is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu'** within the sample **150**. As with the first Raman signal, the Raman signal **160c** is centered at or near the frequency ν_2 of the Stokes beam **120S**, and the frequency ν_3 of the probe beam **120pr** overlaps the Raman signal **160c** and is relatively close to the frequency ν_2 of the Stokes beam of light **120S** (e.g., the probe-beam frequency ν_3 may be within 200 GHz of the Stokes-beam frequency ν_2). The probe beam **120pr** and a spectral portion of the Raman signal **160c** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency ν_3 of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160c** to measure multiple signal characteristics **162** associated with the Raman signal.

[0106] In FIG. 15, the first Raman signal (which may be similar to Raman signal **160a** in FIG. 14) and the second Raman signal **160c** may correspond to two different peaks of a Raman spectrum of a material that is in the sample **150**. One Raman peak, associated with the first Raman signal, is located at a frequency of approximately Ω_1 , and the frequency Ω_1 may correspond to a vibrational frequency of the material. The other Raman peak, associated with Raman signal **160c**, is located at a frequency of approximately Ω_2 , and the frequency Ω_2 may correspond to another vibrational frequency of the material. Based at least in part on the signal characteristics **162** associated with the two frequency offsets Ω_1 and Ω_2 , a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0107] FIG. 16 illustrates an example Raman signal **160** along with two probe beams of light **120pr-1** and **120pr-2**. The Raman signal **160** may be produced by coherent Raman scattering of the pump beam **120pu** and Stokes beam **120S** within a sample **150**. The frequency of the pump beam **120pu** is ν_1 , and the frequency of the Stokes beam **120S** is ν_2 , which corresponds to a frequency offset Ω equal to $\nu_1 - \nu_2$. The Raman signal **160** is centered at or near the frequency ν_2 of the Stokes beam **120S** (e.g., the center frequency of the Raman signal **160** may be within 200 GHz of ν_2). The first probe beam **120pr-1** may be used to measure the Raman signal **160**, and the second probe beam **120pr-2** may be used to measure the pump beam **120pu**.

[0108] The frequency ν_3 of first probe beam **120pr-1** overlaps the Raman signal **160** and is relatively close to the frequency ν_2 of the Stokes beam **120S** (e.g., ν_3 may be within 200 GHz of ν_2). The probe beam **120pr-1** and a spectral portion of the Raman signal **160** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency ν_3 of the probe beam **120pr-1** may be

tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal.

[0109] The frequency ν_4 of the second probe beam **120pr-2** is relatively close to the frequency ν_1 of the pump beam **120pu** (e.g., ν_4 may be within 50 GHz of ν_1). For example, the frequency ν_4 of the second probe beam **120pr-2** may be offset from the frequency ν_1 of the pump beam **120pu** by approximately 10 GHz, 5 GHz, or 1 GHz. Alternatively, the frequency ν_4 of the second probe beam **120pr-2** may be approximately equal to the frequency ν_1 of the pump beam **120pu**. After the pump beam **120pu** has interacted with the sample, the probe beam **120pr-2** may be coherently mixed with the pump beam. For example, after the pump and Stokes beams have produced the Raman signal **160** and after the pump beam has exited the sample, the pump and probe beams may be coherently mixed together at a detector **220** to produce an electronic signal, from which a signal characteristic may be determined.

[0110] FIG. 17 illustrates an example optical receiver **200** for measuring the Raman signal **160** and pump beam of light **120pu** from FIG. 16. The combined pump-Stokes beam **140** (which includes pump beam **120pu** and Stokes beam **120S**) is directed to a sample **150**, which produces a Raman signal **160** in response to the pump and Stokes beams. The Raman signal **160** may be collected by one or more optical elements and directed to the optical receiver **200**. Additionally, residual light from the pump beam **120pu** may be collected and directed to the optical receiver **200** as a residual pump beam **120pu-2**. The residual pump beam **120pu-2** may include light from the pump beam **120pu** after the pump beam has interacted with and exited the sample **150**. The optical receiver **200** in FIG. 17 may be referred to as a two-channel optical receiver that includes two parallel measurement channels for separately detecting and measuring the Raman signal **160** and the residual pump beam **120pu-2**. The first measurement channel detects the Raman signal **160** and includes probe laser **110pr-1**, detector **220-1**, and detection electronics **230-1**. The second measurement channel detects the residual pump beam **120pu-2** and includes probe laser **110pr-2**, detector **220-2**, and detection electronics **230-2**.

[0111] The Raman signal **160** and the residual pump beam **120pu-2** are directed to the combiner **130c**, which may be a dichroic beamsplitter, and the combiner **130c** reflects the Raman signal **160** and transmits the residual pump beam **120pu-2**. The combiner **130c** also transmits at least a portion of the probe beam **120pr-1** produced by the probe laser **110pr-1** and combines the probe beam **120pr-1** with the Raman signal **160** to produce a combined probe-Raman signal **210-1**. The probe-Raman signal **210-1** is sent to the detector **220-1**, where the probe beam **120pr-1** and a spectral portion of the Raman signal **160** are coherently mixed to produce a photocurrent signal i_1 . The detection electronics **230-1** receives the photocurrent signal i_1 and produces a digital output signal **240-1** that corresponds to the photocurrent signal i_1 .

[0112] The combiner **130d** (which may be a dichroic or a non-dichroic beamsplitter) reflects at least a portion of the residual pump beam **120pu-2** and transmits at least a portion of the probe beam **120pr-2** produced by the probe laser **110pr-2**. The combiner **130d** combines the probe beam **120pr-2** with the residual pump beam **120pu-2** to produce a combined probe-pump signal **210-2**, which is sent to the

detector **220-2**. The probe beam **120pr-2** and the residual pump beam **120pu-2** are coherently mixed at the detector **220-2** to produce a photocurrent signal i_2 , and the detection electronics **230-2** produces a digital output signal **240-2** that corresponds to the photocurrent signal i_2 .

[0113] The two digital output signals **240-1** and **240-2** may be sent to a processor which determines a signal characteristic **162** of each of the photocurrent signals i_1 and i_2 based on the digital output signals. Additionally, the frequency of the first probe beam **120pr-1** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal. If the frequency ν_1 of the pump beam **120pu** remains fixed, the frequency ν_4 of the second probe beam **120pr-2** may also remain fixed. Alternatively, if the frequency ν_1 of the pump beam **120pu** is changed (e.g., to switch to a different frequency offset Ω), the frequency of the probe beam **120pr-2** may also be switched to maintain a particular frequency offset between the pump and probe frequencies.

[0114] Measurement of the residual pump beam **120pu-2** may be performed two or more times to determine how the power of the pump beam changes when the Raman signal **160** is produced. For example, the residual pump beam **120pu-2** may be measured once when the Stokes beam **120S** is turned off (and no Raman signal **160** is produced) and another time when the Stokes beam is turned on (and the Raman signal **160** is produced). A processor may determine the change in the power of the residual pump beam **120pu-2** associated with the Stokes beam **120S** being turned off and on. Since at least part of the Raman signal **160** may be produced by Stokes-shifted photons from the pump beam **120pu**, a decrease in the power of the residual pump beam **120pu-2** may correspond to the power of the Raman signal.

[0115] In another embodiment of a two-channel optical receiver, the optical receiver may not include a second probe laser **110pr-2**. Instead, the residual pump beam **120pu-2** may be sent to a detector **220** for direct detection without mixing the residual pump beam with another signal.

[0116] FIG. 18 illustrates an example Raman spectroscopy system **100** for measuring a Raman signal **160** produced by spontaneous Raman scattering. Instead of producing a Raman signal by coherent Raman scattering of pump and Stokes beams within a sample (e.g., as illustrated in FIGS. 1-2), the Raman signal **160** in FIG. 18 is produced by spontaneous Raman scattering of the pump beam of light **120pu** within the sample **150**. While the production of the Raman signal **160** in FIG. 18 is different from that of the Raman spectroscopy systems of FIGS. 1-2, the optical receiver **200** and the Raman-signal detection technique in FIG. 18 is similar to that of FIGS. 1-2. In FIG. 18, the Raman signal **160** is detected by coherently mixing the Raman signal with a probe beam of light **120pr**.

[0117] The Raman spectroscopy system **100** in FIG. 18 includes a pump light source **110pu** that produces a pump beam of light **120pu** at a pump frequency ν_1 . The pump beam **120pu** is directed to a sample **150** (e.g., by one or more optical elements), and the sample **150** produces a Raman signal **160** by spontaneous Raman scattering of light from the pump beam. The spontaneous Raman signal **160** is collected (e.g., by one or more optical elements) and directed to the optical receiver **200**. The optical receiver **200** includes a probe light source **110pr** that produces a probe beam of light **120pr** at a probe frequency ν_3 , where the probe frequency overlaps the Raman signal. The optical combiner

130b, which may be a dichroic or a non-dichroic beamsplitter, combines the Raman signal **160** and the probe beam **120_{pr}** to produce a combined probe-Raman signal **210** that is directed to a detector **220**. The detector **220** coherently mixes a spectral portion of the Raman signal **160** with the probe beam **120_{pr}** to produce a photocurrent signal **i**. The detection electronics **230** may produce (i) an analog voltage signal that corresponds to the photocurrent signal **i** and (ii) a digital output signal **240** that corresponds to the photocurrent signal or the voltage signal. The digital output signal **240** may be sent to a processor, and the processor may determine a signal characteristic **162** of the photocurrent signal or voltage signal based on the digital output signal **240**.

[0118] FIG. 19 illustrates an example Raman signal produced by the Raman spectroscopy system of FIG. 18. The Raman signal **160** produced by spontaneous Raman scattering of the pump beam **120_{pu}** has a peak frequency of ν_{RP} . The frequency offset Ω between the pump beam and the peak frequency of the Raman signal equals $\nu_1 - \nu_{RP}$, and the frequency offset Ω may correspond to a vibrational frequency of a material that is part of the sample **150**. In FIG. 19, the probe beam **120_{pr}** may be coherently mixed with a spectral portion of the Raman signal **160** that is within a particular frequency range of the probe frequency ν_3 to measure one signal characteristic **162**. Additionally, the probe light source **110_{pr}** in FIG. 18 may include a wavelength-tunable laser that tunes the probe beam **120_{pr}** to multiple frequencies across at least a portion of the Raman signal **160**, and the optical receiver **200** may measure multiple respective signal characteristics **162** associated with the Raman signal.

[0119] FIG. 20 illustrates an example laser diode **110** that produces a free-space beam of light **120**. The laser diode **110** in FIG. 20 may be part of a pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}**, and the free-space beam **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**. For example, the probe laser **110_{pr}** in FIG. 2 may be similar to the laser diode **110** in FIG. 20, and the probe laser **110_{pr}** may produce a free-space probe beam **120_{pr}** that is combined with the Raman signal **160** by a free-space beam combiner **130b**. In FIG. 20, the electronic driver **112** supplies laser-diode drive current **I** to the laser diode **110**, and the laser diode produces output light that is collimated by a lens **114** to produce a collimated free-space beam **120**. In other embodiments, a lens **114** may produce a focused free-space beam **120** (e.g., the lens may focus the free-space beam onto a sample **150**). The laser current **I** supplied to the laser diode **110** may be a substantially constant DC current resulting in an output beam of light **120** having a substantially constant optical power. Additionally or alternatively, the laser current **I** may include pulses of current resulting in an output beam **120** that includes corresponding pulses of light.

[0120] The output beam of light **120** produced by the laser diode **110** may have a spectral linewidth of less than approximately 200 MHz, 100 MHz, 50 MHz, 10 MHz, 1 MHz, or 100 kHz. The laser diode **110** in FIG. 20 may be a wavelength-tunable laser diode where the wavelength of the output beam **120** is adjustable over a wavelength range having a width between approximately 10 nm and approximately 100 nm. For example, the operating wavelength of the laser diode **110** may be tunable over at least a portion of one of the following wavelength ranges: 1000 nm to 1100

nm; 1220 nm to 1450 nm; 1490 nm to 1570 nm; 1600 nm to 1690 nm. Alternatively, the laser diode **110** in FIG. 20 may be a fixed-wavelength laser diode. For example, the laser diode **110** in FIG. 20 may be a distributed feedback (DFB) laser diode with a spectral linewidth of less than 1 MHz, and the output beam **120** may have any suitable substantially fixed wavelength between approximately 600 nm and approximately 2000 nm.

[0121] FIG. 21 illustrates an example laser diode **110** that produces seed light **122** that is amplified by a semiconductor optical amplifier (SOA) **124**. Instead of directly emitting an output beam **120** (e.g., as illustrated in FIG. 20), the light from a laser diode **110** may first be amplified by an optical amplifier. The laser diode **110** in FIG. 21 acts as a seed laser that produces seed light **122** that is coupled into the input end of the waveguide **125** of the SOA **124**. The SOA waveguide **125** in FIG. 21 is indicated by the cross-hatched region within the SOA **124**. The SOA **124** amplifies the seed light as it propagates within the waveguide from the input end to the output end, and the output beam of light **120** is emitted from the output end of the SOA. The optical gain provided by the SOA may come from electrical current that is supplied to the SOA by an electronic driver (not illustrated in FIG. 21). For example, an electronic driver **112** may supply substantially constant DC electrical current to the laser diode **110** and to the SOA **124**, and the resulting output beam **120** may have substantially constant optical power. The output beam **120** may be a free-space beam, or the output beam **120** may be coupled into an optical fiber or into a waveguide of a photonic integrated circuit (PIC).

[0122] The laser diode **110** and the SOA **124** in FIG. 21 may be fabricated or integrated together on the same chip so that seed light **122** from the laser diode is directly coupled into the waveguide **125** of the SOA. The waveguide **125** of the SOA **124** may be a tapered optical waveguide (as illustrated in FIG. 21) with a width that increases along a lateral direction from the input end that receives the seed light **122** to the output end that emits the output beam **120**. A light source that includes a seed laser diode **110** that supplies seed light **122** that is amplified by a SOA **124** (as illustrated in FIG. 21) may be referred to as a master-oscillator power-amplifier laser (MOPA laser). The seed laser diode **110** may be referred to as a master oscillator, and the SOA **124** may be referred to as a power amplifier. The MOPA laser in FIG. 21 may be part of a pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}**, and the output beam of light **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**.

[0123] FIG. 22 illustrates an example laser diode **110** that produces seed light **122** that is amplified by a fiber-optic amplifier **126**. The laser diode **110** in FIG. 22 acts as a seed laser that produces seed light **122** that is coupled into an optical fiber **116**. The optical fiber **116** directs the seed light **122** to the fiber-optic amplifier **126**, and the fiber-optic amplifier amplifies the seed light as it propagates through an optical gain fiber of the fiber-optic amplifier. The optical gain fiber may be doped with rare-earth ions (e.g., neodymium, erbium, or ytterbium) or bismuth that provide the optical gain to the seed light. One or more pump lasers may optically pump the active material (e.g., rare-earth ions or bismuth ions) in the optical gain fiber, which in turn provide optical amplification to the seed light **122** propagating through the gain fiber. The amplified seed light produced by the fiber-optic amplifier **126** propagates in an optical fiber as

a fiber-coupled output beam 120. The laser diode 110 and fiber-optic amplifier 126 in FIG. 22 may be part of a pump light source 110_{pu}, a Stokes light source 110_S, or a probe light source 110_{pr}, and the fiber-coupled output beam 120 may be directed to an optical combiner 130, a sample 150, or a detector 220.

[0124] A pump light source 110_{pu}, a Stokes light source 110_S, or a probe light source 110_{pr} may include a seed laser diode 110 followed by an optical amplifier. The seed laser diode 110 may produce seed light 122 that is amplified by the optical amplifier to produce an output beam of light 120. An optical amplifier may include a SOA 124 (e.g., as illustrated in FIG. 21) or a fiber-optic amplifier 126 (e.g., as illustrated in FIG. 22). In some embodiments, an optical amplifier may include a SOA 124 followed by a fiber-optic amplifier 126. For example, a seed laser diode 110 may produce seed light 122 that is first amplified by a SOA 124 and then further amplified by a fiber-optic amplifier 126.

[0125] In some embodiments, a pump light source 110_{pu}, a Stokes light source 110_S, or a probe light source 110_{pr} may include a laser diode 110 and an optical fiber 116 and may not include a fiber-optic amplifier. For example, light produced by a laser diode 110 may be coupled into an optical fiber 116 to produce a fiber-coupled beam 120, and the optical fiber may direct the laser-diode light to an optical combiner 130, a sample 150, or a detector 220. A laser diode 110 that produces a fiber-coupled beam 120 may be referred to as a fiber-coupled laser diode.

[0126] FIG. 23 illustrates an example sampled-grating distributed Bragg reflector (SG-DBR) laser 110. An SG-DBR laser 110 is a wavelength-tunable laser diode that produces an output beam 120 that can be tuned over a wavelength range having a width of between 20 nm and 50 nm. For example, an SG-DBR laser 110 may have a 40-nm wavelength-tuning range from approximately 1530 nm to approximately 1570 nm or from approximately 1630 nm to approximately 1670 nm. The SG-DBR laser 110 in FIG. 23 may be part of a pump light source 110_{pu}, a Stokes light source 110_S, or a probe light source 110_{pr}, and the output beam 120 may be a pump beam 120_{pu}, a Stokes beam 120_S, or a probe beam 120_{pr}. For example, an SG-DBR laser 110 may be part of a probe light source 110_{pr} that produces the probe beam 120_{pr} in FIG. 6, and the SG-DBR laser may tune the frequency ν_3 of the probe beam across at least a portion of the Raman signal 160. An SG-DBR laser 110 may produce a free-space beam 120, a fiber-coupled beam 120, or a beam that is coupled into a waveguide of a PIC. Alternatively, an SG-DBR laser 110 may be integrated with a SOA 124 that amplifies the light produced by the SG-DBR laser (e.g., as illustrated in FIG. 21).

[0127] The SG-DBR laser 110 in FIG. 23 includes a back mirror 180, a phase section 182, a gain section 184, and a front mirror 186, where the phase and gain sections are located between the front and back mirrors. The laser diode current I supplied to the SG-DBR laser 110 includes the following: current I_b supplied to the back mirror 180, current I_p supplied to the phase section 182, current I_g supplied to the gain section 184, and current I_f supplied to the front mirror 186. The gain current I_g provides optical gain to the optical waveguide 188 of the SG-DBR laser 110, and the other currents may be used to set the wavelength of the output beam 120 produced by the SG-DBR laser. The electronic driver 112 may supply particular combinations of electrical currents to the back mirror 180, phase section 182,

gain section 184, and front mirror 186, where each particular combination of electrical currents causes the SG-DBR laser 110 to produce an output beam 120 at a particular wavelength. For example, an SG-DBR laser 110 may be part of a wavelength-tunable probe light source 110_{pr} that produces the probe beam 120_{pr} in FIG. 11, and the electronic driver 112 may supply particular different combinations of the electrical currents I_b , I_p , I_g , and I_f to produce the different probe frequencies ν_{3-1} , ν_{3-2} , ν_{3-3} , . . . and ν_{3-n} to tune the probe beam across at least a portion of the Raman signal 160 in FIG. 11.

[0128] FIG. 24 illustrates an example light source 110 with multiple laser diodes 110 and an optical multiplexer 118 that combines light produced by the laser diodes into a single output beam of light 120. Each of the laser diodes 110-1, 110-2, . . . 110-N produces a respective output beam 120-1, 120-2, . . . 120-N, and the optical multiplexer 118 combines the output beams into the output beam of light 120. The light source 110 may be configured to switch between operating the N laser diodes one at a time so that, at any given time, only one laser diode produces light, and the output beam 120 includes just the light produced by that one laser diode.

[0129] The optical multiplexer 118 may be a free-space device, a fiber-optic device, a waveguide-based device, or a metamaterial-based device, and the multiplexer may combine N different wavelengths of light from the N laser diodes into a single output beam 120. The optical multiplexer 118 may include one or more of the following: a free-space diffraction grating; an arrayed waveguide grating (AWG); a metamaterial that acts as a diffractive optical element; one or more optical filters; one or more optical combiners; one or more optical switches (e.g., thermo-optic switches, liquid crystal switches, or microelectromechanical systems (MEMS) switches); a series of two or more fiber Bragg gratings with optical circulators.

[0130] The light source 110 in FIG. 24 may be a pump light source 110_{pu}, a Stokes light source 110_S, or a probe light source 110_{pr}, and the output beam 120 may be a pump beam 120_{pu}, a Stokes beam 120_S, or a probe beam 120_{pr}. The output beam 120 may be a free-space beam, or the output beam 120 may be coupled into an optical fiber or into a waveguide of a photonic integrated circuit (PIC). For example, each of the laser diodes 110-1, 110-2, . . . 110-N may produce a fiber-coupled beam 120-1, 120-2, . . . 120-N, and the optical multiplexer 118 may be a fiber-optic device that produces a fiber-coupled output beam 120. The output beam 120 may be sent to an optical combiner 130, a sample 150, or a detector 220. Alternatively, the light source 110 may include an optical amplifier (e.g., an SOA or a fiber-optic amplifier) located after the optical multiplexer 118, and the output beam 120 may be coupled from the multiplexer to an optical amplifier that provides optical amplification to the output beam.

[0131] The light source 110 in FIG. 24 includes N laser diodes 110-1, 110-2, . . . 110-N, where N is an integer greater than or equal to 2. Each of the N laser diodes 110 in FIG. 24 may be a wavelength-tunable laser diode or a fixed-wavelength laser diode. For example, the light source in FIG. 24 may include (i) N wavelength-tunable laser diodes, (ii) N fixed-wavelength laser diodes, or (iii) one or more wavelength-tunable laser diodes and one or more fixed-wavelength laser diodes. The light source 110 in FIG. 24 may be referred to as a wavelength-tunable light source, a fre-

quency-tunable light source, or a tunable light source. A wavelength-tunable light source may include one or more continuously tunable laser diodes (e.g., SG-DBR laser diodes); multiple fixed-wavelength laser diodes (e.g., multiple DFB laser diodes), each laser diode operating at a different wavelength; or any combination thereof.

[0132] The light source **110** in FIG. **24** may be a pump light source **110_{pu}** or a Stokes light source **110_S** that includes N fixed-wavelength laser diodes, each laser diode having a different operating wavelength. The wavelength of the output beam **120** produced by the wavelength-tunable light source **110** in FIG. **24** may be adjustable to any wavelength of N different wavelengths by selecting one of the N fixed-wavelength laser diodes for operation. The frequency offset Ω between the pump beam **120_{pu}** and Stokes beam **120_S** may be adjustable by selecting one of the fixed-wavelength laser diodes for operation. For example, the Stokes beams **120_S** and **120_{S'}** in FIG. **14** may be produced by the light source **110** in FIG. **24**. Laser diode **110-1** may be a fixed-wavelength laser diode operating at the frequency ν_2 , and laser diode **110-2** may be a fixed-wavelength laser diode operating at the frequency ν'_2 . Selecting laser diode **110-1** for operation produces the frequency offset Ω_1 in FIG. **14**, and selecting laser diode **110-2** for operation produces the frequency offset Ω_2 .

[0133] The light source **110** in FIG. **24** may operate only one of the N laser diodes **110** at any given time. Each of the laser diodes **110** may operate at a particular wavelength or over a particular range of wavelengths, and one of the laser diodes may be selected for operation based on the wavelength that is needed to perform a particular measurement. For example, the pump beams **120_{pu}** and **120_{pu'}** in FIG. **15** may be produced by the light source **110** in FIG. **24**. Laser diode **110-1** may be a fixed-wavelength laser diode operating at the frequency ν_1 , and laser diode **110-2** may be a fixed-wavelength laser diode operating at the frequency ν'_1 . During a first measurement period, laser diode **110-1** may be operated to produce output beam **120-1** at the frequency ν_1 , and the other laser diodes **110-2** to **110-N** may be turned off or otherwise configured to not produce light. The multiplexer **118** receives the output beam **120-1** from the laser diode **110-1** and directs it to the output of the multiplexer to produce the output beam **120** having a frequency ν_1 . During a second measurement period, the Raman signal **160_c** in FIG. **15** may be measured, and laser diode **110-2** may be operated to produce output beam **120-2** at the frequency ν'_1 . The other laser diodes (i.e., laser diodes **110-1** to **110-N**, excluding laser diode **110-2**) may be turned off or otherwise configured to not produce light. The multiplexer **118** receives the output beam **120-2** from the laser diode **110-2** and directs it to the output of the multiplexer to produce the output beam **120** having a frequency ν'_1 .

[0134] The light source **110** in FIG. **24** may be a probe light source that includes N wavelength-tunable laser diodes. The probe light source may be configured to tune over one or more wavelength ranges having a total width between $p \cdot N \cdot \Delta\lambda_{av}$ and $N \cdot \Delta\lambda_{av}$, where $\Delta\lambda_{av}$ is an average wavelength-tuning range of the N laser diodes **110**, and p is a wavelength-overlap parameter between 0.5 and 1. For example, if the overlap parameter p has a value of 0.7, then the combined wavelength-tuning range of the N wavelength-tunable laser diodes may be between $(0.7)N \cdot \Delta\lambda_{av}$ and $N \cdot \Delta\lambda_{av}$. The wavelength-overlap parameter p represents the amount of wavelength overlap between adjacent wave-

length-tuning ranges (e.g., an overlap value p of 1 indicates that there is no wavelength overlap between the wavelength-tuning ranges). For example, the light source **110** in FIG. **24** may be a probe light source that includes three SG-DBR laser diodes having respective wavelength-tuning ranges of 1490-1530 nm, 1520-1560 nm, and 1550-1590 nm. Each of the SG-DBR laser diodes has a 40-nm tuning range with a 10-nm overlap between adjacent tuning ranges, which results in the light source **110** in FIG. **24** having a 100-nm wavelength-tuning range from 1490 nm to 1590 nm. In this case, the average wavelength-tuning range $\Delta\lambda_{av}$ of the three laser diodes is 40 nm, and the total wavelength-tuning range has a width of 100 nm, which corresponds to the wavelength-overlap parameter p having a value of approximately 0.83. The probe light source **110** may produce an output beam **120** having any wavelength from 1490 nm to 1590 nm by selecting one of the three SG-DBR laser diodes for operation and tuning the laser diode to the desired wavelength. As another example, the light source **110** in FIG. **24** may be a probe light source **110** that includes three SG-DBR laser diodes having respective wavelength-tuning ranges of 1490-1530 nm, 1530-1560, and 1640-1680 nm. Each of the SG-DBR laser diodes has a 40-nm tuning range with no wavelength overlap between adjacent tuning ranges. The average wavelength-tuning range $\Delta\lambda_{av}$ of the three laser diodes is 40 nm, and the total wavelength-tuning range has a width of 120 nm. This corresponds to the wavelength-overlap parameter p having a value of 1, indicating that there is no wavelength overlap between the tuning ranges of the three laser diodes.

[0135] FIG. **25** illustrates an example pump laser **110_{pu}** and Stokes laser **110_S** with a fiber-optic combiner **130** that produces a combined pump-Stokes beam **140** coupled into an optical fiber **116**. A Raman spectroscopy system **100** may include one or more optical elements that direct the pump and Stokes beams to a sample **150**. The optical elements may include a combiner **130** that combines the pump beam **120_{pu}** and the Stokes beam **120_S** to produce a combined pump-Stokes beam **140** that is directed to a sample **150**. In FIG. **1**, the combiner **130_a** may be a free-space optical combiner, and the pump beam **120_{pu}**, Stokes beam **120_S**, and combined beam **140** may each be free-space beams. The combiner **130** in FIG. **25** is a fiber-optic combiner that receives the pump and Stokes beams via two input optical fibers **116** and combines the two beams into a combined pump-Stokes beam **140** that propagates in an output optical fiber **116**. The pump laser **110_{pu}** may be a fiber-coupled laser diode that produces a pump beam **120_{pu}** that is directed to the fiber-optic combiner **130** via an input optical fiber **116**. Similarly, the Stokes laser **110_S** may be a fiber-coupled laser diode that produces a Stokes beam **120_S** that is directed to the fiber-optic combiner **130** via another input optical fiber **116**. The pump laser **110_{pu}** or the Stokes laser **110_S** may be followed by an optical amplifier (not illustrated in FIG. **25**) that amplifies the pump beam **120_{pu}** or Stokes beam **120_S** prior to directing the light to the combiner **130**. After the fiber-optic combiner **130** combines the pump and Stokes beams, the output optical fiber **116** may direct the combined pump-Stokes beam **140** to a sample **150**.

[0136] The fiber-optic combiner **130** in FIG. **25** may include a fiber-optic wavelength division multiplexer (WDM) with two input optical fibers (for the pump and Stokes beam) and one output optical fiber for the output beam **120**. The WDM may include a dichroic beamsplitter or

a fused fiber coupler. Each of the input or output optical fiber **116** may be a single-mode optical fiber or a multi-mode optical fiber.

[0137] In some embodiments, instead of using a single pump laser or a single Stokes laser (as illustrated in FIG. 25), a Raman spectroscopy system may use a pump or Stokes light source **110** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. 24). For example, the light source **110** in FIG. 24 may be a pump light source that produces a fiber-coupled output beam **120** that is coupled to an input fiber of the fiber-optic combiner **130** in FIG. 25. One of the laser diodes of the pump light source may be selected for operation, and the light from the selected laser diode may be directed by the multiplexer **118** to an optical fiber that is coupled to the combiner **130**. Additionally or alternatively, the light source **110** in FIG. 24 may be a Stokes light source that produces a fiber-coupled output beam **120** that is coupled to an input fiber of the fiber-optic combiner **130** in FIG. 25.

[0138] FIG. 26 illustrates an example laser diode **110** coupled to a waveguide **172** of a photonic integrated circuit (PIC) **170**. A PIC **170** (which may be referred to as a planar lightwave circuit (PLC), a waveguide-based device, an integrated-optic device, an integrated optoelectronic device, or a silicon optical bench) may be fabricated from a substrate that includes silicon, indium phosphide, glass (e.g., silica), a polymer, or an electro-optic material (e.g., lithium niobate (LiNbO₃) or lithium tantalate (LiTaO₃)). A PIC **170** may include one or more optical waveguides **172** that confine and guide a beam of light. An optical waveguide **172** that is part of a PIC **170** may be a passive optical waveguide formed in the PIC, and the waveguide may convey light from one optical element to another with relatively low optical loss.

[0139] In FIG. 26, light from the laser diode **110** is coupled into the PIC waveguide **172** to produce a waveguide-coupled beam **120**. The waveguide **172** may convey the beam of light **120** from the laser diode **110** to another optical element (e.g., an optical combiner **130**, a sample **150**, or a detector **220**). Light from the laser diode **110** in FIG. 26 may be coupled into the waveguide **172** using one or more lenses, or the laser diode **110** may be butt-coupled to an input of the waveguide so that the light from the laser diode is directly coupled into the waveguide. The laser diode **110** may be mechanically attached or connected to the PIC **170** or to a substrate to which the PIC is also attached. For example, the laser diode **110** may be attached using epoxy, adhesive, or solder. Alternatively, the laser diode **110** in FIG. 26 may be located apart from the PIC **170**, and the laser diode may send a beam of light **120** to the PIC **170** via optical fiber. An output end of the optical fiber may be attached or connected to the PIC **170** so that the light is coupled into the PIC waveguide **172**. Light from the laser diode **110** may be amplified by an optical amplifier (not illustrated in FIG. 26) prior to being coupled into the PIC waveguide **172**. For example, the laser diode **110** in FIG. 26 may be a MOPA laser similar to that illustrated in FIG. 21, and the output beam **120** produced by the MOPA laser may be directly coupled into the waveguide **172**.

[0140] FIG. 27 illustrates an example pump laser **110_{pu}** and Stokes laser **110_S** with a photonic integrated circuit (PIC) **170** that produces a combined pump-Stokes beam **140** coupled into an optical waveguide **172** of the PIC. The PIC **170** includes a waveguide combiner **130** and three optical waveguides **172** (two input waveguides for the pump and

Stokes beams and one output waveguide for the combined pump-Stokes beam **140**). The waveguide combiner **130** is a waveguide-based optical combiner that combines the pump beam **120_{pu}** and the Stokes beam **120_S** to produce a combined pump-Stokes beam **140**. The combined pump-Stokes beam **140** is coupled to an output optical waveguide **172** of the PIC **170**, and the output waveguide may direct the beam to a sample **150**. The pump laser **110_{pu}** or the Stokes laser **110_S** may be a laser diode that is mechanically attached or connected to the PIC **170** or to a substrate to which the PIC is also attached. Alternatively, the pump laser **110_{pu}** or the Stokes laser **110_S** may be a fiber-coupled laser diode that sends a beam of light to the PIC via optical fiber (e.g., an output end of the optical fiber may be attached to the PIC so that the light is coupled into an input optical waveguide **172**).

[0141] In some embodiments, instead of using a single pump laser or a single Stokes laser (as illustrated in FIG. 27), a Raman spectroscopy system may use a pump or Stokes light source **110** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. 24). For example, the light source **110** in FIG. 24 may be a pump or Stokes light source that produces a fiber-coupled output beam **120** that is coupled to an input optical waveguide **172** of the PIC **170** in FIG. 27. Alternatively, the multiplexer **118** in FIG. 24 may be a waveguide-based device and the output beam **120** may propagate in a PIC waveguide that directs the light to an input optical waveguide **172** of the PIC **170** in FIG. 27.

[0142] FIG. 28 illustrates an example fiber-optic combiner **130** that combines a Raman signal **160** with a probe beam **120_{pr}**. An optical receiver **200** may include an optical combiner **130** that combines a Raman signal **160** and a probe beam of light **120_{pr}** to produce one or more combined probe-Raman signals **210** that are each directed to a detector **220**. In FIG. 1, the combiner **130_b** may be a free-space optical combiner, and the Raman signal **160**, probe beam **120_{pr}**, and the combined probe-Raman signal **210** may each be free-space beams. The combiner **130** in FIG. 28 is a fiber-optic combiner that receives the Raman signal **160** and the probe beam **120_{pr}** via two input optical fibers **116** and combines the two beams into a combined probe-Raman signal **210** that is directed to a detector **220** via an output optical fiber **116**. Each of the input or output optical fiber **116** in FIG. 28 may be a single-mode optical fiber or a multi-mode optical fiber.

[0143] The Raman signal **160** in FIG. 28 may be a free-space beam that is coupled into an input optical fiber **116** using one or more lenses. The probe laser **110_{pr}** may be a fiber-coupled laser diode that directs the probe beam **120_{pr}** to the fiber-optic combiner **130** via optical fiber **116**. The probe beam **120_{pr}** may be amplified by an optical amplifier (not illustrated in FIG. 28) prior to being directed to the combiner **130**. In some embodiments, instead of using a single probe laser **110_{pr}** (as illustrated in FIG. 28), an optical receiver **200** may use a light source **110** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. 24). For example, the light source **110** in FIG. 24 may be a probe light source that produces a fiber-coupled output beam **120** that is coupled to an input optical fiber **116** of the fiber-optic combiner **130** in FIG. 28.

[0144] FIG. 29 illustrates an example photonic integrated circuit (PIC) **170** with a waveguide combiner **130** that combines a Raman signal **160** with a probe beam **120_{pr}**. An optical combiner **130** may be a waveguide combiner **130**

that is part of a PIC 170 and may combine a Raman signal 160 and a probe beam of light 120_{pr} to produce one or more combined probe-Raman signals 210 that are each directed to a detector 220. The waveguide combiner 130 in FIG. 29 receives the Raman signal 160 and the probe beam 120_{pr} via two input waveguides 172. A waveguide combiner 130 may produce 1, 2, or 4 combined output beams 210. The waveguide combiner 130 in FIG. 29 combines the Raman signal 160 and the probe beam 120_{pr} to produce two combined probe-Raman signals 210a and 210b which are each directed to a respective detector 220a and 220b via two output waveguides 172. The PIC 170 in FIG. 29 may be part of an optical receiver 200.

[0145] The Raman signal 160 in FIG. 29 may be a free-space beam that is coupled into an input waveguide 172 using one or more lenses. The probe laser 110_{pr} may be a laser diode that is mechanically attached or connected to the PIC 170 or to a substrate to which the PIC is also attached. Alternatively, the probe laser 110_{pr} may be a fiber-coupled laser diode that sends the probe beam 120_{pr} to the PIC 170 via optical fiber (e.g., an output end of the optical fiber may be attached to the PIC so that the light is coupled into an input optical waveguide 172). The probe beam 120_{pr} may be amplified by an optical amplifier (not illustrated in FIG. 29) prior to being coupled into an input optical waveguide 172. In some embodiments, instead of using a single probe laser 110_{pr} (as illustrated in FIG. 29), an optical receiver 200 may use a light source 110 with multiple laser diodes and an optical multiplexer 118 (e.g., as illustrated in FIG. 24), and the probe beam 120_{pr} may be delivered from the multiplexer to the PIC 170 via optical fiber or via a waveguide 172 of the PIC.

[0146] A Raman spectroscopy system 100 may include one or more optical elements that (i) direct a pump beam 120_{pu} and a Stokes beam 120_S to a sample 150 and (ii) direct a Raman signal 160 and a probe beam 120_{pr} to one or more detectors 220. The optical elements may include one or more PICs 170 that each include one or more optical waveguides 172. One or more of the optical waveguides 172 may direct the pump beam 120 and the Stokes beam 120_S to the sample 150. For example, a PIC 170 may include an optical combiner 130 that produces a combined pump-Stokes beam 140 that is directed to the sample by an optical waveguide 172 of the PIC 170. One or more other optical waveguides 172 may direct the Raman signal 160 and the probe beam 120_{pr} to one or more detectors 220. For example, a PIC 170 may include an optical combiner 130 that combines the Raman signal 160 and the probe beam 120_{pr} to produce one or more combined probe-Raman signals 210 that are each directed to a detector 220 by an optical waveguide 172 of the PIC 170.

[0147] FIGS. 30-35 each illustrate example frequency ranges of a pump beam 120_{pu} and a Stokes beam 120_S. The pump beam 120_{pu} may be produced by a pump light source 110_{pu}, and the Stokes beam 120_S may be produced by a Stokes light source 110_S. The pump and Stokes light sources may each include one or more fixed wavelength laser diodes or one or more wavelength-tunable laser diodes. In each of FIGS. 30-35, the pump beam 120_{pu} and the Stokes beam 120_S each have one or more fixed frequencies or one or more frequencies that are adjustable over a particular frequency range. The corresponding frequency offset Ω between the pump and Stokes beams is indicated as a range of frequencies or a set of discrete frequencies that the

frequency offset can be set to, based on the frequencies that are available to the pump and Stokes beams. The frequency offset Ω is determined from $\nu_1 - \nu_2$, where ν_1 is the range or set of fixed frequencies for the pump beam 120_{pu}, and ν_2 is the range or set of fixed frequencies for the Stokes beam 120_S. The frequency offsets Ω_1 , Ω_2 , Ω_3 , and Ω_4 in FIGS. 30-35 may have any suitable value between approximately 5 THz and approximately 100 THz. The frequency range $\Delta\Omega$ over which a frequency offset may be varied may have any suitable value between approximately 5 THz and approximately 80 THz.

[0148] In FIG. 30, the pump beam 120_{pu} has a single fixed frequency ν_1 , and the Stokes beam 120_S has a frequency ν_2 that is adjustable over a frequency range of width $\Delta\nu_2$. The adjustable frequency range $\Delta\nu_2$ of the Stokes beam 120_S extends from a low frequency ν_{2L} to a high frequency ν_{2H} , where $\Delta\nu_2 = \nu_{2H} - \nu_{2L}$. The pump laser 110_{pu} that produces the pump beam 120_{pu} in FIG. 30 may be a fixed-wavelength laser diode, and the Stokes laser 110_S that produces the Stokes beam 120_S may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between Ω_1 and Ω_2 , and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_2 - \Omega_1$. In FIG. 30, the frequency range $\Delta\Omega$ is also equal to the frequency range $\Delta\nu_2$ of the Stokes beam 120_S. When the Stokes beam 120_S is set to the lower frequency ν_{2L} , the frequency offset between the pump and Stokes beams is $\nu_1 - \nu_{2L}$, which is equal to Ω_2 . Similarly, when the Stokes beam 120_S is set to the upper frequency ν_{2H} , the frequency offset between the pump and Stokes beams is $\nu_1 - \nu_{2H}$, which is equal to Ω_1 .

[0149] For example, the pump beam 120_{pu} in FIG. 30 may have a frequency ν_1 of 250 THz (corresponding to a wavelength of approximately 1200 nm), and the Stokes beam 120_S may be adjustable from a low frequency ν_{2L} of 195 THz to a high frequency ν_{2H} of 200 THz. This corresponds to a frequency-tuning range $\Delta\nu_2$ of the Stokes beam 120_S of 5 THz (167 cm^{-1} in wavenumbers) and a 38-nm wavelength-tuning range from approximately 1499 nm to approximately 1537 nm. The resulting frequency offset Ω between the pump and Stokes beams may be set to a value between the lower value Ω_1 of 50 THz (1668 cm^{-1} in wavenumbers) and the upper value Ω_2 of 55 THz (1835 cm^{-1} in wavenumbers), corresponding to a frequency range $\Delta\Omega$ of 5 THz (167 cm^{-1} in wavenumbers).

[0150] In FIG. 31, the pump beam 120_{pu} has a single fixed frequency ν_1 , and the Stokes beam 120_S has a frequency ν_2 that is adjustable over a frequency range of width $\Delta\nu_2$. The adjustable frequency range $\Delta\nu_2$ of the Stokes beam 120_S extends from a low frequency ν_{2L} to a high frequency ν_{2H} , where $\Delta\nu_2 = \nu_{2H} - \nu_{2L}$. The pump laser 110_{pu} that produces the pump beam 120_{pu} in FIG. 31 may be a fixed-wavelength laser diode. The Stokes light source 110_S that produces the Stokes beam 120_S may include two wavelength-tunable laser diodes. For example, the Stokes light source 110_S may be similar to the light source in FIG. 24 where two wavelength-tunable laser diodes are combined by a multiplexer 118. A first wavelength-tunable laser diode may operate from frequency ν_{2L} to frequency ν_{2M} , and a second wavelength-tunable laser diode may operate from frequency ν_{2M} to frequency ν_{2H} . The total tuning range $\Delta\nu_2$ of the Stokes laser 110_S equals the sum of the tuning ranges $\Delta\nu_{2a}$ and $\Delta\nu_{2b}$ of the two wavelength-tunable laser diodes. In other embodiments, if the tuning ranges of the two wavelength-

tunable laser diodes overlap, the total tuning range Δv_2 of the Stokes light source **110S** will be reduced by the amount of frequency overlap between the two lasers.

[0151] In FIG. **31**, the frequency offset Ω between the pump and Stokes beams may be set to any value between Ω_1 and Ω_2 , and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_2 - \Omega_1$. When the first wavelength-tunable laser diode (with a frequency range from v_{2L} to v_{2M}) is selected to operate, the frequency offset Ω may be set to any value between Ω_M and Ω_2 . For example, when the Stokes beam **120S** is set to the lower frequency v_{2L} , the frequency offset between the pump and Stokes beams is $v_1 - v_{2L}$, which is equal to Ω_2 . When the second wavelength-tunable laser diode (with a frequency range from v_{2M} to v_{2H}) is selected to operate, the frequency offset Ω may be set to any value between Ω_1 and Ω_M . For example, when the Stokes beam **120S** is set to the upper frequency v_{2H} , the frequency offset between the pump and Stokes beams is $v_1 - v_{2H}$, which is equal to Ω_1 . The total frequency range $\Delta\Omega$ of the frequency offset Ω is equal to the sum of the two frequency ranges $\Delta\Omega_{2a}$ and $\Delta\Omega_{2b}$. The frequency range $\Delta\Omega$ is also equal to the overall frequency range Δv_2 of the Stokes beam **120S**.

[0152] In FIG. **32**, the pump beam **120pu** can be set to two fixed frequencies v_{1a} and v_{1b} , and the Stokes beam **120S** has a frequency v_2 that is adjustable over a frequency range of width Δv_2 . The adjustable frequency range Δv_2 of the Stokes beam **120S** extends from a low frequency v_{2L} to a high frequency v_{2H} , where $\Delta v_2 = v_{2H} - v_{2L}$. The pump light source **110pu** that produces the two pump beams **120pu-a** and **120pu-b** may include two fixed-wavelength laser diodes. For example, the pump light source **110pu** may be similar to the light source in FIG. **24** where two fixed-wavelength laser diodes are combined by a multiplexer **118**. The Stokes laser **110S** that produces the Stokes beam **120S** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between Ω_1 and Ω_2 , and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_2 - \Omega_1$. When the pump laser **110pu** produces the pump beam **120pu-a** at frequency v_{1a} , the Stokes beam **120S** may be tuned to a frequency between v_{2L} and v_{2H} to produce a frequency offset Ω between Ω_1 and Ω_M . For example, with the Stokes beam **120S** set to the upper frequency v_{2H} , the frequency offset between the pump and Stokes beams is $v_{1a} - v_{2H}$, which is equal to Ω_1 . When the pump laser **110pu** produces the pump beam **120pu-b** at frequency v_{1b} , the Stokes beam **120S** may be tuned to a frequency between v_{2L} and v_{2H} to produce a frequency offset Ω between Ω_M and Ω_2 . For example, with the Stokes beam **120S** set to the lower frequency v_{2L} , the frequency offset between the pump and Stokes beams is $v_{1b} - v_{2L}$, which is equal to Ω_2 .

[0153] FIG. **33** is similar to FIG. **30**, except in FIG. **33**, the Stokes beam **120S** has a fixed frequency v_2 and the frequency of the pump beam **120pu** is adjustable. The adjustable frequency range Δv_1 of the pump beam **120pu** extends from a low frequency v_{1L} to a high frequency v_{1H} , where $\Delta v_1 = v_{1H} - v_{1L}$. The Stokes laser **110S** that produces the Stokes beam **120pu** in FIG. **33** may be a fixed-wavelength laser diode, and the pump laser **110pu** that produces the pump beam **120pu** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between Ω_1 and Ω_2 , and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_2 - \Omega_1$. In FIG. **33**, the frequency range $\Delta\Omega$ is also equal to the frequency range Δv_1 of the pump beam **120pu**. When the

pump beam **120pu** is set to the lower frequency v_{1L} , the frequency offset between the pump and Stokes beams is $v_{1L} - v_2$, which is equal to Ω_1 . Similarly, when the pump beam **120pu** is set to the upper frequency v_{1H} , the frequency offset between the pump and Stokes beams is $v_{1H} - v_2$, which is equal to Ω_2 .

[0154] In FIG. **34**, both the pump beam **120pu** and the Stokes beam **120S** have adjustable frequencies. The pump laser **110pu** that produces the pump beam **120pu** and the Stokes laser **110S** that produces the Stokes beam **120S** may each include a wavelength-tunable laser diode. The adjustable frequency range Δv_1 of the pump beam **120pu** extends from a low frequency v_{1L} to a high frequency v_{1H} , where $\Delta v_1 = v_{1H} - v_{1L}$. The adjustable frequency range Δv_2 of the Stokes beam **120S** extends from a low frequency v_{2L} to a high frequency v_{2H} , where $\Delta v_2 = v_{2H} - v_{2L}$. The frequency offset Ω between the pump and Stokes beams may be set to any value between Ω_1 and Ω_2 , and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_2 - \Omega_1$. When the Stokes beam **120S** is set to the lower Stokes frequency v_{2L} and the pump beam **120pu** is set to the upper pump frequency v_{1H} , the frequency offset between the pump and Stokes beams is $v_{1H} - v_{2L}$, which is equal to Ω_2 . When the Stokes beam **120S** is set to the upper Stokes frequency v_{2H} and the pump beam **120pu** is set to the lower pump frequency v_{1L} , the frequency offset between the pump and Stokes beams is $v_{1L} - v_{2H}$, which is equal to Ω_1 .

[0155] In FIG. **35**, both the pump beam **120pu** and the Stokes beam **120S** can be set to two different fixed frequencies. A pump light source **110pu** and a Stokes light source **110S** may each include two or more fixed wavelength laser diodes, and each light source may be similar to the light source in FIG. **24** where multiple fixed-wavelength laser diodes are combined by a multiplexer **118**. In FIG. **35**, the pump light source **110pu** that produces the two pump beams **120pu-a** and **120pu-b** may include two fixed-wavelength laser diodes, and the Stokes light source **110S** that produces the two Stokes beams **120S-a** and **120S-b** may include two fixed-wavelength laser diodes. The frequencies of the pump and Stokes beams may be selected to produce four different frequency offsets Ω_1 , Ω_2 , Ω_3 , and Ω_4 . For example, selecting Stokes beam **120S-b** at frequency v_{2b} and pump beam **120pu-a** at frequency v_{1a} produces a frequency offset Ω_1 , which is equal to $v_{1a} - v_{2b}$. As another example, selecting Stokes beam **120S-a** at frequency v_{2a} and pump beam **120pu-b** at frequency v_{1b} produces a frequency offset Ω_4 , which is equal to $v_{1b} - v_{2a}$. The frequency offset Ω_2 may be produced by selecting Stokes beam **120S-a** at frequency v_{2a} and pump beam **120pu-a** at frequency v_{1a} , and the frequency offset Ω_3 may be produced by selecting Stokes beam **120S-b** at frequency v_{2b} and pump beam **120pu-b** at frequency v_{1b} .

[0156] FIG. **36** illustrates an example optical receiver **200** with two detectors **220a** and **220b**. The optical receivers **200** in FIGS. **29** and **36** are similar, except one difference is that the optical receiver in FIG. **29** is a waveguide-based optical receiver, while the optical receiver in FIG. **36** is a free-space optical receiver **200**. The optical combiner **130** in FIG. **36** may be a 50/50 free-space beamsplitter that reflects approximately 50% of an incident beam of light and transmits approximately 50% of the beam. The optical combiner **130** splits the Raman signal **160** and the probe beam **120pr** into two beams to produce two combined probe-Raman signals **210a** and **210b**. The combined probe-Raman signal **210a** is directed to detector **220a** and includes a transmitted portion

of the probe beam **120pr** and a reflected portion of the Raman signal **160** (e.g., approximately 50% of the probe beam and approximately 50% of the Raman signal). Similarly, the combined probe-Raman signal **210b** is directed to detector **220b** and includes a reflected portion of the probe beam **120pr** and a transmitted portion of the Raman signal **160**. The portions of the probe beam **120pr** and the Raman signal **160** that make up the combined probe-Raman signal **210a** may be coherently mixed at detector **220a** to produce the photocurrent signal i_a . Similarly, the portions of the probe beam **120pr** and the Raman signal **160** that make up the combined probe-Raman signal **210b** may be coherently mixed at detector **220b** to produce the photocurrent signal i_b .

[0157] The two detectors **220a** and **220b** are arranged so that their respective photocurrents i_a and i_b are subtracted. The anode of detector **220a** is electrically connected to the cathode of detector **220b**, and the subtracted photocurrent signal $i_a - i_b$ from the anode-cathode connection is sent to the detection electronics **230**, which produces a digital output signal **240** that corresponds to the subtracted photocurrent signal. The subtracted photocurrent signal may be expressed as $i_a - i_b = 2E_R E_{pr} \cos[2\pi(v_R - v_3)t + \Delta\phi]$, which corresponds to the coherent-mixing term discussed herein. The subtracted photocurrent signal does not include the terms E_R^2 and E_{pr}^2 corresponding to the respective optical powers of the Raman signal **160** and the probe beam **120pr**. By subtracting the two photocurrents i_a and i_b , the common-mode terms E_R^2 and E_{pr}^2 (as well as common-mode noise) that appear in each of the photocurrent signals i_a and i_b are substantially removed, leaving the coherent-mixing term, which is the quantity of interest. Since subtraction may remove common-mode noise, the subtracted photocurrent signal $i_a - i_b$ may have a reduced noise compared to each of the photocurrent signals i_a and i_b alone. The dual-detector arrangement in FIG. 36 in which the photocurrents are subtracted may be referred to as a balanced optical detector. A balanced detector may be implemented as a free-space device, a fiber-optic-based device, or a waveguide-based device.

[0158] FIG. 37 illustrates an example optical receiver **200** configured for polarization-sensitive detection of a Raman signal **160**. A polarization-sensitive optical receiver **200** may be used to determine the polarization of a Raman signal **160**. The polarization of a Raman signal **160** may be determined by a processor based on one or more digital output signals **240** produced by the polarization-sensitive optical receiver **200**. Determining the polarization of a Raman signal **160** may include determining a relative size or ratio of two orthogonal polarization components of the Raman signal (e.g., horizontal and vertical polarization components of the Raman signal). For example, if the ratio of the horizontal and vertical polarization components of a Raman signal **160** is 100:1, then the Raman signal may be determined to be substantially horizontally polarized. As another example, if the ratio of the horizontal and vertical polarization components of a Raman signal **160** is 1:1, then the horizontal and vertical polarization components of the Raman signal may be determined to be approximately equal (e.g., the Raman signal may be circularly polarized or linearly polarized at a 45-degree angle to the horizontal and vertical directions).

[0159] A polarization-sensitive optical receiver **200** may include a polarization beamsplitter (PBS) **135** that splits an input beam into two output beams, where one output beam is horizontally polarized, and the other output beam is vertically polarized. The horizontally polarized output beam

includes the horizontal polarization component of the input beam, and the vertically polarized output beam includes the vertical polarization component of the input beam. The Raman signal **160** in FIG. 37 is directed to a Raman-signal PBS **135R** that splits the Raman signal into a horizontal-polarization Raman signal **160-h** and a vertical-polarization Raman signal **160-v**. Similarly, the probe beam **120pr** is directed to a probe-beam PBS **135pr** that splits the probe beam into a horizontal-polarization probe beam **120pr-h** and a vertical-polarization probe beam **120pr-v**.

[0160] A polarization-sensitive optical receiver **200** may include a waveplate **132** that changes the polarization of the probe beam **120pr** so that the probe beam is split into two polarization components. The two polarization components may each have approximately one-half the power of the probe beam **120pr**. The waveplate **132c** in FIG. 37 may be (i) a half-wave plate that rotates the polarization of the probe beam **120pr** or (ii) a quarter-wave plate that converts the probe beam **120pr** to a circular or elliptical polarization. For example, the probe laser **110pr** may produce a probe beam **120pr** that is vertically polarized, and the waveplate **132c** may be a half-wave plate that rotates the probe-beam polarization by 45 degrees so that the horizontal-polarization and vertical-polarization probe beams **120pr-h** and **120pr-v** each have approximately equal optical powers.

[0161] The optical combiner **130h** in FIG. 37 combines the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** to produce a horizontal probe-Raman signal **210h** that is directed to a horizontal-polarization optical receiver **200h**. Similarly, the optical combiner **130v** combines the vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-v** to produce a vertical probe-Raman signal **210v** that is directed to a vertical-polarization optical receiver **200v**. The horizontal-polarization optical receiver **200h** may include one or more optical detectors **200**, where each detector is configured to coherently mix at least a portion of the horizontal-polarization Raman signal **160-h** and at least a portion of the horizontal-polarization probe beam **120pr-h** to produce a horizontal-polarization electronic signal. Similarly, the vertical-polarization optical receiver **200v** may include one or more optical detectors **200**, where each detector is configured to coherently mix at least a portion of the vertical-polarization Raman signal **160-v** and at least a portion of the vertical-polarization probe beam **120pr-v** to produce a vertical-polarization electronic signal. The horizontal-polarization optical receiver **200h** and the vertical-polarization optical receiver **200v** may each include: (i) a single detector **220** (e.g., similar to the optical receiver **200** in FIG. 2), (ii) two detectors **220** (e.g., similar to the balanced optical detector arrangement in FIG. 36), or (iii) four detectors **220** (e.g., similar to the arrangement in FIG. 39). The electronic signals may include a photocurrent signal i , and each optical receiver may include an electronic amplifier **232** that produces a corresponding voltage signal **234**. The h-polarization optical receiver **200h** may produce a digital output signal **240-h** corresponding to the horizontal-polarization electronic signal, and the v-polarization optical receiver **200v** may produce a digital output signal **240-v** corresponding to the vertical-polarization electronic signal.

[0162] A processor may determine one or more characteristics of the horizontal-polarization and vertical-polarization electronic signals based on the digital output signals **240-h** and **240-v**. Additionally, a processor may determine a

polarization of the Raman signal **160** based on the characteristics of the horizontal-polarization and vertical-polarization electronic signals. For example, the characteristics of the electronic signals may include an amplitude or an area associated with the electronic signals, and the polarization of the Raman signal **160** may be expressed as a relative size or ratio of the amplitudes or areas associated with the horizontal and vertical polarization components of the Raman signal. If the horizontal digital output signal **240-h** includes an amplitude characteristic with value 100 and the vertical digital output signal **240-v** includes a corresponding amplitude characteristic with value 1, then the Raman signal **160** may be determined to be substantially horizontally polarized. If the horizontal and vertical digital output signals each include amplitude characteristics having approximately equal values, then the Raman signal **160** may be determined to have approximately equal horizontal and vertical polarization components.

[0163] A polarization-sensitive optical receiver **200** as illustrated in FIG. 37 may be implemented with free-space optical elements, fiber-optic components, waveguide-based optical elements, a metamaterial-based device, or any suitable combination thereof. For example, the two PBSs **135** in FIG. 37 may be free-space polarization beamsplitter cubes, and the Raman signal **160** and the probe beam **120_{pr}** may be free-space optical beams. Alternatively, the two PBSs **135** may be fiber-optic components, and the Raman signal **160** and the probe beam **120_{pr}** may be conveyed to the PBSs **135** via optical fiber (e.g., single-mode optical fiber or polarization-maintaining optical fiber). Additionally, the horizontally and vertically polarized probe-Raman signals **210_h** and **210_v** may be conveyed to the respective h-polarization and v-polarization optical receivers via polarization-maintaining optical fiber. The h-polarization and v-polarization optical receivers may each preserve the polarization of the respective horizontally and vertically polarized probe-Raman signals. For example, the h-polarization and v-polarization optical receivers may each include polarization-maintaining optical fiber that maintains the polarization of the beams. Alternatively, the h-polarization and v-polarization receivers may each include a PIC with optical waveguides configured to maintain the polarization of the beams.

[0164] FIG. 38 illustrates an example optical receiver **200** configured to detect in-phase and quadrature components of a Raman signal **160**. The optical receiver **200** includes a 90-degree optical hybrid **250** and four detectors **220I+**, **220I-**, **220Q+**, and **220Q-**. A 90-degree optical hybrid **250** is an optical-combiner component with two input ports and four output ports. Input light received at each of the two input ports is split, combined, and directed to each of the four output ports, and a 90-degree phase shift is imparted to one of the split beams before the Raman signal **160** and probe beam **120_{pr}** are combined. The 90-degree optical hybrid **250** in FIG. 38 combines a Raman signal **160** and a probe beam **120_{pr}** to produce four combined output beams: two in-phase combined beams **210I+** and **210I-**, and two quadrature combined beams **210Q+** and **210Q-**. Each of the four combined beams **210** may include a portion of the Raman signal **160** and a portion of the probe beam **120_{pr}**, and each of the combined beams is directed to one of the four detectors of the optical receiver **200**. In FIG. 38, each of the four detectors produces a photocurrent signal that corresponds to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120_{pr}**.

[0165] A 90-degree optical hybrid **250** may be configured so that the combined beams directed to each of the output ports have approximately the same optical power or energy. For example, the 90-degree optical hybrid **250** in FIG. 38 may split the Raman signal **160** into four approximately equal portions and direct each of the Raman-signal portions to one of the detectors. Similarly, the probe beam **120_{pr}** may be split into four approximately equal portions directed to each of the four detectors. In the example of FIG. 38, the combined beam **210I+**, which is directed to detector **220I+**, may include approximately one-quarter of the power of the Raman signal **160** and approximately one-quarter of the power of the probe beam **120_{pr}**. Similarly, each of the three other combined beams (**210I-**, **210Q+**, **210Q-**) in FIG. 38 may also include approximately one-quarter of the Raman signal **160** and approximately one-quarter of the probe beam **120_{pr}**.

[0166] A 90-degree optical hybrid **250** may be implemented as a waveguide-based device in a PIC. The 90-degree optical hybrid **250** in FIG. 38 is a waveguide-based optical device that includes two waveguide-based optical splitters (**252a**, **252b**) and two waveguide-based optical combiners (**130I**, **130Q**). Splitter **252a** may split the Raman signal **160** into two portions having substantially equal optical power, a first portion directed to combiner **130I** and a second portion directed to combiner **130Q**. Similarly, splitter **252b** may split the probe beam **120_{pr}** into two portions having substantially equal power, a first portion directed to combiner **130I** and a second portion directed to combiner **130Q**. Each optical combiner **130** combines a portion of the Raman signal **160** with a portion of the probe beam **120_{pr}**, and the combined portions are split into a first combined beam (e.g., combined beam **210I+**) and a second combined beam (e.g., combined beam **210I-**). The combined beam **210I+** is directed to detector **220I+** and includes portions of the Raman signal **160** and the probe beam **120_{pr}** (e.g., approximately 25% of the Raman signal **160** and approximately 25% of the probe beam **120_{pr}**). The combined beam **210I-** is directed to detector **220I-** and may include approximately 25% of the Raman signal **160** and approximately 25% of the probe beam **120_{pr}**.

[0167] In other embodiments, all or part of a 90-degree optical hybrid **250** may be implemented as a free-space optical device. For example, a free-space 90-degree optical hybrid **250** may include one or more free-space beamsplitters or combiners that receive the Raman signal **160** and probe beam **120_{pr}** as free-space beams and produce four free-space combined beams (**210I+**, **210I-**, **210Q+**, **210Q-**). Alternatively, all or part of a 90-degree optical hybrid **250** may be implemented as a fiber-optic device. For example, a 90-degree optical hybrid **250** may be contained in a package with two input optical fibers that direct the Raman signal **160** and probe beam **120_{pr}** into the package and four output optical fibers that direct the four combined beams to four respective detectors.

[0168] A 90-degree optical hybrid **250** may include an optical phase shifter **254** that imparts a 90-degree phase change ($\Delta\phi$) to a portion of the probe beam **120_{pr}** or to a portion of the Raman signal **160**. The phase shifter **254** may apply the 90-degree phase change after a beam of light is split by an optical splitter **252** and prior to combining the Raman signal with the probe beam at an optical combiner **130**. For example, a splitter **252a** may split the Raman signal **160** into two portions, and a phase shifter **254** may impart a

90-degree phase change to one portion of the Raman signal with respect to the other portion, after which the two portions are sent to two different optical combiners. As another example, a splitter **252b** may split the probe beam **120pr** into two portions, and a phase shifter **254** may impart a 90-degree phase change to one portion of the probe beam with respect to the other portion. In FIG. 38, the phase shifter is located after the splitter **252b** and before the combiner **130Q**. The splitter **252b** splits the probe beam **120pr** into two portions, and the phase shifter **254** imparts a 90-degree phase change to the probe-beam portion directed to combiner **130Q**. The other portion of the probe beam **120pr** directed to combiner **130I** does not pass through the phase shifter **254** and does not receive a phase shift from the phase shifter **254**.

[0169] An optical phase shifter **254** may be implemented as a part of a waveguide-based 90-degree optical hybrid **250**. For example, a phase shifter **254** may be implemented as part of an optical waveguide that only one portion of the probe beam **120pr** propagates through. That part of the optical waveguide may be temperature controlled to adjust the refractive index of the waveguide and produce a relative phase delay of approximately 90 degrees between two portions of the probe beam **120pr**. Additionally or alternatively, the 90-degree optical hybrid **250** as a whole may be temperature controlled to set and maintain a 90-degree phase delay. As another example, a phase shifter **254** may be implemented by applying an external electric field to part of an optical waveguide to change the refractive index of the waveguide and produce a 90-degree phase delay. In other embodiments, a phase shifter **254** may be implemented as a part of a free-space or fiber-coupled 90-degree optical hybrid **250**. For example, the input and output beams in a free-space 90-degree optical hybrid **250** may be reflected by or transmitted through the optical surfaces of a free-space optical hybrid **250** so that a relative phase shift of 90 degrees is imparted to one portion of the probe beam **120pr** with respect to another portion of the probe beam.

[0170] In FIG. 38, each of the four detectors produces a photocurrent signal that corresponds to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120pr**. The photocurrents are subtracted in a manner similar to that illustrated in FIG. 36. The photocurrents i_{I+} and i_{I-} from detectors **220I+** and **220I-** are subtracted to produce the subtracted in-phase photocurrent signal i_I which is equal to $i_{I+} - i_{I-}$. Similarly, the photocurrents i_{Q+} and i_{Q-} from detectors **220Q+** and **220Q-** are subtracted to produce the subtracted quadrature photocurrent signal i_Q which is equal to $i_{Q+} - i_{Q-}$. Each of the subtracted photocurrent signals represents a coherent-mixing term corresponding to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120pr**. The two subtracted photocurrent signals i_I and i_Q are similar, except the in-phase photocurrent signal i_I includes a cosine function, while the quadrature photocurrent signal i_Q includes a sine function. This difference between the two subtracted photocurrent signals arises from the 90-degree phase shift provided by the phase shifter **254**. Because a 90-degree phase shift is imparted to the probe beam **120pr** directed to the combiner **130Q**, the subtracted quadrature photocurrent signal i_Q includes a sine function (which has a 90-degree phase offset with respect to a cosine function).

[0171] Each of the subtracted photocurrent signals i_I and i_Q may be sent to detection electronics **230** that produce

voltage signals and digital output signals corresponding to the subtracted photocurrent signals. Based on the digital output signals (which result from the four photocurrent signals i_{I+} , i_{I-} , i_{Q+} , and i_{Q-}), a processor may determine an in-phase portion IP associated with the Raman signal **160** and a quadrature portion Q associated with the Raman signal. Additionally or alternatively, the processor may determine a phase associated with the Raman signal **160**. For example, the processor may determine a phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr**. A phase difference may be referred to as a phase offset or a relative phase between the Raman signal **160** and the probe beam **120pr**.

[0172] The in-phase portion IP associated with the Raman signal **160** may be determined from a characteristic (e.g., an amplitude or an area) of an electronic signal associated with the in-phase photocurrent signal i_I and the quadrature portion Q may be determined from a characteristic associated with the quadrature photocurrent signal i_Q . The in-phase portion IP may correspond to an amount of the Raman signal **160** that is in-phase with the probe beam **120pr**, and the quadrature portion Q may represent an amount of the Raman signal that is out of phase (i.e., 90-degrees phase-shifted) with the probe beam. For example, the in-phase portion IP and the quadrature portion Q, may each have values from -1 to 1. If the Raman signal **160** is in-phase with the probe beam **120pr**, then the in-phase portion IP may have a value of approximately 1, and the quadrature portion Q may have a value of approximately 0. Similarly, if the Raman signal **160** is out of phase by ± 90 degrees with respect to the probe beam **120pr**, then the in-phase portion IP may have a value of 0, and the quadrature portion Q may have a value of ± 1 . The phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr** may be determined from the expression $\Delta\phi = \arctan(Q/IP)$. For example, if Q is 0 and IP is 1, then the Raman signal and the probe beam are substantially in phase, with a phase difference $\Delta\phi$ of 0 degrees. As another example, if Q is 1 and IP is 0, then the Raman signal and the probe beam are substantially out of phase, with a phase difference $\Delta\phi$ of 90 degrees.

[0173] FIG. 39 illustrates an example optical receiver **200** configured to detect polarization as well as in-phase and quadrature components of a Raman signal **160**. The optical receiver **200** in FIG. 39 is similar the optical receiver **200** in FIG. 37, where the horizontal-polarization optical receiver **200h** and the vertical-polarization optical receiver **200v** each includes a 90-degree optical hybrid **250**. Each of the 90-degree optical hybrids in FIG. 39 may be similar to the 90-degree optical hybrid **250** in FIG. 38. In FIG. 39, the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** are directed to a horizontal-polarization optical receiver **200h** that includes a 90-degree optical hybrid **250h** and four detectors **220h-I+**, **220h-I-**, **220h-Q+**, and **220h-Q-**. The vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-v** are directed to a vertical-polarization optical receiver **200v** that includes a 90-degree optical hybrid **250v** and four detectors **220v-I+**, **220v-I-**, **220v-Q+**, and **220v-Q-**. The h-polarization optical receiver **200h** may be used to determine the relative size of the horizontal-polarization Raman signal **160-h** as well as the in-phase and quadrature components of the horizontal-polarization Raman signal **160-h**. Similarly, the v-polarization optical receiver **200v**

may be used to determine the relative size of the vertical-polarization Raman signal **160-v** as well as the in-phase and quadrature components of the vertical-polarization Raman signal **160-v**.

[0174] The 90-degree optical hybrid **250h** in FIG. 39 combines the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** to produce four horizontally polarized combined output beams: two in-phase combined beams **210h-I+** and **210h-I-**, and two quadrature combined beams **210h-Q+** and **210h-Q-**. Each of the four horizontally polarized combined beams includes a portion of the horizontal-polarization Raman signal **160-h** and a portion of the horizontal-polarization probe beam **120pr-h**. The four combined beams are directed to four respective detectors (**220h-I+**, **220h-I-**, **220h-Q+**, **220h-Q-**), and each detector produces a respective photocurrent signal (i_{h-I+} , i_{h-I-} , i_{h-Q+} , i_{h-Q-}) that corresponds to the coherent mixing of a portion of the horizontal Raman signal **160-h** and a portion of the horizontal probe beam **120pr-h**. Each of the four photocurrents i_{h-I+} , i_{h-I-} , i_{h-Q+} , and i_{h-Q-} may be referred to as a horizontal-polarization electronic signal. The photocurrents from the detectors are subtracted to produce a subtracted horizontal in-phase photocurrent signal i_{h-I} which is equal to $i_{h-I+} - i_{h-I-}$ and a subtracted horizontal quadrature photocurrent signal i_{h-Q} which is equal to $i_{h-Q+} - i_{h-Q-}$.

[0175] The 90-degree optical hybrid **250v** in FIG. 39 combines the vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-v** to produce four vertically polarized combined output beams: two in-phase combined beams **210v-I+** and **210v-I-**, and two quadrature combined beams **210v-Q+** and **210v-Q-**. Each of the four vertically polarized combined beams includes a portion of the vertical-polarization Raman signal **160-v** and a portion of the vertical-polarization probe beam **120pr-v**. The four combined beams are directed to four respective detectors (**220v-I+**, **220v-I-**, **220v-Q+**, **220v-Q-**), and each detector produces a respective photocurrent signal (i_{v-I+} , i_{v-I-} , i_{v-Q+} , i_{v-Q-}) that corresponds to the coherent mixing of a portion of the vertical Raman signal **160-v** and a portion of the vertical probe beam **120pr-v**. Each of the four photocurrents i_{v-I+} , i_{v-I-} , i_{v-Q+} , and i_{v-Q-} may be referred to as a vertical-polarization electronic signal. The photocurrents from the detectors are subtracted to produce a subtracted vertical in-phase photocurrent signal i_{v-I} which is equal to $i_{v-I+} - i_{v-I-}$ and a subtracted vertical quadrature photocurrent signal i_{v-Q} which is equal to $i_{v-Q+} - i_{v-Q-}$.

[0176] Each of the subtracted photocurrent signals i_{h-I} , i_{h-Q} , i_{v-I} , and i_{v-Q} may be sent to detection electronics **230** that produces voltage signals and digital output signals corresponding to the subtracted photocurrent signals. Based on the digital output signals (which are determined from the four horizontal-polarization electronic signals and the four vertical-polarization electronic signals), a processor may determine (i) the polarization of the Raman signal **160** and (ii) a phase associated with the Raman signal (e.g., a phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr**). Determining the polarization of a Raman signal **160** may include determining a relative size or ratio of the horizontal and vertical polarization components of the Raman signal. For example, the relative size of the horizontal polarization component of the Raman signal **160** may be determined by adding characteristics (e.g., areas or amplitudes) associated with the two horizontal photocurrent sig-

nals i_{h-I} and i_{h-Q} . Similarly, the relative size of the vertical polarization component of the Raman signal **160** may be determined by adding characteristics associated with the two vertical photocurrent signals i_{v-I} and i_{v-Q} . As an example, if the relative size of the horizontal polarization component of the Raman signal **160** is 1 and the relative size of the vertical polarization component of the Raman signal **160** is 100, then the Raman signal **160** may be determined to be substantially vertically polarized.

[0177] Based on the digital output signals, a processor may determine a phase associated with the Raman signal. For example, the processor may determine (i) a phase difference $\Delta\phi_h$ between the horizontal Raman signal **160-h** and the horizontal probe beam **120pr-h** and (ii) a phase difference $\Delta\phi_v$ between the vertical Raman signal **160-v** and the vertical probe beam **120pr-v**. Based on the digital output signals, a processor may determine in-phase and quadrature portions associated with each of the horizontal Raman signal **160-h** and vertical Raman signal **160-v**. The phase difference $\Delta\phi_h$ between the horizontal Raman signal **160-h** and the horizontal probe beam **120pr-h** may be determined from the expression $\Delta\phi_h = \arctan(Q_h/IP_h)$, where Q_h and IP_h are the quadrature and in-phase portions associated with the horizontal Raman signal. The phase difference $\Delta\phi_v$ between the vertical Raman signal **160-v** and the vertical probe beam **120pr-v** may be determined from the expression $\Delta\phi_v = \arctan(Q_v/IP_v)$, where Q_v and IP_v are the quadrature and in-phase portions associated with the vertical Raman signal.

[0178] An optical receiver **200** may include one or more detectors **220**. An optical receiver **200** may include one detector **220** (e.g., as illustrated in FIGS. 1, 2, and 18), or an optical receiver **200** may include multiple detectors **220** (e.g., as illustrated in FIGS. 17, 36, 38, and 39). An optical receiver **200** with multiple detectors **220** may include 2, 3, 4, 8, 16, or any other suitable number of detectors. For example, an optical receiver **200** may include two detectors **220** arranged so that their respective photocurrents are subtracted (e.g., as illustrated in FIG. 36). As another example, an optical receiver **200** may include four detectors **220** (e.g., as illustrated in FIG. 38) or eight detectors **220** (e.g., as illustrated in FIG. 39). In an optical receiver **200** with multiple detectors **220**, portions of a probe beam **120pr** and a Raman signal **160** may be coherently mixed together at one or more of the multiple detectors **220**, and each of these one or more detectors may produce a photocurrent signal i corresponding to the coherent mixing of the probe beam and the Raman signal. Any of the optical receivers **200** described herein as having a single detector **220** may also be configured to have two or more detectors. For example, the optical receiver in FIG. 1 (which includes one detector **220**) may include a second detector (not illustrated in FIG. 1), and the detection electronics **230** may be configured to receive and process photocurrent signals from each of the two detectors.

[0179] FIG. 40 illustrates an example Raman spectroscopy system **100** with counterpropagating pump and Stokes beams. The Raman spectroscopy system **100** in FIG. 40 includes a pump light source **110pu** that produces a pump beam of light **120pu** and a Stokes light source **110S** that produces a Stokes beam of light **120S**. The pump light source **110pu** produces the pump beam of light **120pu** at a pump frequency ν_{pu} , and the Stokes light source **110S** produces the Stokes beam of light **120S** at a Stokes frequency ν_S . The pump and Stokes frequencies may be offset

by a frequency offset Ω , where Ω equals $\nu_{pu} - \nu_S$. The pump light source **110_{pu}** may be referred to as a pump laser and may include one or more lasers (e.g., one or more fixed-wavelength laser diodes or one or more wavelength-tunable laser diodes). Similarly, the Stokes light source **110_S** may be referred to as a Stokes laser and may include one or more lasers (e.g., one or more fixed-wavelength laser diodes or one or more wavelength-tunable laser diodes).

[0180] In FIG. 40, the pump beam **120_{pu}** and the Stokes beam **120_S** are directed to a sample **150** along counterpropagating beam paths. Counterpropagating beams refer to two beams of light that are directed to a sample **150** along separate beam paths that do not overlap until the beams reach the sample. In FIGS. 1-2, the pump and Stokes beams are combined at the optical combiner **130_a** and then directed to the sample as co-propagating beams. In FIG. 40, the pump and Stokes beams are directed to the sample **150** along different, counterpropagating beam paths, and the two counterpropagating beams enter the sample from opposite sides of the sample. The pump and Stokes beams are overlapped at the sample **150**, and the sample produces a Raman signal **160** in response to the pump and Stokes beams. For example, the Raman signal **160** may be produced by coherent Raman scattering of the pump and Stokes beams at the sample **150**. The Raman signal **160** being produced at the sample **150** may refer to the Raman signal being produced within the sample or at the surface of the sample. For example, the Raman signal **160** may be produced by the pump and Stokes beams interacting with a particular material located within the sample or on the surface of the sample.

[0181] A Raman spectroscopy system may include one or more optical elements that direct the pump beam **120_{pu}** and the Stokes beam **120_S** to a sample **150** along counterpropagating beam paths. Additionally, the optical elements may collect the Raman signal **160** produced by the sample **150** in response to the pump and Stokes beams and direct the Raman signal **160** to an optical receiver **200**. The optical elements may include free-space optics, fiber-optic components, waveguide-based optics, metamaterials, or any suitable combination thereof. For example, the optical elements may include a mirror, lens, optical combiner (e.g., beam-splitter), optical fiber, photonic integrated circuit (PIC), optical waveguide, or metamaterial-based optic.

[0182] The lens **114_a** in FIG. 40 is an optical element that directs the pump beam **120_{pu}** to the sample **150**. The lens **114_a** receives the pump beam **120_{pu}** produced by the pump laser **110_{pu}** and focuses the pump beam into the sample **150**. For example, the pump beam **120_{pu}** received from the pump laser **110_{pu}** may be a collimated optical beam with a beam diameter of 1-2 mm, and the lens **114_a** may focus the pump beam to a 50-100 μm -diameter beam waist located within the sample **150**. The lens **114_b** is an optical element that (i) directs the Stokes beam **120_S** to the sample **150** and (ii) collects the Raman signal **160** produced by the sample. The lens **114_b** receives the Stokes beam **120_S** produced by the Stokes laser **110_S** and focuses the Stokes beam into the sample **150**. For example, the Stokes beam **120_S** received from the Stokes laser **110_S** may be a collimated optical beam with a beam diameter of 1-2 mm, and the lens **114_b** may focus the Stokes beam to a 50-100 μm -diameter beam waist located within the sample **150**. Additionally, the lens **114_b** collects the Raman signal **160** from the sample **150** and produces a Raman-signal beam that is directed to the optical receiver **200**. The Raman-signal beam directed to the optical

receiver **200** may be a collimated or focused optical beam that includes the light from the Raman signal **160** collected by the lens **114_b**.

[0183] The Raman spectroscopy system **100** in FIG. 40 is similar to the system in FIG. 2, except in FIG. 2, the pump and Stokes beams are combined so that they co-propagate to the sample **150** in the same direction along approximately the same optical axis. In FIG. 40, the pump and Stokes beams are directed to the sample **150** separately and along counterpropagating beam paths. The counterpropagating pump and Stokes beams in FIG. 40 propagate to the sample **150** in opposite directions along approximately the same optical axis, and the two beams enter the sample from opposite sides of the sample. The pump and Stokes beams enter the sample **150** separately, and the beams are spatially overlapped or combined within the sample **150** to produce the Raman signal **160**. While the pump and Stokes beams may propagate along approximately the same optical axis so that their beam paths are approximately collinear, the pump and Stokes beams in FIG. 40 (as well as other figures herein) are offset laterally for clarity to allow the individual beams to be visualized. Similarly, the Raman signal **160**, which may propagate along approximately the same optical axis as the pump and Stokes beams, is offset laterally in FIG. 40 (as well as other figures herein) for clarity to allow the beams to be visualized.

[0184] The Raman spectroscopy system **100** in FIG. 40 includes an optical receiver **200** that detects the Raman signal **160** produced by the sample **150**. The optical receiver **200** detects the Raman signal **160** using an optical heterodyne technique in which the Raman signal **160** is coherently mixed with a probe beam of light **120_{pr}** at a detector **220**. The optical receiver **200** in FIG. 40 (which is similar to that illustrated in FIG. 2) includes a probe light source **110_{pr}** that produces a probe beam of light **120_{pr}** at a probe frequency ν_{pr} . The probe light source **110_{pr}** may be referred to as a probe laser and may include one or more lasers (e.g., one or more fixed-wavelength laser diodes or one or more wavelength-tunable laser diodes). The probe beam **120_{pr}** and the Raman signal **160** are combined at the optical combiner **130_b** to produce a combined probe-Raman signal **210** that is directed to a detector **220**. The detector **220** coherently mixes a portion of the Raman signal **160** with at least a portion of the probe beam **120_{pr}** to produce a corresponding photocurrent signal **i**. The detection electronics **230** (which may be referred to as an electronic circuit) receives the photocurrent signal **i** and produces a digital output signal **240** that corresponds to the photocurrent signal.

[0185] The digital output signal **240** may be sent to a processor, and the processor may determine a characteristic of the analog photocurrent signal **i** based on the digital output signal **240**. For example, the characteristic of the analog photocurrent signal (or the characteristic of an analog voltage signal **234** that corresponds to the photocurrent signal) determined by the processor may include one or more of: a peak amplitude (e.g., a peak amplitude of the photocurrent signal or a peak amplitude of the corresponding voltage signal), an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization (e.g., a polarization of a Raman signal).

[0186] A Raman spectroscopy system with counterpropagating pump and Stokes beams may include one or more

optical filters **134**. The Raman spectroscopy system **100** in FIG. **40** includes three optical filters **134a**, **134b**, and **134c**. Optical filter **134a** is located between the pump light source **110pu** and the sample **150** and may transmit light at one or more wavelengths associated with the pump beam **120pu** and block light at one or more wavelengths associated with the Stokes beam **120S**. For example, the filter **134a** may (i) transmit greater than 90% of the pump beam **120pu** to allow most of the pump beam to be directed to the sample **150** and (ii) block greater than 90% of the Stokes beam **120S** to prevent most of the residual Stokes beam that is transmitted through the sample from being directed to the pump laser **110pu** (e.g., to prevent the Stokes light from interfering with or damaging the pump laser). Optical filter **134b** is located between the Stokes light source **110S** and the sample **150** and may transmit light at one or more wavelengths associated with the Stokes beam **120S** and block light at one or more wavelengths associated with the pump beam **120pu**. For example, the filter **134b** may (i) transmit greater than 90% of the Stokes beam **120S** to allow most of the Stokes beam to be directed to the combiner **130a** and then to the sample **150** and (ii) block greater than 90% of the pump beam **120pu** to prevent most of the residual pump beam that is transmitted through the sample and reflected by the combiner from being directed to the Stokes laser **110S** (e.g., to prevent the pump light from interfering with or damaging the Stokes laser). In other embodiments, in addition to or instead of using an optical filter, a Raman spectroscopy system may include an optical isolator. For example, an optical isolator located between the pump laser **110pu** and the sample **150** may be configured to transmit the pump beam **120pu** and block the residual Stokes beam from being directed to the pump laser.

[0187] The optical filter **134c** in FIG. **40** is located between the sample **150** and the optical receiver **200** and may transmit light at one or more wavelengths associated with the Raman signal **160** and block light at one or more wavelengths associated with the pump beam **120pu**. For example, the filter **134c** may (i) transmit greater than 90% of the incident Raman signal **160** to allow most of the Raman signal to be directed to the optical receiver **200** and (ii) block greater than 90% of the pump beam **120pu** to prevent most of the residual pump beam that is transmitted through the sample from being directed to the optical receiver (e.g., to prevent the pump light from interfering with or adding noise to the detection of the Raman signal).

[0188] A Raman spectroscopy system with counterpropagating pump and Stokes beams may include one or more optical polarizers **136**. The Raman spectroscopy system **100** in FIG. **40** includes a polarizer **136** located between the sample **150** and the optical receiver **200**. The polarizer **136** may be oriented to transmit light with a polarization associated with the Raman signal **160**. Additionally, the polarizer **136** may block a polarization associated with the pump or Stokes beams. For example, the pump beam **120pu** in FIG. **40** may be horizontally polarized, and the Raman signal **160** may be at least partially vertically polarized. The polarizer **136** may be oriented to block horizontally polarized light and transmit vertically polarized light so that the pump beam **120pu** is blocked and the Raman signal **160** is at least partially transmitted by the polarizer.

[0189] A Raman spectroscopy system **100** with counterpropagating pump and Stokes beams may include one or more optical waveplates **132** that change or rotate the

polarization of a beam of light. For example, a Raman spectroscopy system may include a half-wave plate that rotates a polarization of the pump or Stokes beam of light prior to the beam of light being directed to a sample **150**. As another example a Raman spectroscopy system may include a quarter-wave plate that converts a polarization of the pump or Stokes beam of light to a circular or elliptical polarization prior to the beam of light being directed to a sample **150**. The Raman spectroscopy system **100** in FIG. **40** may include an optical waveplate (not illustrated in FIG. **40**) located between the pump light source **110pu** and the sample **150** and configured to change or rotate the polarization of the pump beam **120pu**. Similarly, the Raman spectroscopy system **100** in FIG. **40** may include an optical waveplate (not illustrated in FIG. **40**) located between the Stokes light source **110S** and the sample **150** and configured to change or rotate the polarization of the Stokes beam **120S**.

[0190] FIGS. **41-42** each illustrate example beam angles Θ between counterpropagating pump and Stokes beams. A beam angle between a pump beam **120pu** and a Stokes beam **120S** refers to the angle between the beam paths of the two beams as they are directed to a sample **150**. In FIGS. **41-42**, the beam path of the pump beam **120pu** is represented by a solid-line arrow, and the beam path of the Stokes beam **120S** is represented by a dashed-line arrow. A beam angle Θ may have a value from 0 to 180 degrees. In FIGS. **1-2**, the pump and Stokes beams are co-propagating and have a beam angle of approximately 0 degrees (e.g., the beams propagate to the sample in the same direction and along approximately the same optical axis). A pump beam **120pu** and a Stokes beam **120S** may be referred to as counterpropagating beams if the beam angle Θ between the beams as they propagate to the sample **150** is between 90 degrees and 180 degrees. For example, the beam angle Θ between counterpropagating pump and Stokes beams may be between 170 and 180 degrees. In FIG. **41**, the beam angle Θ between the counterpropagating pump and Stokes beams is approximately 140 degrees. In FIG. **42**, the beam angle is 180 degrees, which indicates that the pump and Stokes beams propagate in opposite directions along approximately the same optical axis. In FIGS. **41-42**, the sample **150** is held in a sample container **152**. For example, the container **152** may be made from glass or a transparent plastic material. The container **152** has a rectangular shape, and in each of FIGS. **41-42**, the counterpropagating pump and Stokes beams enter the container from opposite sides of the container.

[0191] A technical advantage of a Raman spectroscopy system with counterpropagating pump and Stokes beams is the ability to excite and measure molecular vibrational states that may not be detectable or accessible with a Raman spectroscopy system with co-propagating pump and Stokes beams. Coherent Raman spectroscopy is based on a Raman transition from a vibrational state excited by the absorption of a photon from a pump beam and relaxed into a ground state via stimulated emission of another photon into a Stokes beam. A change to the excited vibrational state, and the resulting stimulated emission, may require momentum transfer. In a configuration where the pump and Stokes beams are co-propagating, there may be little net momentum transfer, and this co-propagating configuration may not be able to excite certain vibrational states. A Raman spectroscopy system with counterpropagating pump and Stokes beams provides a directional variance resulting in a net momentum change that allows for additional vibrational

states to be excited. The resulting Raman signals measured by the Raman spectroscopy system may include additional details and vibrational sidebands that are not present in a system with co-propagating pump and Stokes beams. By measuring these Raman spectra that contain additional details and vibrational sidebands, a Raman spectroscopy system may be able to detect and identify various chemical species with an improved sensitivity. Additionally, based on the additional details and vibrational sidebands present in a Raman spectrum, a Raman spectroscopy system with counterpropagating beams may be able to identify or distinguish between multiple chemical species contained within a sample.

[0192] FIGS. 43-46 illustrate four different example beam configurations for a Raman spectroscopy system 100 with counterpropagating pump and Stokes beams. Each of the Raman spectroscopy systems 100 includes an optical combiner 130a located between the sample 150 and the optical receiver 200. Depending on the particular beam configuration, the optical combiner 130a is configured to reflect or transmit a portion of the pump beam 120pu, Stokes beam 120S, or Raman signal 160. In FIGS. 43-44, the combiner 130a reflects or transmits the Stokes beam 120S and the Raman signal 160, and the pump beam 120pu does not encounter the combiner until after interacting with the sample 150. In FIGS. 45-46, the combiner 130a reflects or transmits the pump beam 120pu and the Raman signal 160, and the Stokes beam 120S does not encounter the combiner until after interacting with the sample 150. An optical combiner 130 may be a free-space optical element, a fiber-optic component, or a waveguide-based optical element. For example, the optical combiner 130a in FIGS. 43-46 may be a free-space non-dichroic beamsplitter that reflects approximately R % of an incident beam of light and transmits approximately T % of the incident beam of light, where R+T is approximately equal to 100. An optical combiner 130 may reflect any suitable percentage (R %) of incident light (e.g., 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, or 90%) and may transmit any suitable percentage (T %) of incident light, where R+T is approximately 100. For example, an optical combiner 130 may be a 50/50 beamsplitter that reflects approximately 50% of an incident beam of light and transmits approximately 50% of the incident beam of light.

[0193] The Raman signal 160 that is measured by a Raman spectroscopy system 100 with counterpropagating beams, may propagate in approximately the same direction as the pump beam of light 120pu or the Stokes beam of light 120S. The propagation direction of each of the beams in FIGS. 43-46 corresponds to the direction of the arrow associated with the beam. In FIGS. 43-44, the propagation direction of the Raman signal 160 that is directed from the sample 150 to the optical receiver 200 is approximately equal to the propagation direction of the pump beam of light 120pu. For example, the angle between the propagation directions of the Raman signal 160 and the pump beam of light 120pu may be less than 10 degrees. In FIGS. 45-46, the propagation direction of the Raman signal 160 that is directed from the sample 150 to the optical receiver 200 is approximately equal to the propagation direction of the Stokes beam of light 120S. For example, the angle between the propagation directions of the Raman signal 160 and the Stokes beam of light 120S may be less than 10 degrees.

[0194] In FIG. 43, the optical combiner 130a (i) reflects a portion of the Stokes beam of light 120S received from the

Stokes light source 110S to direct the reflected Stokes beam of light to the sample 150 and (ii) transmits a portion of the Raman signal 160 received from the sample 150 to direct the transmitted Raman signal to the optical receiver 200. The optical combiner 130a may be a 50/50 beamsplitter that reflects approximately 50% of the Stokes beam and transmits approximately 50% of the Stokes beam. The reflected portion of the Stokes beam is directed to the sample 150, and the transmitted portion of the Stokes beam may be discarded or ignored. Similarly, the optical combiner 130a may reflect approximately 50% of the Raman signal and transmit approximately 50% of the Raman signal. The transmitted portion of the Raman signal is directed to the optical receiver 200, and the reflected portion of the Raman signal may be discarded or ignored. Herein, optical beams that may be discarded or ignored may not be included in their corresponding figures for clarity.

[0195] In FIG. 44, the optical combiner 130a (i) transmits a portion of the Stokes beam of light 120S received from the Stokes light source 110S to direct the transmitted Stokes beam of light to the sample 150 and (ii) reflects a portion of the Raman signal 160 received from the sample 150 to direct the reflected Raman signal to the optical receiver 200. The optical combiner 130a may be a 50/50 beamsplitter that reflects approximately 50% of the Stokes beam and transmits approximately 50% of the Stokes beam. The transmitted portion of the Stokes beam is directed to the sample 150, and the reflected portion of the Stokes beam may be discarded or ignored. Similarly, the optical combiner 130a may reflect approximately 50% of the Raman signal and transmit approximately 50% of the Raman signal. The reflected portion of the Raman signal is directed to the optical receiver 200, and the transmitted portion of the Raman signal may be discarded or ignored.

[0196] In FIG. 45, the optical combiner 130a (i) reflects a portion of the pump beam of light 120pu received from the pump light source 110pu to direct the reflected pump beam of light to the sample 150 and (ii) transmits a portion of the Raman signal 160 received from the sample 150 to direct the transmitted Raman signal to the optical receiver 200. The optical combiner 130a may be a 50/50 beamsplitter that reflects approximately 50% of the pump beam and transmits approximately 50% of the pump beam. The reflected portion of the pump beam is directed to the sample 150, and the transmitted portion of the pump beam may be discarded or ignored. Similarly, the optical combiner 130a may reflect approximately 50% of the Raman signal and transmit approximately 50% of the Raman signal. The transmitted portion of the Raman signal is directed to the optical receiver 200, and the reflected portion of the Raman signal may be discarded or ignored.

[0197] In FIG. 46, the optical combiner 130a (i) transmits a portion of the pump beam of light 120pu received from the pump light source 110pu to direct the transmitted pump beam of light to the sample 150 and (ii) reflects a portion of the Raman signal 160 received from the sample 150 to direct the reflected Raman signal to the optical receiver 200. The optical combiner 130a may be a 50/50 beamsplitter that reflects approximately 50% of the pump beam and transmits approximately 50% of the pump beam. The transmitted portion of the pump beam is directed to the sample 150, and the reflected portion of the pump beam may be discarded or ignored. Similarly, the optical combiner 130a may reflect approximately 50% of the Raman signal and transmit

approximately 50% of the Raman signal. The reflected portion of the Raman signal is directed to the optical receiver 200, and the transmitted portion of the Raman signal may be discarded or ignored.

[0198] FIGS. 47-48 each illustrate an example Raman spectroscopy system 100 with counterpropagating pump and Stokes beams and with an optical circulator 131. Instead of using an optical combiner to direct the pump, Stokes, and Raman beams (e.g., as illustrated in FIGS. 43-46), a Raman spectroscopy system with counterpropagating pump and Stokes beams may include an optical circulator 131 to direct the optical beams. The optical circulator 131 in FIGS. 47-48 is a three-port optical element that receives input light at one port and directs that light to exit the circulator from another port. Light that enters the optical circulator 131 at port 1 is directed to exit the circulator from port 2, and light that enters the circulator at port 2 is directed to exit the circulator from port 3. In FIG. 47, the optical circulator 131 receives the Stokes beam 120S at port 1 and directs the Stokes beam to exit the circulator via port 2, where port 2 directs the Stokes beam to the sample 150. Additionally, the optical circulator 131 receives the Raman signal 160 at port 2 and directs the Raman signal to exit the circulator via port 3, where port 3 directs the Raman signal to the optical receiver 200. At least a portion of the pump beam 120pu may also be directed from port 2 to port 3 along with the Raman signal 160, and an optical filter (not illustrated in FIG. 47) located between the circulator 131 and the optical receiver 200 may be used to block the pump beam and transmit the Raman signal. In FIG. 48, the optical circulator 131 receives the pump beam 120pu at port 1 and directs the pump beam to exit the circulator via port 2, where port 2 directs the pump beam to the sample 150. Additionally, the optical circulator 131 receives the Raman signal 160 at port 2 and directs the Raman signal to exit the circulator via port 3, where port 3 directs the Raman signal to the optical receiver 200.

[0199] An optical circulator 131 may be a free-space optical element, a fiber-optic component, or a waveguide-based optical element. For example, the optical circulator 131 in FIG. 47 may be a fiber-optic component, and the Stokes beam 120S may be directed from the Stokes light source 110S to port 1 via an optical fiber. Additionally, the Stokes beam 120S may be directed from port 2 to the sample 150 via an optical fiber, and the Raman signal 160 produced at the sample may be coupled into the same optical fiber and directed to port 2 of the circulator 131. Alternatively, the optical circulator 131 in FIG. 47 may be a free-space optical element that includes one or more bulk optical elements, and the Stokes beam 120S or the Raman signal 120S may be directed to or from the optical circulator as a free-space optical beam.

[0200] An optical circulator 131 may include one or more of the following optical elements: Faraday rotator, polarization optic (e.g., polarization beamsplitter, polarization beam displacer, or birefringent wedge), and optical waveplate. A Faraday rotator may include a permanent magnet and a magneto-optic material (e.g., a yttrium-iron-garnet (YIG)) crystal or a bismuth-doped YIG (Bi:YIG) crystal). The permanent magnet supplies a static magnetic field to the magneto-optic material, which causes the magneto-optic material to alter the polarization of light traveling through the Faraday rotator. The Faraday rotator may rotate the polarization of light traveling through the rotator in the same angular direction (e.g., clockwise or counterclockwise about

the longitudinal axis of the rotator) regardless of the direction of travel of the light through the rotator. For example, an optical circulator 131 may include a Faraday rotator that rotates the polarization of linearly polarized light by 45 degrees, and this polarization rotation may be in the same angular direction regardless of the direction of propagation of the light through the rotator. This non-reciprocal nature of the Faraday effect as provided by the 45-degree polarization rotation of the Faraday rotator creates a one-way path that allows an optical circulator 131 to route light from one port to another as described above and as illustrated in FIGS. 47-48.

[0201] FIG. 49 illustrates an example Raman spectroscopy system 100 with counterpropagating pump and Stokes beams and with a single optical combiner 130. The Raman spectroscopy system 100 in FIG. 40 uses two optical combiners: combiner 130a directs the Stokes beam 120S to the sample 150 and transmits the Raman signal 160, and combiner 130b combines the Raman signal with the probe beam 120pr. The single optical combiner 130 in FIG. 49 performs each of these functions: (i) a portion of the Stokes beam 120S is reflected and directed to the sample 150, (ii) a portion of the Raman signal 160 is transmitted through the combiner, and (iii) a portion of the probe beam 120pr is reflected and combined with the transmitted portion of the Raman signal to produce a combined probe-Raman signal 210 that is sent to a detector 220. A Raman spectroscopy system 100 with a single optical combiner 130 may allow more of the Raman signal to reach the detector 220, which may result in an improved sensitivity compared to a system with two optical combiners. For example, if each of the optical combiners 130 in FIG. 40 is a 50/50 beamsplitter, then approximately 25% of the Raman signal 160 produced at the sample 150 may reach the detector 220. If the optical combiner 130 in FIG. 49 is a 50/50 beamsplitter, then approximately 50% of the Raman signal 160 produced at the sample 150 may reach the detector 220, which corresponds to twice the amount of signal compared to the two-combiner configuration in FIG. 40.

[0202] FIGS. 50-53 illustrate four different example beam configurations for a Raman spectroscopy system 100 with counterpropagating pump and Stokes beams and with a single optical combiner 130. Each of the Raman spectroscopy systems 100 includes a single optical combiner 130 located between the sample 150 and the optical receiver 200. The optical combiner 130 combines a portion of the Raman signal 160 with a portion of the probe beam 120pr to produce a combined probe-Raman signal 210 that is sent to an optical receiver 200. The combined probe-Raman signal 210 may be directed to an optical detector 220 and may be coherently mixed together at the detector. The optical receiver 200 may not include an optical combiner, since the probe beam and Raman signal arrive at the receiver as an already combined beam 210. Depending on the particular beam configuration, the optical combiner 130 is configured to reflect or transmit a portion of the pump beam 120pu, Stokes beam 120S, Raman signal 160, or probe beam 120pr. The optical combiner 130 may be a free-space optical element, a fiber-optic component, or a waveguide-based optical element. For example, the optical combiner 130 may be a free-space non-dichroic beamsplitter that transmits approximately 50% of an incident beam of light and reflects approximately 50% of the incident beam of light.

[0203] In FIG. 50, the optical combiner 130 (i) reflects a portion of the Stokes beam of light 120S to direct the reflected Stokes beam of light to the sample 150 and (ii) transmits a portion of the Raman signal 160 to direct the transmitted Raman signal to the optical receiver 200. Additionally, the optical combiner 130 reflects a portion of the probe beam of light 120_{pr} and combines the reflected probe beam with the transmitted Raman signal to produce a combined probe-Raman signal 210 that is directed to the optical receiver 200. The optical combiner 130 may be a 50/50 beamsplitter that reflects approximately 50% of the Stokes beam and transmits approximately 50% of the Stokes beam. The reflected portion of the Stokes beam is directed to the sample 150, and the transmitted portion of the Stokes beam may be discarded or ignored. Similarly, the optical combiner 130 may reflect approximately 50% of the Raman signal and transmit approximately 50% of the Raman signal. The transmitted portion of the Raman signal is directed to the optical receiver 200, and the reflected portion of the Raman signal may be discarded or ignored. Additionally, the optical combiner 130 may reflect approximately 50% of the probe beam 120_{pr} and transmit approximately 50% of the probe beam. The reflected portion of the probe beam is combined with the transmitted Raman signal and directed to the optical receiver, and the transmitted portion of the probe beam may be discarded or ignored.

[0204] In FIG. 51, the optical combiner 130 (i) transmits a portion of the Stokes beam of light 120S to direct the transmitted Stokes beam of light to the sample 150 and (ii) reflects a portion of the Raman signal 160 to direct the reflected Raman signal to the optical receiver 200. Additionally, the optical combiner 130 transmits a portion of the probe beam of light 120_{pr} and combines the transmitted probe beam with the reflected Raman signal to produce a combined probe-Raman signal 210 that is directed to the optical receiver 200. The optical combiner 130 may be a 50/50 beamsplitter that reflects approximately 50% of the Stokes beam and transmits approximately 50% of the Stokes beam. The transmitted portion of the Stokes beam is directed to the sample 150, and the reflected portion of the Stokes beam may be discarded or ignored. Similarly, the optical combiner 130 may reflect approximately 50% of the Raman signal and transmit approximately 50% of the Raman signal. The reflected portion of the Raman signal is directed to the optical receiver 200, and the transmitted portion of the Raman signal may be discarded or ignored. Additionally, the optical combiner 130 may reflect approximately 50% of the probe beam 120_{pr} and transmit approximately 50% of the probe beam. The transmitted portion of the probe beam is combined with the reflected Raman signal and directed to the optical receiver, and the reflected portion of the probe beam may be discarded or ignored.

[0205] In FIG. 52, the optical combiner 130 (i) reflects a portion of the pump beam of light 120_{pu} to direct the reflected pump beam of light to the sample 150 and (ii) transmits a portion of the Raman signal 160 to direct the transmitted Raman signal to the optical receiver 200. Additionally, the optical combiner 130 reflects a portion of the probe beam of light 120_{pr} and combines the reflected probe beam with the transmitted Raman signal to produce a combined probe-Raman signal 210 that is directed to the optical receiver 200. The optical combiner 130 may be a 50/50 beamsplitter that reflects approximately 50% of the pump beam and transmits approximately 50% of the pump

beam. The reflected portion of the pump beam is directed to the sample 150, and the transmitted portion of the pump beam may be discarded or ignored. Similarly, the optical combiner 130 may reflect approximately 50% of the Raman signal and transmit approximately 50% of the Raman signal. The transmitted portion of the Raman signal is directed to the optical receiver 200, and the reflected portion of the Raman signal may be discarded or ignored. Additionally, the optical combiner 130 may reflect approximately 50% of the probe beam 120_{pr} and transmit approximately 50% of the probe beam. The reflected portion of the probe beam is combined with the transmitted Raman signal and directed to the optical receiver, and the transmitted portion of the probe beam may be discarded or ignored.

[0206] In FIG. 53, the optical combiner 130 (i) transmits a portion of the pump beam of light 120_{pu} to direct the transmitted pump beam of light to the sample 150 and (ii) reflects a portion of the Raman signal 160 to direct the reflected Raman signal to the optical receiver 200. Additionally, the optical combiner 130 transmits a portion of the probe beam of light 120_{pr} and combines the transmitted probe beam with the reflected Raman signal to produce a combined probe-Raman signal 210 that is directed to the optical receiver 200. The optical combiner 130 may be a 50/50 beamsplitter that reflects approximately 50% of the pump beam and transmits approximately 50% of the pump beam. The transmitted portion of the pump beam is directed to the sample 150, and the reflected portion of the pump beam may be discarded or ignored. Similarly, the optical combiner 130 may reflect approximately 50% of the Raman signal and transmit approximately 50% of the Raman signal. The reflected portion of the Raman signal is directed to the optical receiver 200, and the transmitted portion of the Raman signal may be discarded or ignored. Additionally, the optical combiner 130 may reflect approximately 50% of the probe beam 120_{pr} and transmit approximately 50% of the probe beam. The transmitted portion of the probe beam is combined with the reflected Raman signal and directed to the optical receiver, and the reflected portion of the probe beam may be discarded or ignored.

[0207] FIG. 54 illustrates an example Raman spectroscopy system 100 with counterpropagating pump and Stokes beams. The system 100 is configured for polarization-sensitive detection of a Raman signal 160 and includes a polarization-sensitive optical receiver 200. The polarization-sensitive optical receiver 200 includes a horizontal-polarization optical receiver 200_h which detects the horizontal-polarization Raman signal 160-_h and a vertical-polarization optical receiver 200_v which detects the vertical-polarization Raman signal 160-_v.

[0208] The polarization beamsplitter (PBS) 135 splits the Raman signal 160 into a horizontally polarized component (horizontal-polarization Raman signal 160-_h) and a vertically polarized component (vertical-polarization Raman signal 160-_v). Additionally, the PBS 135 splits the probe beam 120_{pr} into a horizontally polarized component (horizontal-polarization probe beam 120_{pr}-_h) and a vertically polarized component (vertical-polarization probe beam 120_{pr}-_v). In other embodiments, a system may include two PBSs (e.g., as illustrated in FIG. 37), where one PBS splits the Raman signal 160 and the other PBS splits the probe beam 120_{pr}.

[0209] The optical combiner 130 in FIG. 54 combines the reflected probe beam 120_{pr} and the transmitted Raman signal 160 to produce a combined probe-Raman signal 210.

The PBS **135** splits the probe-Raman signal **210** into a horizontal probe-Raman signal **210_h** and a vertical probe-Raman signal **210_v**. The horizontal probe-Raman signal **210_h** includes the horizontally polarized portion of the Raman signal (horizontal-polarization Raman signal **160_{-h}**) and the horizontally polarized portion of the probe beam (horizontal-polarization probe beam **120_{pr-h}**). The vertical probe-Raman signal **210_v** includes the vertically polarized portion of the Raman signal (vertical-polarization Raman signal **160_{-v}**) and the vertically polarized portion of the probe beam (vertical-polarization probe beam **120_{pr-v}**). The horizontal probe-Raman signal **210_h** is sent to the horizontal-polarization optical receiver **200_h** for detection of the horizontal-polarization Raman signal **160_{-h}**, and the vertical probe-Raman signal **210_v** is sent to the vertical-polarization optical receiver **200_v** for detection of the vertical-polarization Raman signal **160_{-v}**.

[0210] The horizontal-polarization optical receiver **200_h** may include an optical detector **220** that coherently mixes the horizontal-polarization Raman signal **160_{-h}** and the horizontal-polarization probe beam **120_{pr-h}** to produce a horizontal-polarization photocurrent signal. Similarly, the vertical-polarization optical receiver **200_v** may include an optical detector **220** that coherently mixes the vertical-polarization Raman signal **160_{-v}** and the vertical-polarization probe beam **120_{pr-v}** to produce a vertical-polarization photocurrent signal. Each optical receiver may include an electronic amplifier **232** that produces a voltage signal **234** corresponding to the photocurrent signal. The h-polarization optical receiver **200_h** may produce a digital output signal **240_{-h}** corresponding to the horizontal-polarization photocurrent signal, and the v-polarization optical receiver **200_v** may produce a digital output signal **240_{-v}** corresponding to the vertical-polarization photocurrent signal. A processor may determine one or more characteristics of the horizontal-polarization and vertical-polarization photocurrent signals based on the digital output signals **240_{-h}** and **240_{-v}**. Additionally, a processor may determine a polarization of the Raman signal **160** based on the characteristics of the horizontal-polarization and vertical-polarization photocurrent signals.

[0211] The waveplate **132a** in FIG. **54**, which is located between the pump light source **110_{pu}** and the sample **150**, may be an optical waveplate (e.g., a half-wave plate or a quarter-wave plate) that changes or rotates the polarization of the pump beam **120_{pu}**. Similarly, the waveplate **132b**, which is located between the Stokes light source **110_S** and the sample **150**, may be an optical waveplate that changes or rotates the polarization of the Stokes beam **120_S**. The waveplate **132c** may be (i) a half-wave plate that rotates the polarization of the probe beam **120_{pr}** or (ii) a quarter-wave plate that converts the probe beam **120_{pr}** to a circular or elliptical polarization. For example, the probe light source **110_{pr}** may produce a probe beam **120_{pr}** that is vertically polarized, and the waveplate **132c** may be a half-wave plate that rotates the probe-beam polarization by 45 degrees so that the horizontal-polarization and vertical-polarization probe beams **120_{pr-h}** and **120_{pr-v}** each have approximately equal optical powers.

[0212] FIG. **55** illustrates an example method **400** for measuring a Raman signal **160**. The method may be performed by a Raman spectroscopy system with counterpropagating pump and Stokes beams as described herein. For

example, the method may be performed by the Raman spectroscopy system **100** illustrated in any of FIGS. **40** and **43-54**.

[0213] The method may begin at step **410** in which a pump light source **110_{pu}** produces a pump beam of light **120_{pu}** at a pump frequency (ν_{pu}). At step **420**, a Stokes light source **110_S** produces a Stokes beam of light **120_S** at a Stokes frequency (ν_S), where the pump and Stokes frequencies are offset by a frequency offset Ω (e.g., $\Omega = \nu_{pu} - \nu_S$). For example, the pump and Stokes frequencies may each correspond to a wavelength between approximately 300 nm and approximately 5,000 nm, and the frequency offset Ω may be between approximately 5 THz and approximately 100 THz. The pump and Stokes beams of light may each be produced by a light source that includes one or more laser diodes, where each laser diode is a fixed-wavelength laser diode or a wavelength-tunable laser diode. The pump beam of light may be referred to as a first beam of light at a first frequency (ν_1), and the Stokes beam of light may be referred to as a second beam of light at a second frequency (ν_2).

[0214] At step **430**, the pump and Stokes beams of light are directed to a sample **150** along counterpropagating beam paths. The beam angle Θ between the counterpropagating pump and Stokes beams of light may be between 90 degrees and 180 degrees. For example, the pump beam of light **120_{pu}** and the Stokes beam of light **120_S** may be directed to the sample **150** in approximately opposite directions so that the beam angle Θ between the counterpropagating pump and Stokes beams of light is approximately 180 degrees (e.g., the beam angle may be between 170 and 180 degrees). As another example, the sample **150** may be held in a sample container **152**, and the counterpropagating pump and Stokes beams may enter the container from opposite sides of the container. At step **440**, a Raman signal **160** is collected, where the Raman signal is produced by coherent Raman scattering of the pump and Stokes beams of light at the sample **150**. The pump and Stokes beams of light may be directed to the sample by one or more optical elements that include a free-space optical element, an optical fiber, or an optical waveguide. Similarly, one or more optical elements (which may include a free-space optic, an optical fiber, or an optical waveguide) may collect the Raman signal and direct the Raman signal to an optical receiver.

[0215] At step **450**, an optical receiver **200** detects the Raman signal **160**, where detection of the Raman signal includes steps **452**, **454**, and **456**. At step **452**, a probe light source **110_{pr}** produces a probe beam of light **120_{pr}** at a probe frequency ν_{pr} . The probe light source **110_{pr}** may include a wavelength-tunable laser, where the probe frequency ν_{pr} is adjustable by changing the wavelength of light produced by the wavelength-tunable laser. The probe beam of light may be referred to as a third beam of light at a third frequency (ν_3). At step **454**, a portion of the Raman signal **160** is coherently mixed with at least a portion of the probe beam of light **120_{pr}** to produce a corresponding photocurrent signal **i**. The Raman signal and the probe beam of light may be coherently mixed at a detector **220**, and the detector may produce a photocurrent signal in response to the Raman signal and probe beam. The portion of the Raman signal that is coherently mixed with the probe beam may include optical frequency components of the Raman signal that are within a particular frequency range of the probe frequency, and the particular frequency range may depend on the electronic bandwidth of the detector. For example, the

particular frequency range may extend from approximately $v_{pr}-\Delta f$ to approximately $v_{pr}+\Delta f$, where v_{pr} is the probe frequency, and Δf is the electronic bandwidth of the detector. At step 456, an electronic circuit 230 produces a digital output signal 240 that corresponds to the photocurrent signal i. For example, the electronic circuit 230 may include: (i) an electronic amplifier 232 that amplifies the photocurrent signal to produce a corresponding voltage signal 234 and (ii) a digitizer 236 that produces a digital representation of the voltage signal.

[0216] At step 460, a processor determines a characteristic of the photocurrent signal i based on the digital output signal 240, at which point the method may end. The digital output signal may include a digital representation of a voltage signal 234 that corresponds to the photocurrent signal i, and the characteristic of the photocurrent signal may be determined based on the digital representation of the voltage signal. The characteristic of the photocurrent signal may be associated with the Raman signal 160 and may include a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, or a polarization. Additionally, the characteristic of the photocurrent signal may be associated with a Raman shift at a frequency $v_{pu}-v_{pr}$. Based on one or more determined signal characteristics 162, a processor may determine (i) whether a particular material is present in a sample 150 or (ii) an amount or a concentration of the particular material in the sample. A processor of a Raman spectroscopy system 100 may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. A processor may be similar to the computer system 1000 illustrated in FIG. 56 and described herein.

[0217] FIG. 56 illustrates an example computer system 1000. One or more computer systems 1000 may perform one or more steps of one or more methods described or illustrated herein. One or more computer systems 1000 may provide functionality described or illustrated herein. Software running on one or more computer systems 1000 may perform one or more steps of one or more methods described or illustrated herein or may provide functionality described or illustrated herein. A computer system 1000 may include or may be referred to as a processor, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. Herein, reference to a computer system may encompass one or more computer systems, where appropriate. A processor of a Raman spectroscopy system 100 may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus.

[0218] Computer system 1000 may take any suitable physical form. As an example, computer system 1000 may be an embedded computer system, a system-on-chip (SOC), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a single-board computer system (SBC), a desktop computer system, a laptop or notebook computer system, a graphics processing unit (GPU), a mainframe, a mesh of computer systems, a server, a tablet computer system, or any suitable combination of two or more of these. As another example, all or part of computer system 1000 may be combined with, coupled to, or integrated into a variety of devices, including, but not limited to,

a medical device, wearable device, camera, camcorder, personal digital assistant (PDA), mobile telephone, smartphone, electronic reading device (e.g., an e-reader), game console, smart watch, clock, calculator, television monitor, flat-panel display, computer monitor, eyewear, or head-mounted display. Where appropriate, computer system 1000 may include one or more computer systems 1000; be unitary or distributed; span multiple locations; span multiple machines; span multiple data centers; or reside in a cloud, which may include one or more cloud components in one or more networks. Where appropriate, one or more computer systems 1000 may perform without substantial spatial or temporal limitation one or more steps of one or more methods described or illustrated herein. As an example, one or more computer systems 1000 may perform in real time or in batch mode one or more steps of one or more methods described or illustrated herein. One or more computer systems 1000 may perform at different times or at different locations one or more steps of one or more methods described or illustrated herein, where appropriate.

[0219] As illustrated in the example of FIG. 56, computer system 1000 may include a processor 1010, memory 1020, storage 1030, an input/output (I/O) interface 1040, a communication interface 1050, or a bus 1060. Computer system 1000 may include any suitable number of any suitable components in any suitable arrangement.

[0220] Processor 1010 may include hardware for executing instructions, such as those making up a computer program. As an example, to execute instructions, processor 1010 may retrieve (or fetch) the instructions from an internal register, an internal cache, memory 1020, or storage 1030; decode and execute them; and then write one or more results to an internal register, an internal cache, memory 1020, or storage 1030. A processor 1010 may include one or more internal caches for data, instructions, or addresses. A processor 1010 may include one or more internal registers for data, instructions, or addresses. Processor 1010 may include any suitable number of any suitable internal registers, where appropriate. Where appropriate, processor 1010 may include one or more arithmetic logic units (ALUs); may be a multi-core processor; or may include one or more processors 1010.

[0221] Memory 1020 may include main memory for storing instructions for processor 1010 to execute or data for processor 1010 to operate on. As an example, computer system 1000 may load instructions from storage 1030 or another source (such as, for example, another computer system 1000) to memory 1020. Processor 1010 may then load the instructions from memory 1020 to an internal register or internal cache. To execute the instructions, processor 1010 may retrieve the instructions from the internal register or internal cache and decode them. During or after execution of the instructions, processor 1010 may write one or more results (which may be intermediate or final results) to the internal register or internal cache. Processor 1010 may then write one or more of those results to memory 1020. One or more memory buses (which may each include an address bus and a data bus) may couple processor 1010 to memory 1020. Bus 1060 may include one or more memory buses. Memory 1020 may include random access memory (RAM). This RAM may be volatile memory, where appropriate. Where appropriate, this RAM may be dynamic RAM (DRAM) or static RAM (SRAM). Memory 1020 may include one or more memories 1020, where appropriate.

[0222] Storage **1030** may include mass storage for data or instructions. As an example, storage **1030** may include a hard disk drive (HDD), a floppy disk drive, flash memory, an optical disc, a magneto-optical disc, magnetic tape, or a Universal Serial Bus (USB) drive or a combination of two or more of these. Storage **1030** may include removable or non-removable (or fixed) media, where appropriate. Storage **1030** may be internal or external to computer system **1000**, where appropriate. Storage **1030** may be non-volatile, solid-state memory. Storage **1030** may include read-only memory (ROM). Where appropriate, this ROM may be mask ROM (MROM), programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), flash memory, or a combination of two or more of these. Storage **1030** may include one or more storage control units facilitating communication between processor **1010** and storage **1030**, where appropriate. Where appropriate, storage **1030** may include one or more storages **1030**.

[0223] I/O interface **1040** may include hardware, software, or both, providing one or more interfaces for communication between computer system **1000** and one or more I/O devices. Computer system **1000** may include one or more of these I/O devices, where appropriate. One or more of these I/O devices may enable communication between a person and computer system **1000**. As an example, an I/O device may include a keyboard, keypad, microphone, monitor, mouse, printer, scanner, speaker, camera, stylus, tablet, touch screen, trackball, another suitable I/O device, or any suitable combination of two or more of these. An I/O device may include one or more sensors. Where appropriate, I/O interface **1040** may include one or more device or software drivers enabling processor **1010** to drive one or more of these I/O devices. I/O interface **1040** may include one or more I/O interfaces **1040**, where appropriate.

[0224] Communication interface **1050** may include hardware, software, or both providing one or more interfaces for communication (such as, for example, packet-based communication) between computer system **1000** and one or more other computer systems **1000** or one or more networks. As an example, communication interface **1050** may include a network interface controller (NIC) or network adapter for communicating with an Ethernet or other wire-based network or a wireless NIC (WNIC); a wireless adapter for communicating with a wireless network, such as a WI-FI network; or an optical transmitter (e.g., a laser or a light-emitting diode) or an optical receiver (e.g., a photodetector) for communicating using fiber-optic communication or free-space optical communication. Computer system **1000** may communicate with an ad hoc network, a personal area network (PAN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), or one or more portions of the Internet or a combination of two or more of these. One or more portions of one or more of these networks may be wired or wireless. As an example, computer system **1000** may communicate with a wireless PAN (WPAN) (such as, for example, a BLUETOOTH WPAN), a WI-FI network, a Worldwide Interoperability for Microwave Access (WiMAX) network, a cellular telephone network (such as, for example, a Global System for Mobile Communications (GSM) network), or other suitable wireless network or a combination of two or more of these. As another example, computer system **1000** may communicate using fiber-optic communication based on 100 Gigabit Ethernet (100 GbE), 10 Gigabit Ethernet (10 GbE), or Synchro-

nous Optical Networking (SONET). Computer system **1000** may include any suitable communication interface **1050** for any of these networks, where appropriate. Communication interface **1050** may include one or more communication interfaces **1050**, where appropriate.

[0225] Bus **1060** may include hardware, software, or both coupling components of computer system **1000** to each other. As an example, bus **1060** may include an Accelerated Graphics Port (AGP) or other graphics bus, an Enhanced Industry Standard Architecture (EISA) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an Industry Standard Architecture (ISA) bus, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCIe) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local bus (VLB), or another suitable bus or a combination of two or more of these. Bus **1060** may include one or more buses **1060**, where appropriate.

[0226] Various modules, circuits, systems, methods, or algorithm steps described in connection with the implementations disclosed herein may be implemented as electronic hardware, computer software, or any suitable combination of hardware and software. Computer software (which may be referred to as software, computer-executable code, computer code, a computer program, computer instructions, or instructions) may be used to perform various functions described or illustrated herein, and computer software may be configured to be executed by or to control the operation of computer system **1000**. As an example, computer software may include instructions configured to be executed by processor **1010**. Owing to the interchangeability of hardware and software, the various illustrative logical blocks, modules, circuits, or algorithm steps have been described generally in terms of functionality. Whether such functionality is implemented in hardware, software, or a combination of hardware and software may depend upon the particular application or design constraints imposed on the overall system.

[0227] A computing device may be used to implement various modules, circuits, systems, methods, or algorithm steps disclosed herein. As an example, all or part of a module, circuit, system, method, or algorithm disclosed herein may be implemented or performed by a general-purpose single- or multi-chip processor, a digital signal processor (DSP), an ASIC, a FPGA, a GPU, any other suitable programmable-logic device, discrete gate or transistor logic, discrete hardware components, or any suitable combination thereof. A general-purpose processor may be a microprocessor, or any conventional processor, controller, microcontroller, GPU, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0228] One or more implementations of the subject matter described herein may be implemented as one or more computer programs (e.g., one or more modules of computer-program instructions encoded or stored on a computer-readable non-transitory storage medium). As an example, the steps of a method or algorithm disclosed herein may be implemented in a processor-executable software module

which may reside on a computer-readable non-transitory storage medium. A computer-readable non-transitory storage medium may include any suitable storage medium that may be used to store or transfer computer software and that may be accessed by a computer system. Herein, a computer-readable non-transitory storage medium or media may include one or more semiconductor-based or other integrated circuits (ICs) (such, as for example, field-programmable gate arrays (FPGAs) or application-specific ICs (ASICs)), hard disk drives (HDDs), hybrid hard drives (HHDs), optical discs (e.g., compact discs (CDs), CD-ROM, digital versatile discs (DVDs), Blu-ray discs, or laser discs), optical disc drives (ODDs), magneto-optical discs, magneto-optical drives, floppy diskettes, floppy disk drives (FDDs), magnetic tapes, flash memories, solid-state drives (SSDs), RAM, RAM-drives, ROM, SECURE DIGITAL cards or drives, any other suitable computer-readable non-transitory storage media, or any suitable combination of two or more of these, where appropriate. A computer-readable non-transitory storage medium may be volatile, non-volatile, or a combination of volatile and non-volatile, where appropriate.

[0229] Certain features described herein in the context of separate implementations may also be combined and implemented in a single implementation. Conversely, various features that are described in the context of a single implementation may also be implemented in multiple implementations separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

[0230] While operations may be depicted in the drawings as occurring in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all operations be performed. Further, the drawings may schematically depict one more example processes or methods in the form of a flow diagram or a sequence diagram. However, other operations that are not depicted may be incorporated in the example processes or methods that are schematically illustrated. For example, one or more additional operations may be performed before, after, simultaneously with, or between any of the illustrated operations. Moreover, one or more operations depicted in a diagram may be repeated, where appropriate. Additionally, operations depicted in a diagram may be performed in any suitable order. Furthermore, although particular components, devices, or systems are described herein as carrying out particular operations, any suitable combination of any suitable components, devices, or systems may be used to carry out any suitable operation or combination of operations. In certain circumstances, multitasking or parallel processing operations may be performed. Moreover, the separation of various system components in the implementations described herein should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems may be integrated together in a single software product or packaged into multiple software products.

[0231] Various embodiments have been described in connection with the accompanying drawings. However, it should be understood that the figures may not necessarily be

drawn to scale. As an example, distances or angles depicted in the figures are illustrative and may not necessarily bear an exact relationship to actual dimensions or layout of the devices illustrated.

[0232] One or more of the figures described herein may include example data that is prophetic. For example, the graphs illustrated FIGS. 3-16 and 19 may include or may be referred to as prophetic examples.

[0233] The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes or illustrates respective embodiments herein as including particular components, elements, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend.

[0234] The term “or” as used herein is to be interpreted as an inclusive or meaning any one or any combination, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, the expression “A or B” means “A, B, or both A and B.” As another example, herein, “A, B, or C” means at least one of the following: A; B; C; A and B; A and C; B and C; A, B, and C. An exception to this definition will occur if a combination of elements, devices, steps, or operations is in some way inherently mutually exclusive.

[0235] As used herein, words of approximation such as, without limitation, “approximately,” “substantially,” or “about” refer to a condition that when so modified is understood to not necessarily be absolute or perfect but would be considered close enough to those of ordinary skill in the art to warrant designating the condition as being present. The extent to which the description may vary will depend on how great a change can be instituted and still have one of ordinary skill in the art recognize the modified feature as having the required characteristics or capabilities of the unmodified feature. In general, but subject to the preceding discussion, a numerical value herein that is modified by a word of approximation such as “approximately” may vary from the stated value by $\pm 0.5\%$, 1% , 2% , 3% , 4% , 5% , 10% , 12% , or $\pm 15\%$. The term “substantially constant” refers to a value that varies by less than a particular amount over any suitable time interval. For example, a value that is substantially constant may vary by less than or equal to 20% , 10% , 1% , 0.5% , or 0.1% over a time interval of approximately 104 s, 103 s, 10^2 s, 10 s, 1 s, 100 ms, 10 ms, 1 ms, 100 μ s, 10 μ s, or 1 μ s. The term “substantially constant” may be applied to any suitable value, such as for example, an optical power, an electrical current, a wavelength, an optical or electrical frequency, or an optical or electrical phase.

[0236] As used herein, the terms “first,” “second,” “third,” etc. may be used as labels for nouns that they precede, and these terms may not necessarily imply a particular ordering (e.g., a particular spatial, temporal, or logical ordering). As an example, a system may be described as determining a “first result” and a “second result,” and the terms “first” and “second” may not necessarily imply that the first result is determined before the second result.

[0237] As used herein, the terms “based on” and “based at least in part on” may be used to describe or present one or more factors that affect a determination, and these terms may not exclude additional factors that may affect a determination. A determination may be based solely on those factors which are presented or may be based at least in part on those factors. The phrase “determine A based on B” indicates that B is a factor that affects the determination of A. In some instances, other factors may also contribute to the determination of A. In other instances, A may be determined based solely on B.

[0238] Various example aspects included in this disclosure may be presented in a range format. It should be understood that a description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of a disclosed aspect. Accordingly, the description of a range should be considered to have specifically disclosed all suitable sub-ranges as well as individual numerical values within that range, unless expressly indicated otherwise or indicated otherwise by context. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed sub-ranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6, etc., as well as individual values within that range, for example, 1.1, 2, 2.3, 5, and 5.9.

[0239] Although the foregoing embodiments have been described in some detail for purposes of clarity of understanding, the invention is not limited to the details provided. There may be numerous alternative ways of implementing the invention. The disclosed embodiments are illustrative and not restrictive.

1. A system comprising:
 - a pump light source configured to produce a pump beam of light at a pump frequency;
 - a Stokes light source configured to produce a Stokes beam of light at a Stokes frequency, wherein the pump and Stokes frequencies are offset by a frequency offset Ω ;
 - one or more optical elements configured to:
 - direct the pump and Stokes beams of light to a sample along counterpropagating beam paths; and
 - collect a Raman signal produced by coherent Raman scattering of the pump and Stokes beams of light at the sample;
 - an optical receiver configured to detect the Raman signal, the optical receiver comprising:
 - a probe light source configured to produce a probe beam of light at a probe frequency;
 - an optical detector configured to coherently mix a portion of the Raman signal with at least a portion of the probe beam of light to produce a corresponding photocurrent signal; and
 - an electronic circuit configured to produce a digital output signal corresponding to the photocurrent signal; and
 - a processor configured to determine a characteristic of the photocurrent signal based on the digital output signal.
2. The system of claim 1, wherein a beam angle between the counterpropagating pump and Stokes beams directed to the sample is between 90 degrees and 180 degrees.
3. The system of claim 2, wherein the beam angle between the counterpropagating pump and Stokes beams is approximately equal to 180 degrees.

4. The system of claim 1, wherein the sample is held in a container, and the counterpropagating pump and Stokes beams enter the container from opposite sides of the container.

5. The system of claim 1, wherein the Raman signal propagates from the sample along a propagation direction that is approximately equal to a propagation direction of the pump beam of light, wherein an angle between the propagation directions of the Raman signal and the pump beam of light is less than 10 degrees.

6. The system of claim 1, wherein the Raman signal propagates from the sample along a propagation direction that is approximately equal to a propagation direction of the Stokes beam of light, wherein an angle between the propagation directions of the Raman signal and the Stokes beam of light is less than 10 degrees.

7. The system of claim 1, wherein the optical elements comprise an optical combiner located between the sample and the optical receiver, the optical combiner configured to:

- reflect a portion of the Stokes beam of light to direct the reflected Stokes beam of light to the sample; and
- transmit a portion of the Raman signal to direct the transmitted Raman signal to the optical receiver.

8. The system of claim 7, wherein the optical combiner is further configured to reflect a portion of the probe beam of light and combine the reflected portion of the probe beam of light with the transmitted portion of the Raman signal to produce a combined probe-Raman signal that is directed to the optical receiver.

9. The system of claim 1, wherein the optical elements comprise an optical combiner located between the sample and the optical receiver, the optical combiner configured to:

- transmit a portion of the Stokes beam of light to direct the transmitted Stokes beam of light to the sample; and
- reflect a portion of the Raman signal to direct the reflected Raman signal to the optical receiver.

10. The system of claim 8, wherein the optical combiner is further configured to transmit a portion of the probe beam of light and combine the transmitted portion of the probe beam of light with the reflected portion of the Raman signal to produce a combined probe-Raman signal that is directed to the optical receiver.

11. The system of claim 1, wherein the optical elements comprise an optical combiner located between the sample and the optical receiver, the optical combiner configured to:

- reflect a portion of the pump beam of light to direct the reflected pump beam of light to the sample; and
- transmit a portion of the Raman signal to direct the transmitted Raman signal to the optical receiver.

12. The system of claim 11, wherein the optical combiner is further configured to reflect a portion of the probe beam of light and combine the reflected portion of the probe beam of light with the transmitted portion of the Raman signal to produce a combined probe-Raman signal that is directed to the optical receiver.

13. The system of claim 1, wherein the optical elements comprise an optical combiner located between the sample and the optical receiver, the optical combiner configured to:

- transmit a portion of the pump beam of light to direct the transmitted pump beam of light to the sample; and
- reflect a portion of the Raman signal to direct the reflected Raman signal to the optical receiver.

14. The system of claim 13, wherein the optical combiner is further configured to transmit a portion of the probe beam

of light and combine the transmitted portion of the probe beam of light with the reflected portion of the Raman signal to produce a combined probe-Raman signal that is directed to the optical receiver.

15. The system of claim 1, wherein the optical elements comprise an optical circulator comprising a first port, second port, and third port, wherein the optical circulator is configured to:

receive the Stokes beam at the first port, and direct the Stokes beam to exit the circulator via the second port, wherein the second port directs the Stokes beam to the sample; and

receive the Raman signal at the second port and direct the Raman signal to exit the circulator via the third port, wherein the third port directs the Raman signal to the optical receiver.

16. The system of claim 1, further comprising an optical filter located between the sample and the optical receiver, wherein the optical filter is configured to transmit the Raman signal and block the pump beam of light.

17. The system of claim 1, wherein the optical elements comprise a lens configured to focus the pump beam of light into the sample.

18. The system of claim 1, wherein the optical elements comprise a lens configured to:

focus the Stokes beam of light into the sample; and

collect the Raman signal to produce a Raman-signal beam that is directed to the optical receiver.

19. The system of claim 1, further comprising an optical polarizer located between the sample and the optical receiver, wherein the polarizer is oriented to transmit light with a polarization associated with the Raman signal.

20. The system of claim 1, further comprising a half-wave plate configured to rotate a polarization of the pump or Stokes beam of light prior to being directed to the sample.

21. The system of claim 1, further comprising a quarter-wave plate configured to convert a polarization of the pump or Stokes beam of light to a circular or elliptical polarization prior to being directed to the sample.

22. The system of claim 1, wherein:

the system further comprises a polarization beamsplitter configured to split the Raman signal into a horizontal-polarization Raman signal and a vertical-polarization Raman signal; and

the optical receiver comprises (i) a horizontal-polarization optical receiver configured to detect the horizontal-polarization Raman signal and (ii) a vertical-polarization optical receiver configured to detect the vertical-polarization Raman signal, wherein:

the optical detector is part of the horizontal-polarization optical receiver and is configured to coherently mix the horizontal-polarization Raman signal with a horizontal-polarization probe beam; and

the vertical-polarization optical receiver comprises another optical detector configured to coherently mix the vertical-polarization Raman signal with a vertical-polarization probe beam.

23. The system of claim 22, wherein the polarization beamsplitter is further configured to split the probe beam of light to produce the horizontal-polarization probe beam and the vertical-polarization probe beam.

24. The system of claim 1, wherein the probe light source comprises a wavelength-tunable laser, wherein the probe frequency is adjustable by changing a wavelength of light produced by the wavelength-tunable laser.

25. The system of claim 1, wherein:

the frequency offset Ω is approximately equal to a vibrational frequency of a particular material; and

the processor is further configured to determine, based on the determined characteristic of the photocurrent signal, (i) whether the particular material is present in the sample or (ii) an amount or a concentration of the particular material in the sample.

26. The system of claim 1, wherein the portion of the Raman signal that is coherently mixed with the probe beam of light comprises optical frequency components of the Raman signal within a particular frequency range of the probe frequency, wherein the particular frequency range is based on an electronic bandwidth of the detector.

27. The system of claim 1, wherein the photocurrent signal comprises a coherent-mixing term that is proportional to a product of (i) an amplitude of an electric field of the Raman signal and (ii) an amplitude of an electric field of the probe beam of light.

28. The system of claim 1, wherein the electronic circuit comprises:

an electronic amplifier configured to amplify the photocurrent signal to produce a voltage signal corresponding to the photocurrent signal; and

a digitizer configured to produce a digital representation of the voltage signal, wherein the digital output signal comprises the digital representation of the voltage signal, wherein:

the processor is configured to determine the characteristic of the photocurrent signal based on the digital representation of the voltage signal, wherein the characteristic comprises one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization.

29. A method for measuring a Raman signal, the method comprising:

producing a pump beam of light at a pump frequency; producing a Stokes beam of light at a Stokes frequency, wherein the pump and Stokes frequencies are offset by a frequency offset Ω ;

directing the pump and Stokes beams of light to a sample along counterpropagating beam paths;

collecting a Raman signal produced by coherent Raman scattering of the pump and Stokes beams of light at the sample;

detecting the Raman signal, comprising:

producing a probe beam of light at a probe frequency; coherently mixing a portion of the Raman signal with at least a portion of the probe beam of light to produce a corresponding photocurrent signal; and producing a digital output signal corresponding to the photocurrent signal; and

determining a characteristic of the photocurrent signal based on the digital output signal.

* * * * *